Subject

Date : / /

MOHSIN IBNA HOSSAIN
AIUB, DLC Note

DIGITAL LOGIC & CIRCUITS (DLC)

Date : / /

#Logic Gates

Logic gate:-

⇒ It is a physical device which perform logic operation on one or more logical inputs, and produces a single logical output.

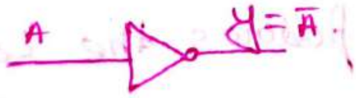
There are three catagory:-

1. Basic Gates: NOT, AND, OR
2. Universal Gates: NAND, NOR
2. Arithmetic Gates: X-OR, X-NOR

NOT Gate:-

⇒ It is a type of of logic gate that Inverts the provided Input. Refferred to as an invetefer.

Symbol for "NOT" gate:-



Truth table:-

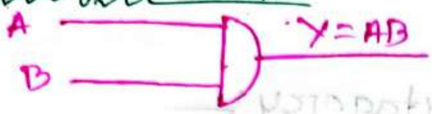
A	$X=\overline{A}$
0	1
1	0

AND Gate:

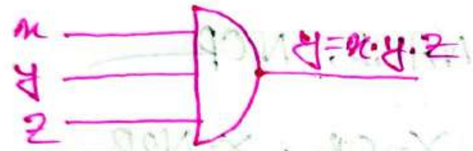
⇒ Output is high (1) when all the inputs are high.

Symbol:

1. For two inputs:



2. For three inputs:



Truth Table:

A	B	Y=A·B
0	0	0
0	1	0
1	0	0
1	1	1

Now, check whether "AND Gate" follows those two laws or not:

Associative Law: $[A\cdot(B\cdot C) = (A\cdot B)\cdot C]$

Let, $A=1$, $B=0$, $C=1$

$A\cdot(B\cdot C) = 1\cdot(0\cdot 1) = 1\cdot 0 = 0$

$(A\cdot B)\cdot C = (1\cdot 0)\cdot 1 = 0\cdot 1 = 0$

∴ It's follows the Law.

Commutative Law: $[A\cdot B = B\cdot A]$:

Let, $A=1$, $B=0$

∴ $A\cdot B = 1\cdot 0 = 0$

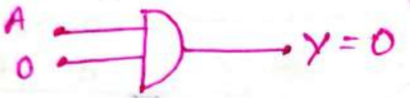
$B\cdot A = 0\cdot 1 = 0$

∴ It's follows the Law.

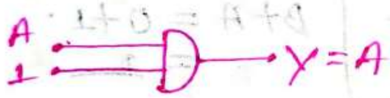
Enable & Disable:—

	A	B	Y
Disable O/P=0	0	0	0
	0	1	0
Enable buffer	1	0	0
	1	1	1

∴ "0" → disable with O/P = 0



"1" → Enable with O/P = ~~Buffer~~.



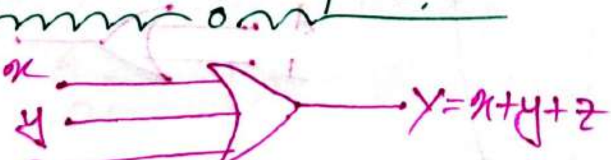
OR-Gate:—

⇒ Output is high when any one or all input are high.

Symbol:—

1. For two inputs:—

2. For three inputs:—



Truth Table:—

A	B	Y=A+B
0	0	0
0	1	1
1	0	1
1	1	1

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Check Whether "OR" Gate Follows Those two laws or Not.

Associative law:- $[(A+B)+C = A+(B+C)]$

Let, $A = 1$, $B = 0$, $C = 1$

$\therefore (A+B)+C = (1+0)+1 = 1+1 = 1$

$A+(B+C) = 1+(0+1) = 1+1 = 1$

$\therefore$ It's follows the law.

Commutative Law:- $[A+B = B+A]$

Let, $A=1$, $B=0$

$\therefore A+B = 1+0 = 1$

$B+A = 0+1 = 1$

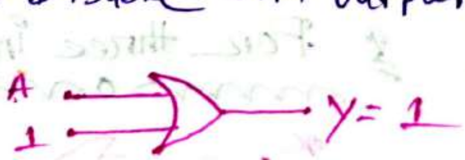
$\therefore$ It's also follows the law.

Enable & Disable:-

	A	B	Y=A+B
Enable Buffer	0	0	0
	0	1	1
Disable O/P=1	1	0	1
	1	1	1

"0" → Enable with act as Buffer.

"1" → Disable with output = 1



Check Whether "OR" Gate Follows Those two laws or Not.

Associative Law:- [(A+B) + C = A + (B+C)]

Let, $A = 1$, $B = 0$, $C = 1$

$\therefore (A+B)+C = (1+0)+1 = 1+1 = 1$

$A+(B+C) = 1+(0+1) = 1+1 = 1$

$\therefore$ It's follows the law.

Commutative Law:- [A+B = B+A]

Let, $A = 1$, $B = 0$

$\therefore A+B = 1+0 = 1$

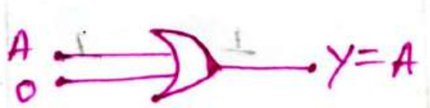
$B+A = 0+1 = 1$

$\therefore$ It's also follows the law.

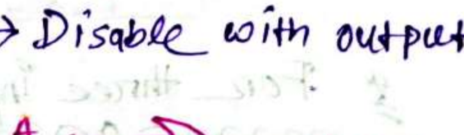
Enable & Disable:-

	A	B	Y=A+B
Enable Buffer	0	0	0
	0	1	1
Disable O/P=1	1	0	1
	1	1	1

"0" → Enable with act as Buffer.



"1" → Disable with output = 1



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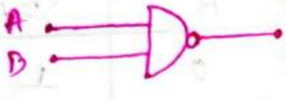
NAND Gate:-

⇒ It is a Combination of digital logic "AND" gate & a "NOT" gate connected together in Series.

Symbol:-



Truth Table:-

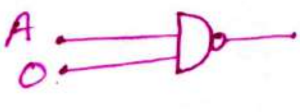
A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

Note:- 1. "NAND" Gate does not follows the "associative" laws.
2. But It's follow the "Commutative" Laws.

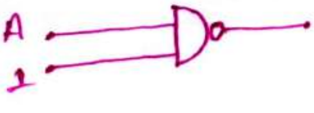
Enable & Disable:-

	A	B	Y
Disable O/P=1	0	0	1
	0	1	1
Enable inverter	1	0	1
	1	1	0

"0" → Disable with O/P = 1



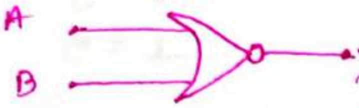
"1" → Enable with Act As Inverter



NOR-Gate:—

⇒ It's a Combination of digital Logic "OR" gate & a "NOT" gate Connected together in Series.

Symbol:—



$Y = \overline{A+B}$

$= \overline{A} \cdot \overline{B}$

Truth Table:—

A	B	$Y = \overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

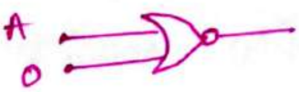
Note:—

1. "NOR"-Gate doesnot follows the "associative Laws".
2. But It's follows the Commutative Laws.

Enable & Disable:—

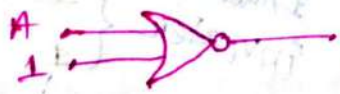
	A	B	Y
Enable Inverter	0	0	1
	0	1	0
Disable o/p = 0	1	0	0
	1	1	0

"0" → Enable with act as Inverter



$Y = \overline{A}$

"1" → Disable with o/p = 0



Y = 0

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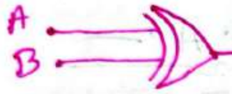
X-OR-Gate:-

⇒ Output is high (1) when the number of 1 inputs is odd.

Symbol:-



$Y = A \oplus B$

$= A \cdot \overline{B} + \overline{A} \cdot B$

Truth Table:-

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Enable & Disable:-

	A	B	Y
Enable Buffer	0	0	0
	0	1	1
Enable Inverter	1	0	1
	1	1	0

Controlled Inverter:-

⇒ Since Both case are Enable so that we called it Controlled Inventer.

Ex:- (1)

$Y = A$ ⇒ $Y = A$

(2)

$Y = \overline{A}$ ⇒ $Y = \overline{A}$

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Properties of X-OR Gate:—

(i)

① $A \oplus A = 0$

② $A \oplus \bar{A} = 1$

③ $A \oplus 0 = A$

④ $A \oplus 1 = \bar{A}$

For:- $A \oplus A$:-

$Y = A\bar{B} + B\bar{A}$

$\therefore A \oplus A = A\bar{A} + A \cdot \bar{A} = A \cdot \bar{A} = 0$

For:- $A \oplus \bar{A}$:—

$Y = A \cdot \bar{\bar{A}} + \bar{A} \cdot \bar{A}$

$= A + \bar{A} = 1$

(ii) if $A \oplus B \oplus C = 0$ then,

$A \oplus B = C$,

$A \oplus C = B$,

$B \oplus C = A$

(iii) $A \oplus A \oplus A \oplus A \oplus \ldots$

For: Even:-

$A \oplus A = 0$

For odd:—

$A \oplus A \oplus A = A$

Problem: 1:—

(a) Find out Y for 20 X-OR gate.

(b) Find out Y for 31 X-OR gate.

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(a)

For y_1:-

$y_1 = \bar{x}$ $[\because y_1 = x \oplus 1]$

For y_2:- $y_2 = \bar{\bar{x}} = x$ $[\because y_2 = \bar{x} \oplus 1]$

$\therefore$ After 2 X-OR gate we got $= x$

$\therefore$ After 4-X-OR " we are going to get $= x$ again.

$\therefore$ for 20 X-OR gate $Y = x$

also for all even number of X-OR gate

$y = x$.

(b)

For y_3:-

$y_3 = x \oplus 1 = \bar{x}$

$\therefore$ After 1 & 3 X-OR gate we got $y = \bar{x}$

$\therefore$ for 31 X-OR gate we get $y = \bar{x}$ again

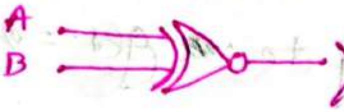
also for all odd number of X-OR gate.

$y = \bar{x}$

X-NOR Gate:-

⇒ output is high (1) when number of 0 & 1 inputs are even. otherwise output is low (0).

Symbol:-



$Y = A \odot B$

$= \bar{A}\cdot\bar{B} + A\cdot B$

Truth Table:-

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Notes:- X-NOR gates follow Commutative & associative laws.

Enable & Disable:-

	A	B	Y
Enable Converter	0	0	1
	0	1	0
Enable Buffer	1	0	0
	1	1	1

①

$\therefore$ $Y = \bar{A}$ ⇒ A —▷o— $\bar{A} = Y$

②

$Y = A$ ⇒ A —▷— $A = Y$

Properties of X-NOR gate:—

① (i) $A \odot A = 1$

(ii) $A \odot \overline{A} = 0$

(iii) $A \odot 0 = \overline{A}$

(iv) $A \odot 1 = A$

For: $A \odot A$:—

$Y = \overline{A} \cdot \overline{A} + A \cdot A$

$= \overline{A} + A = 1$

For $A \odot \overline{A}$:—

$Y = \overline{A} \cdot \overline{\overline{A}} + A \cdot \overline{A}$

$= \overline{A} \cdot A + A \cdot \overline{A}$

$= \overline{A} \cdot A = 0$

② $A \odot A \odot A \odot A \odot \ldots\ldots$

For Even:—

$A \odot A = 1$

For odd:—

$A \odot A \odot A = A$

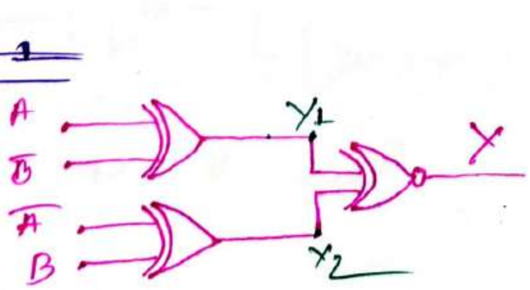
③ XOR and XNOR → Complement each other → inputs even

XOR and XNOR → Same → inputs are odd.

Ex:- ① $A \oplus B \oplus C = A \odot B \odot C$ → odd

② $A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$ → even

problem:- 1



ⓐ Find the Y

⇒ $Y_1 = A \oplus \overline{B}$

$Y_2 = \overline{A} \oplus B$

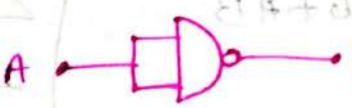
$\therefore Y = (A \oplus \overline{B}) \odot (\overline{A} \oplus B)$

$= \overline{(A \oplus \overline{B}) \oplus (\overline{A} \oplus B)}$

NAND Gate As Universal Gate:-

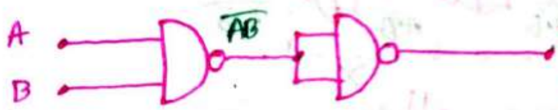
⇒ Using only NAND gates, we can create AND, OR, NOT, NOR, XOR, XNOR gates. That's why it's called Universal gate.

NAND as NOT:- (1)



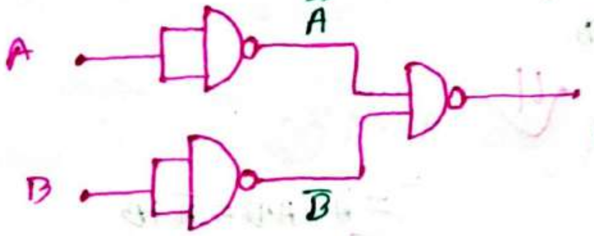
$Y = \overline{A}$

NAND as AND:- (2)



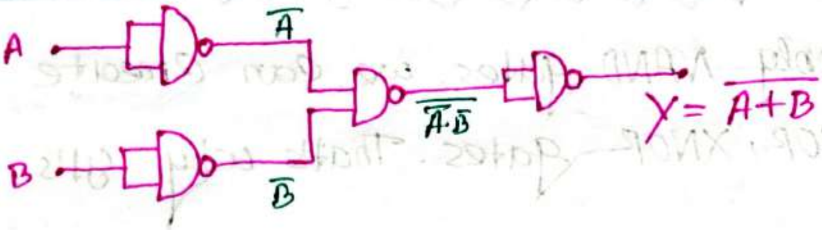
$Y = \overline{\overline{AB}} = AB$

NAND as OR:- (3)



$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

NAND as NOR:—



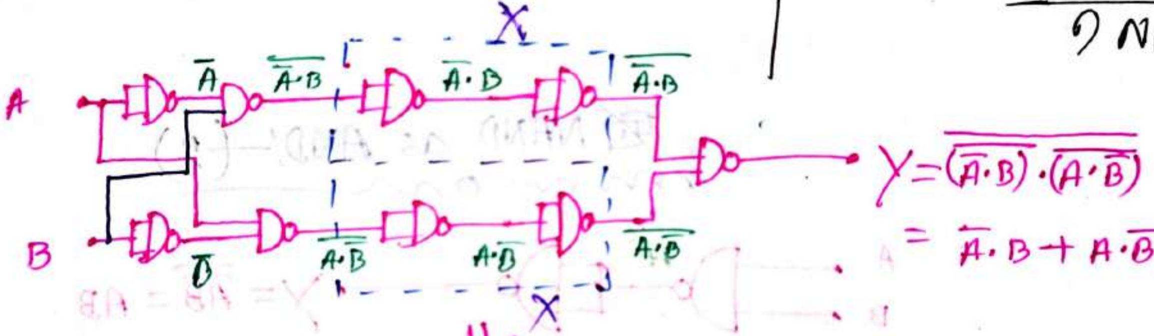
NAND as X-OR:— (4)

For X-OR

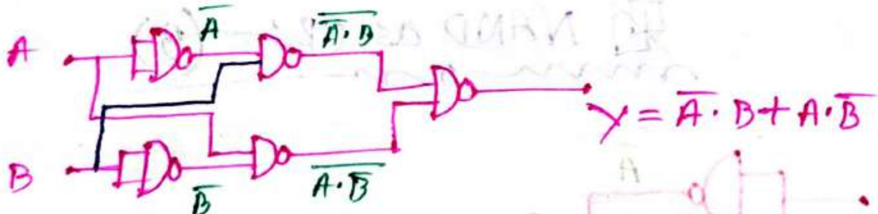
We know:—

$$Y = A\overline{B} + \overline{A}B$$

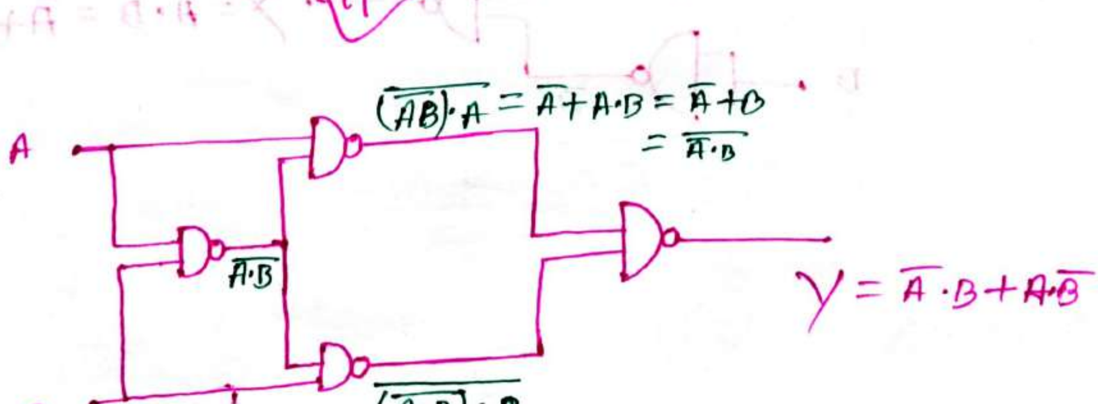
2 NOT → 2 NAND
2 AND → 4 NAND
1 OR → 3 NAND
9 NAND



⇓



⇓



$\overline{(\overline{AB})\cdot A} = \overline{A} + A\cdot B = \overline{A} + B$
$= \overline{A\cdot\overline{B}}$

$\overline{(\overline{A\cdot B})\cdot B}$
$= \overline{B} + A\cdot B$
$= \overline{B} + A = \overline{B\cdot\overline{A}}$

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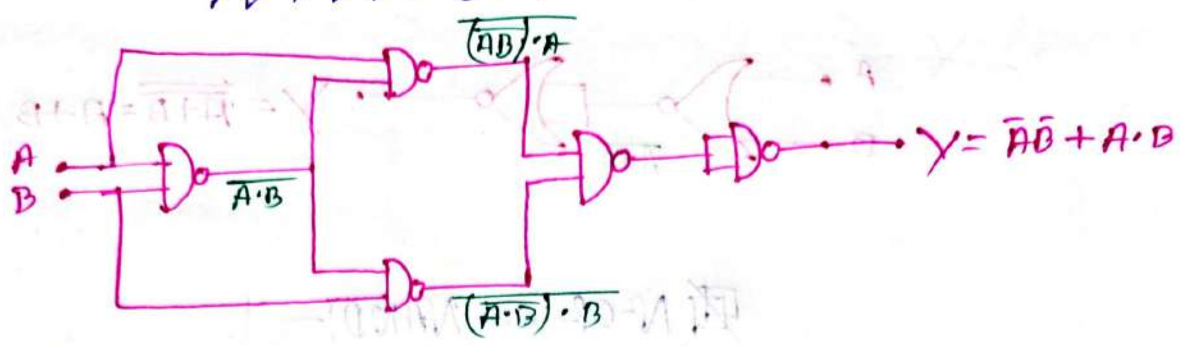
NAND as X-NOR:- (5)

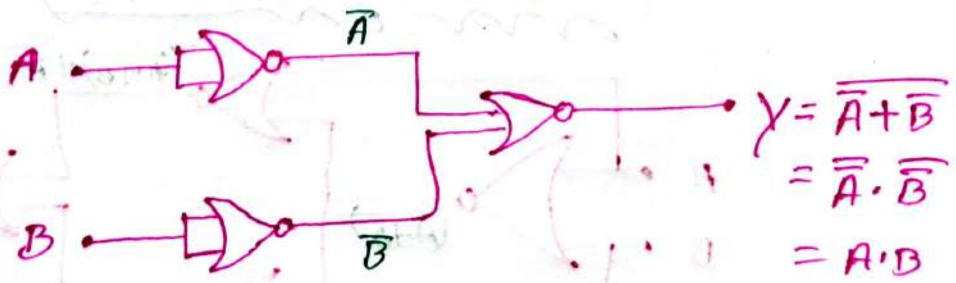


N-OR Gate as Universal Gate:-

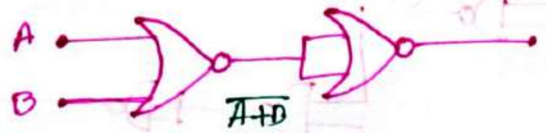
⇒ Using only N-OR gate, we can we can create AND, OR, NOT, NAND, X-OR, X-NOR gates. thats why we called it Universal gate.

N-OR as NOT:- (1)

N-OR as AND:- (2)

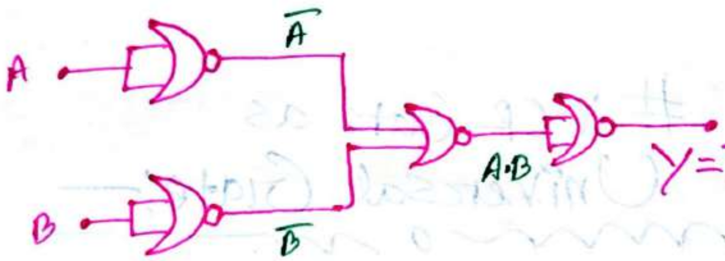


N-OR as OR: — (2)



$Y = \overline{\overline{A+B}} = A+B$

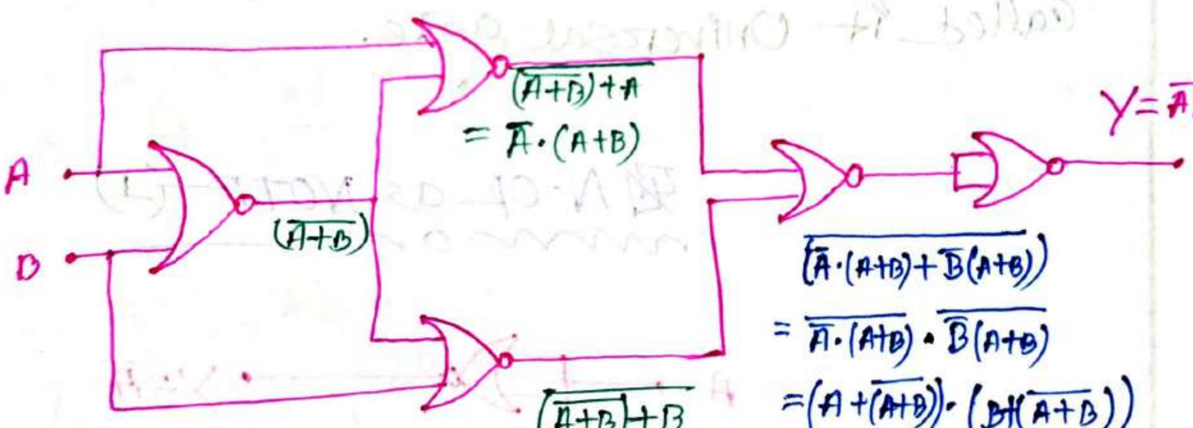
N-OR as NAND! —



$Y = \overline{A \cdot B}$

N-OR as X-OR: — (5)

$Y = \overline{A}B + A\overline{B}$



$$\overline{\left(\overline{A} \cdot (A+B) + \overline{B}(A+B)\right)}$$

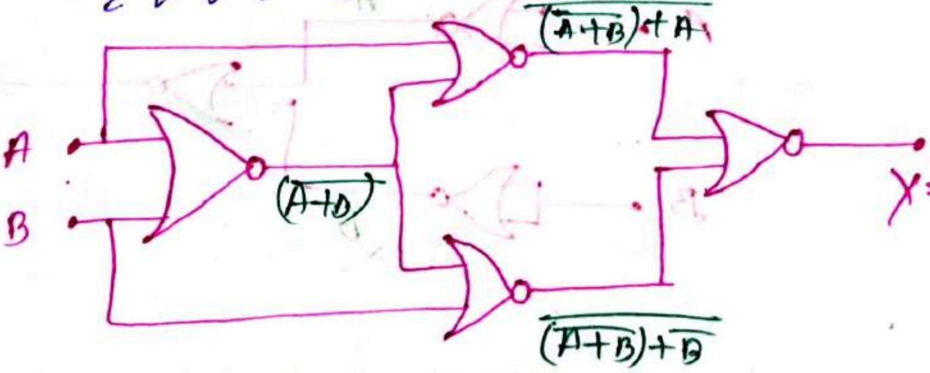
$$= \overline{\overline{A} \cdot (A+B)} \cdot \overline{\overline{B}(A+B)}$$

$$= \left(A + (\overline{A+B})\right) \cdot \left(B + (\overline{A+B})\right)$$

$$= (\overline{A+B}) + A \cdot B$$

$$= \overline{A} \cdot \overline{B} + A \cdot B$$

N-OR as X-NOR! — (4)



$Y = \overline{A} \cdot \overline{B} + A \cdot B$

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problem :1:—

⇒ Implement the following expression with $Y=(AB'+C)'D+E$

(a) NAND Gates Only.

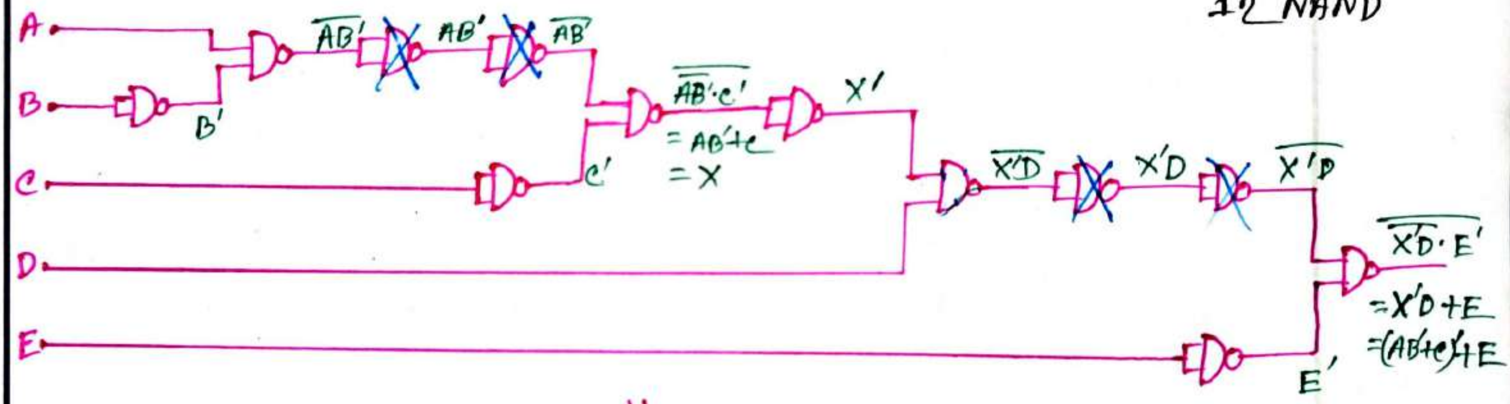
(b) NOR Gates only.

(a)

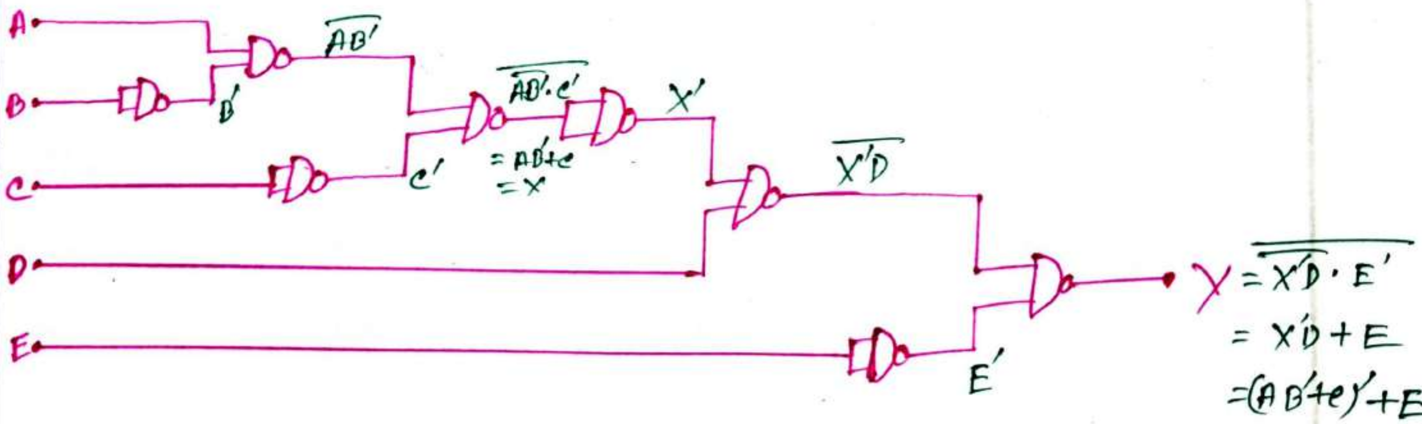
Is given:—

$Y=(AB'+C)'D+E$

2 NOT → 2 NAND
2 AND → 4 NAND
2 OR → 6 NAND
12 NAND



⇓



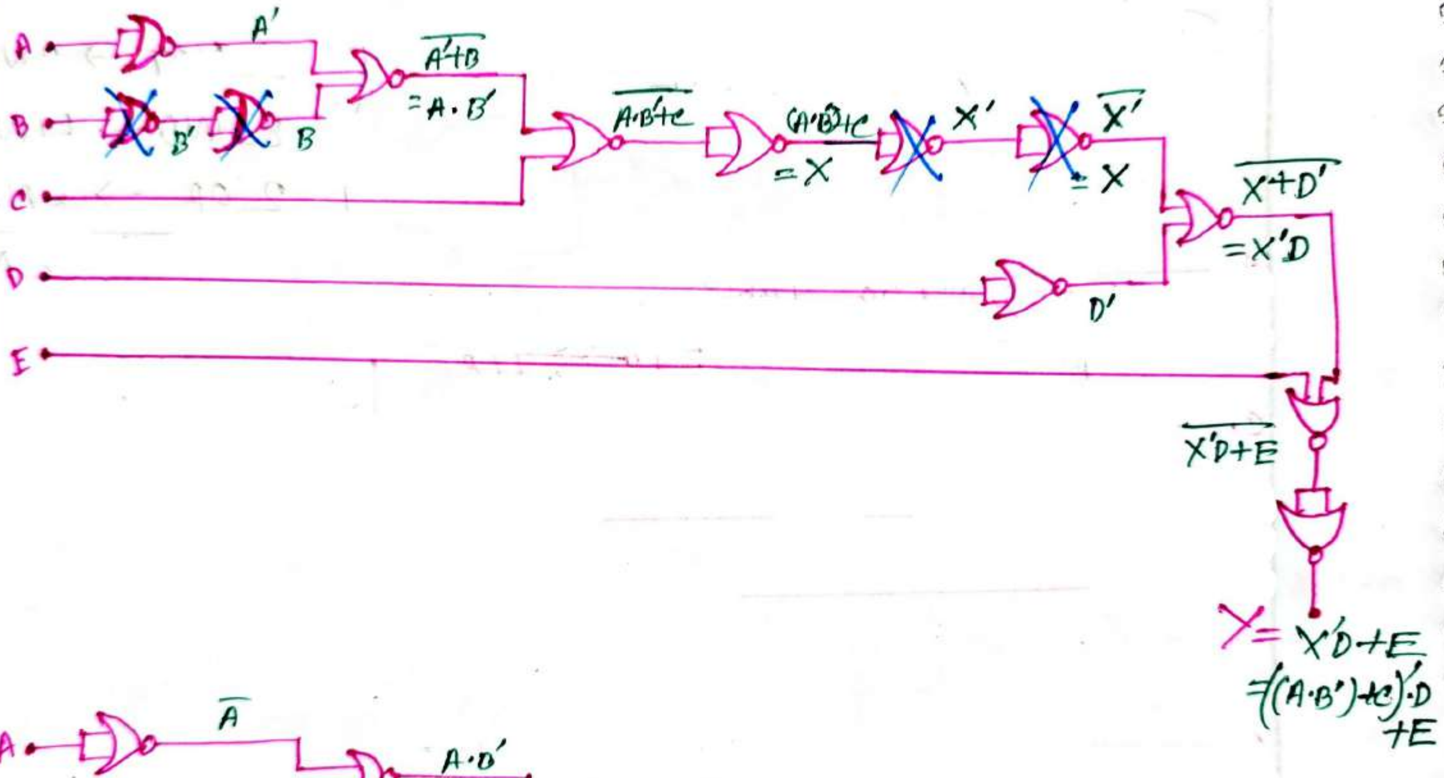
(b)

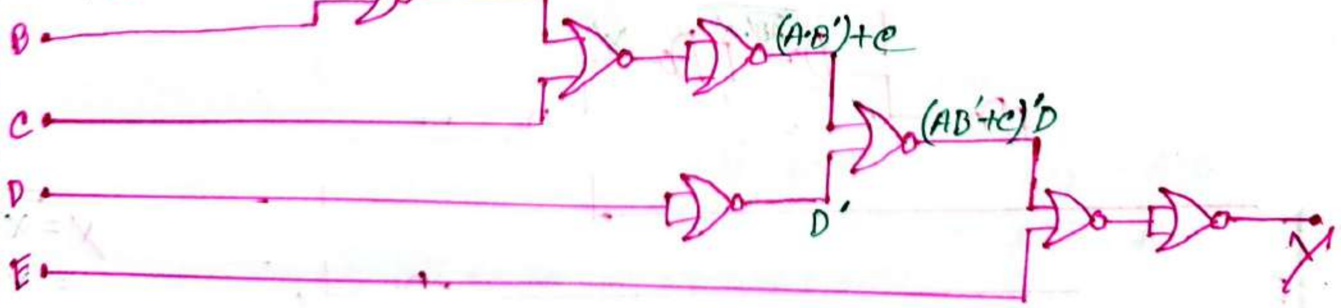
is given:

$$Y = (AB' + C)' \cdot D + E$$

- 2 NOT → 2 NOR
- 2 AND → 2 × 3 = 6 NOR
- 2 OR → 4 NOR

12 NOR





problem : 2 :-

⇒ Implement the following expression $Y = AB + BC + CA$

(a) NAND Gate Only.

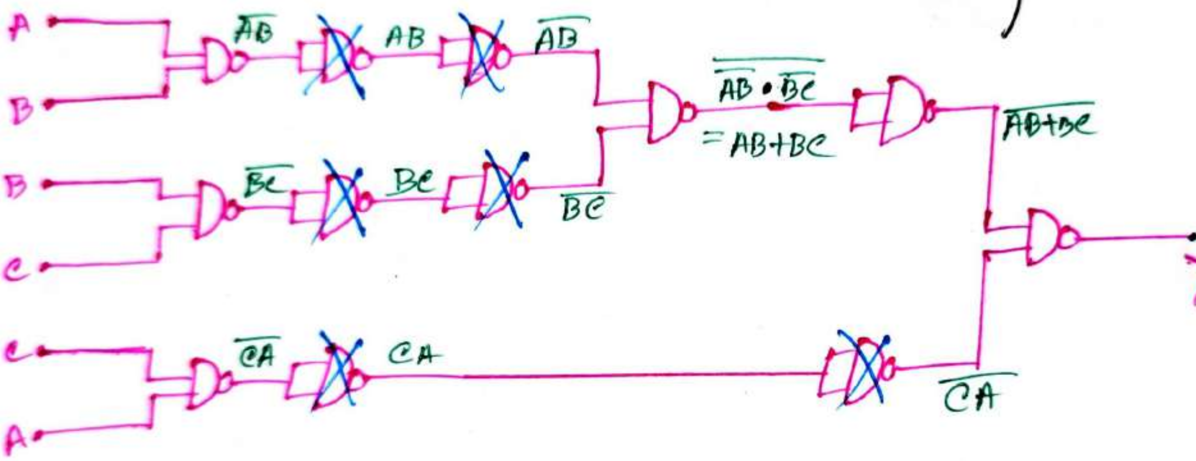
(b) NOR Gate Only.

(a)

is given :-

$F = AB + BC + CA$

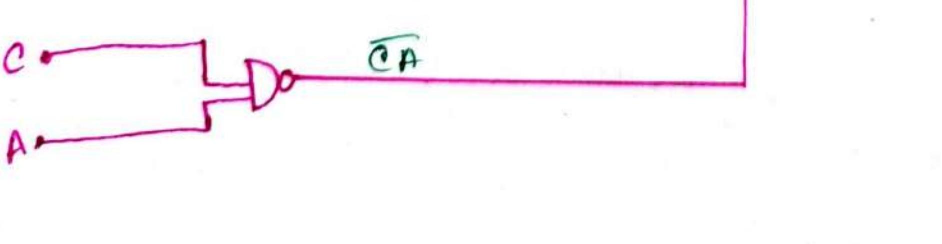
3 AND → 6 NAND
2 OR → 6 NAND
12 NAND



$Y = \overline{(\overline{AB+BC})\cdot\overline{CA}}$

$= AB + BC + CA$

॥



$Y = \overline{\overline{X}\cdot\overline{CA}}$

$= \overline{\overline{X}} + \overline{\overline{CA}}$

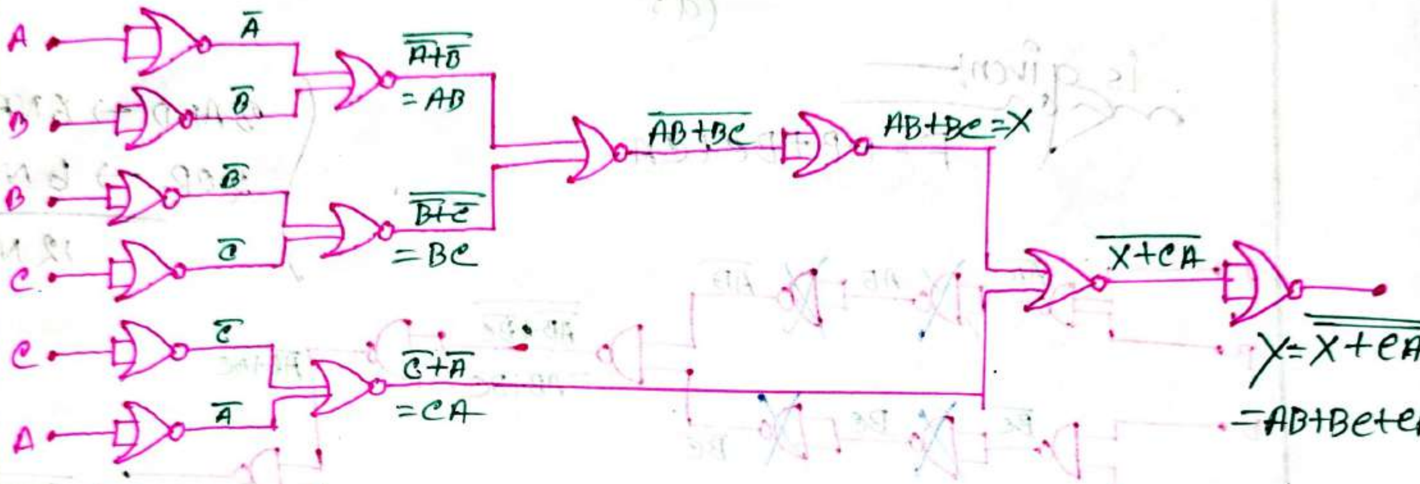
$= AB + BC + CA$

(b)

~~soln~~ is given:—

$F = AB + BC + CA$

$\left\{ \begin{array}{l} 3\,AND \rightarrow 9\,NOR \\ 2\,OR \rightarrow 4\,NOR \\ \hline 13\,NOR \end{array} \right.$



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#BOOLEAN ALGEBRA

Boolean Algebra:—

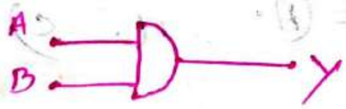
⇒ It is the set of rules, used to simplify the given Expression without changing its functionality.

Complement Rules:—

1. A Complement $= \overline{A} = A'$
2. $(A')' = A$

AND operator

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



① $A \cdot A = A$

② $A \cdot \overline{A} = 0$

③ $A \cdot 0 = 0$

④ $A \cdot 1 = A$

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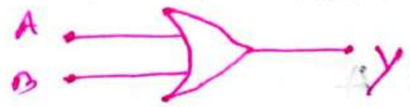
OR-Operation :—

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1



(1) $A + A = A$

(2) $A + \bar{A} = 1$

(3) $A + 0 = A$

(4) $A + 1 = 1$

Distributive Law :—

(1) $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$

(2) $A + (B \cdot C) = (A + B) \cdot (A + C)$

(3) $A + \bar{A}B = (A + \bar{A}) \cdot (A + B)$
$= 1 \cdot (A + B)$
$= A + B$

(3.1) $A + \bar{A}B = A + B$

(3.2) $\bar{A} + AB = \bar{A} + B$

(3.3) $A + AB = A$

(3.4) $(A + B) \cdot (A + C) = A + B \cdot C$

Commutative Law:—

1. $A + B = B + A$

2. $A \cdot B = B \cdot A$

Associative Law:—

1. $(A \cdot B) \cdot C = A \cdot (B \cdot C)$

Priority:—

1. NOT
2. AND
3. OR

De Morgans Law:—

1. $\overline{A+B} = \overline{A} \cdot \overline{B}$

2. $(\overline{A \cdot B}) = \overline{A} + \overline{B}$

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problem: 1:

⇒ Reduce the expression:— $Y = BAC' + B'AC' + BC'$

⇒ Is given:—

$$Y = BA\overline{C} + \overline{B}A\overline{C} + B\overline{C}$$
$$= B\overline{C}(A+1) + \overline{B}A\overline{C}$$
$$= B\overline{C} \cdot 1 + \overline{B}A\overline{C}$$
$$= B\overline{C} + \overline{B}A\overline{C}$$
$$= \overline{C} \cdot (B + \overline{B}A)$$
$$= \overline{C} \cdot (B + A)$$
$$= A\overline{C} + B\overline{C}$$

problem: 2

⇒ Reduce the expression; $Y = AB + AB'$

⇒ Is given:—

$$Y = AB + AB'$$
$$= A(B + B')$$
$$= A \cdot 1$$
$$= A$$

problem: 3:—

⇒ Reduce the expression; $Y = AB + AB'C + AB'C'$

⇒ Is given:—

$$Y = AB + AB'C + AB'C'$$
$$= A(B + B'C) + AB'C'$$
$$= A(B + C) + AB'C'$$
$$= A(B + C + B'C')$$
$$= A(B + C' + C)$$
$$= A \cdot (B + 1)$$
$$= A \cdot 1$$
$$= A$$

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problem: 4:-

⇒ Reduce the expression ; $Y = (A+B+C)\cdot(A+B'+C)\cdot(A+B+C')$

⇒ Is given:-

$$Y = (A+B+C)\cdot(A+B'+C)\cdot(A+B+C')$$

Let, $X = A+B$

$$\therefore Y = (X+C)\cdot(A+B'+C)\cdot(X+C')$$
$$= (X+C)\cdot(X+C')\cdot(A+B'+C)$$
$$= (X+CC')\cdot(A+B'+C)$$
$$= (X+0)\cdot(A+B'+C)$$
$$= X\cdot(A+B'+C)$$
$$= (A+B)\cdot(A+B'+C)$$
$$= A+B\cdot(B'+C)$$
$$= A+BB'+BC$$
$$= A+0+BC$$
$$= A+BC$$

problem: 5:-

⇒ Reduce the expression; $Y = (A+B)\cdot(A+B')\cdot(A'+B)\cdot(A'+B')$

⇒ Is given:-

$$Y = (A+B)\cdot(A+B')\cdot(A'+B)\cdot(A'+B')$$
$$= (A+BB')\cdot(A'+BB')$$
$$= (A+0)\cdot(A'+0)$$
$$= A\cdot A'$$
$$= 0$$

problem:- 6:-

⇒ Reduce the expression ; $Y = AB + A'C + BC$

⇒ In given:-

$$Y = AB + A'C + BC$$
$$= AB + A'C + BC \cdot 1$$
$$= AB + A'C + BC(A + \bar{A})$$
$$= AB + A'C + ABC + \bar{A}BC$$
$$= AB(C+1) + A'C(B+1)$$
$$= AB + A'C$$

problem: 7:-

⇒ Reduce the expression;

(i) $Y = AD + B\bar{C} + AC$

(ii) $Y = A\bar{B} + BC + AC$

⇒ Is given:-

(i)

$$Y = AB + B\bar{C} + AC$$
$$= AB \cdot 1 + B\bar{C} + AC$$
$$= AB(C + \bar{C}) + B\bar{C} + AC$$
$$= ABC + AB\bar{C} + B\bar{C} + AC$$
$$= AC(B+1) + B\bar{C}(A+1)$$
$$= AC + B\bar{C}$$

(ii)

Similar way:-

$$Y = A\bar{B} + BC$$

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problem: 8:

⇒ Reduce the expression.

(i) $Y = (A+B)\cdot(A'+C)\cdot(B+C)$

(ii) $Y = (A+B)\cdot(\bar{B}+C)\cdot(A+C)$

(iii) $Y = \overline{A}\overline{B} + A\bar{C} + \bar{B}\bar{C}$

⇒ Is given:—

(i)

$$Y = (A+B)\cdot(A'+C)\cdot(B+C)$$
$$= (AA'+AC+A'B+BC)\cdot(B+C)$$
$$= (0+AC+A'B+BC)\cdot(B+C)$$
$$= ABC + ACC + A'BB + A'BC + BBC + BCC$$
$$= ABC + AC + A'B + A'BC + BC + BC$$
$$= AC(B+1) + A'B(C+1) + BC$$
$$= AC + A'B + BC$$
$$= AC + A'B$$

⇒ Similar way:—

(ii)

$$Y = (A+B)\cdot(\bar{B}+C)$$
$$= A\bar{B} + AC + B\bar{B} + BC$$
$$= A\bar{B} + AC + BC$$
$$= A\bar{B} + BC$$

⇒ Similar Way:—

(iii)

$$Y = \overline{A}\overline{B} + A\bar{C} + \bar{B}\bar{C}$$
$$\therefore Y = \overline{A}\overline{B} + A\bar{C}$$

problem:-9:-

⇒ Apply DeMorgan's Law for each expression:-

(a) $\overline{(A+B)+\bar{C}}$

(b) $\overline{(\bar{A}+B)+CD}$

(c) $\overline{(A+B)\cdot\bar{C}\cdot\bar{D}+E+F}$

⇒ is given:-

(a)

$$Y=\overline{(\overline{A+B})+\bar{C}}$$
$$=(\overline{\overline{A+B}})\cdot\overline{\bar{C}}$$
$$=(A+B)\cdot C$$
$$=AC+BC$$

(b)

⇒ Is given:-

$$Y=\overline{(\bar{A}+B)+CD}$$
$$=\overline{(\bar{A}+B)}\cdot\overline{CD}$$
$$=(\bar{\bar{A}}\cdot\bar{B})\cdot(\bar{C}+\bar{D})$$
$$=(A\cdot\bar{B})\cdot(\bar{C}+\bar{D})$$

⇒ Is given:-

(c)

$$Y=\overline{(A+D)\cdot\bar{C}\cdot\bar{D}+E+F}$$
$$=\overline{((A+B)\cdot\overline{CD})}\cdot\overline{(E+F)}$$
$$=((\overline{A+B})+(\overline{\bar{C}\cdot\bar{D}}))\cdot(\bar{E}\cdot\bar{F})$$
$$=((\bar{A}\cdot\bar{B})+(\bar{\bar{C}}+\bar{\bar{D}}))\cdot(\bar{E}\cdot F)$$
$$=((\bar{A}\cdot\bar{B})+(C+D))\cdot(\bar{E}\cdot F)$$
$$=(A\bar{B}+C+D)\cdot(E\cdot F)$$

Problem: 10!

⇒ Apply De-Morgan's Law for each expression!

(a) $\overline{(A+B+C)D}$

(b) $\overline{ABC+DEF}$

(c) $\overline{A\overline{B}+\overline{C}D+EF}$

(d) $\overline{ABC}+(\overline{\overline{D}+E})$

(e) $\overline{(A+B)\cdot C}$

(f) $\overline{A+B+C}+\overline{\overline{D}E}$

(a)

⇒ Is given!

$$Y=\overline{(A+B+C)D}$$
$$=\overline{(A+B+C)}+\overline{D}$$
$$=(\overline{A}\cdot\overline{B}\cdot\overline{C})+\overline{D}$$

(b)

⇒ Is given!

$$Y=\overline{ABC+DEF}$$
$$=\overline{ABC}\cdot\overline{DEF}$$
$$=(\overline{A}+\overline{B}+\overline{C})\cdot(\overline{D}+\overline{E}+\overline{F})$$

(c)

⇒ Is given!

$$Y=\overline{A\overline{B}+\overline{C}D+EF}$$
$$=(\overline{A\overline{B}})\cdot(\overline{\overline{C}D+EF})$$
$$=(\overline{A}+\overline{\overline{B}})\cdot(\overline{\overline{C}D}\cdot\overline{EF})$$
$$=(\overline{A}+B)\cdot((C+\overline{D})\cdot(\overline{E}+\overline{F}))$$

(d)

⇒ In given:— mod

$$Y = \overline{ABC} + (\overline{\overline{D}+E})$$

$$= (\overline{A}+\overline{B}+\overline{C}) + (\overline{\overline{D}}\cdot\overline{E})$$

$$= (\overline{A}+\overline{B}+\overline{C}) + (D\cdot\overline{E})$$

(e)

⇒ Is given:— mod

$$Y = \overline{(A+B)C}$$

$$= (\overline{A+B}) + \overline{C}$$

$$= (\overline{A}\cdot\overline{B}) + \overline{C}$$

(f)

⇒ In given:— mod

$$Y = \overline{A+B+C} + \overline{\overline{D}E}$$

$$= \overline{A}\cdot(\overline{B+C}) + (\overline{\overline{D}}+\overline{E})$$

$$= (\overline{A}\cdot(\overline{B}\cdot\overline{C})) + D + \overline{E}$$

$$= (\overline{A}\cdot\overline{B}\cdot\overline{C}) + D + \overline{E}$$

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Problem: 11:

⇒ Simplify the Boolean expression:

(a) $Y = AB + A(B+C) + B(B+C)$

(b) $Y = A\bar{B} + A(\overline{B+C}) + B(\overline{B+C})$

(c) $Y = [A\bar{B}(C+BD) + \bar{A}\bar{B}]C$

(d) $Y = \bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$

(e) $Y = \overline{AB + AC} + \bar{A}\bar{B}C$

(f) $Y = AB + A\bar{B}C$

(g) $Y = (\bar{A}+B)C + ABC$

(h) $Y = A\bar{B}C(BD + CDE) + A\bar{C}$

(a)

⇒ In given:

$$Y = AB + A(B+C) + B(B+C)$$
$$= AB + AB + AC + BB + BC$$
$$= AB + AB + AC + B + BC$$
$$= AB + AC + B + BC$$
$$= A(B+C) + B(C+1)$$
$$= AB + AC + B \cdot 1$$
$$= AB + AC + B$$
$$= B(A+1) + AC$$
$$= B + AC$$

(b)

is given:—

$$Y = A\bar{B} + A(\overline{B+C}) + B(\overline{B+C})$$

$$= A\bar{B} + A(\bar{B}\cdot\bar{C}) + B(\bar{B}\cdot\bar{C})$$

$$= A\bar{B} + A\bar{B}\bar{C} + B\bar{B}\bar{C}$$

$$= A\bar{B}(1+\bar{C}) + 0\cdot\bar{C}$$

$$= A\bar{B}\cdot 1 + 0$$

$$= A\bar{B}$$

(c)

is given:—

$$Y = [A\bar{B}(C+BD) + \bar{A}\bar{B}]C$$

$$= [A\bar{B}C + A\bar{B}BD + \bar{A}\bar{B}]C$$

$$= [A\bar{B}C + 0\cdot AD + \bar{A}\bar{B}]C$$

$$= A\bar{B}C\cdot C + 0 + \bar{A}\bar{B}C$$

$$= A\bar{B}C + \bar{A}\bar{B}C$$

$$= \bar{B}C(A+\bar{A})$$

$$= \bar{B}C\cdot 1$$

$$= \bar{B}C$$

(d)

is given:—

$$Y = \bar{A}BC + AB\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$$

$$= BC(\bar{A}+A) + \bar{B}\bar{C}(A+\bar{A}) + A\bar{B}C$$

$$= BC\cdot 1 + \bar{B}\bar{C}\cdot 1 + A\bar{B}C$$

$$= BC + \bar{B}\bar{C} + A\bar{B}C$$

$$= BC + \bar{B}(\bar{C} + AC)$$

$$= BC + \bar{B}(\bar{C} + A) = BC + \bar{B}\bar{C} + A\bar{B}$$

(e)

⇒ In given:—

$$Y = \overline{AB+AC} + \bar{A}\bar{B}C$$
$$= (\overline{AB}\cdot\overline{AC}) + \bar{A}\bar{B}C$$
$$= ((\bar{A}+\bar{B})\cdot(\bar{A}+\bar{C})) + \bar{A}\bar{B}C$$
$$= (\bar{A}+\bar{B}\bar{C}) + \bar{A}\bar{B}C$$
$$= \bar{A}+\bar{B}(\bar{C}+\bar{A}C)$$
$$= \bar{A}+\bar{B}(\bar{C}+\bar{A})$$
$$= \bar{A}+\bar{B}\bar{C}+\bar{A}\bar{B}$$
$$= \bar{A}(\bar{B}+1)+\bar{B}\bar{C}$$
$$= \bar{A}\cdot 1+\bar{B}\bar{C}$$
$$= \bar{A}+\bar{B}\bar{C}$$

(f)

⇒ In given:—

$$Y = AB + A\bar{B}C$$
$$= A(B+\bar{B}C)$$
$$= A(B+C)$$
$$= AB+AC$$

(g)

⇒ In given:—

$$Y = (\bar{A}+B)C + ABC$$
$$= \bar{A}C + BC + ABC$$
$$= \bar{A}C + BC(1+A)$$
$$= C(\bar{A}+B)$$

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3) In given:—

$$Y = A\bar{B}C(BD + CDE) + A\bar{C}$$

$$= A\bar{B}BCD + A\bar{B}CCDE + A\bar{C}$$

$$= A \cdot 0 \cdot CD + A\bar{B}CDE + A\bar{C}$$

$$= 0 + A + \bar{B}CDE\bar{C}$$

$$= A + \bar{B}C\bar{C}DE$$

$$= A + 0$$

$$= A$$

#SOP & POS

Sum of Products (SOP):-

⇒ An expression in Boolean algebra where multiple product terms are summed together.

	A	B	C	F
0 m_0	0	0	0	0
1 m_1	0	0	1	0
2 m_2	0	1	0	$\boxed{1}$ → High
3 m_3	0	1	1	0
4 m_4	1	0	0	$\boxed{1}$
5 m_5	1	0	1	$\boxed{1}$
6 m_6	1	1	0	$\boxed{1}$
7 m_7	1	1	1	$\boxed{1}$

Representation

if $\bar{A} = 0$, $A = 1$

Then, $\bar{B} = 0$, $B = 1$

also, $\bar{C} = 0$, $C = 1$

→ SOP is written when the function is high

$\therefore$ SOP; $F = \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$

Note:- This particular form is known as "Standard or Canonical SOP" form.

Minterms:-

⇒ SOP uses minterms to create a boolean expression which in product of Boolean Variables.

Ex; $\bar{A} \cdot B \cdot C$, ABC, AB, AC, $\bar{A}C$ etc.

for the above truth table minterms function is:-

$$F(A,B,C) = m_2 + m_4 + m_5 + m_6 + m_7$$

$$\therefore F(A,B,C) = \sum m(2,4,5,6,7)$$

Minimal SOP:-

⇒ The most simplified SOP expression of a function.

Let,

$$F = \bar{A}\cdot B\cdot\bar{C} + A\cdot\bar{B}\cdot\bar{C} + A\cdot\bar{B}\cdot C + A\cdot B\cdot\bar{C} + A\cdot B\cdot C$$

$$= \bar{A}\cdot B\cdot\bar{C} + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C)$$

$$= \bar{A}\cdot B\cdot\bar{C} + A\cdot\bar{B}\cdot(1) + A\cdot B\cdot(1)$$

$$= \bar{A}\cdot B\cdot\bar{C} + A\cdot\bar{B} + AB$$

$$= \bar{A}\cdot B\cdot\bar{C} + A(\bar{B}+B)$$

$$= \bar{A}\cdot B\cdot\bar{C} + A\cdot 1$$

$$= A + B\bar{C} \rightarrow \text{minimal SOP}$$

Cononical/Standard SOP form:-

⇒ Each minterm is having all the variable in Normal or Complimented form.

Ex:- $F = \bar{A}B + AB + \bar{A}\bar{B}$

Minimal SOP form:-

⇒ Each minterm does not have all the variables in Normal or Complimented form.

Ex:- $G = A + \bar{B}C$

Problem: 1 :-

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

⇒ For the given truth table, minimize the SOP expression.

⇒ SOP Expression is :-

$$Y = \bar{A}B + AB$$
$$= B(\bar{A} \cdot A)$$
$$= B \cdot 1$$
$$= B$$

Problem : 2 :-

⇒ Simplify the expression for $Y(A,B) = \sum m(0,2,3)$

⇒ SOP Expression is :-

$$Y = m_0 + m_2 + m_3$$
$$= \bar{A}\bar{B} + A\bar{B} + AB$$
$$= \bar{A}\bar{B} + A(\bar{B} + B)$$
$$= \bar{A}\bar{B} + A \cdot 1$$
$$= A + \bar{B}$$

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Product of Sum:- (POS):-

⇒ POS form is used when the output is "0" or low.

	A	B	C	Y
0 m_0	0	0	0	$\boxed{0}$ → Low
1 m_1	0	0	1	$\boxed{0}$
2 m_2	0	1	0	1
3 m_3	0	1	1	$\boxed{0}$
4 m_4	1	0	0	1
5 m_5	1	0	1	1
6 m_6	1	1	0	1
7 m_7	1	1	1	1

Representation:-

if $A \to 0$
$\bar{A} \to 1$

then $B \to 0$
$\bar{B} \to 1$

also $C \to 0$
$\bar{C} \to 1$

⇒ POS is written when the function is Low.

$$\therefore POS,\ Y = (A+B+C)\cdot(A+B+\bar{C})\cdot(A+\bar{B}+\bar{C})$$

Note:- This particular form is known as "Standard or Canonical POS" form.

Maxterm:-

⇒ POS uses maxterm to create a boolean expression which is the Sum of Boolean Variables.

Ex:- $A+B+C$, $\bar{A}+B+\bar{C}$, $A+\bar{B}+\bar{C}$, etc.

For the above truth table MAXterm function is:-

$$F(A,B,C) = \pi(m_1, m_0, m_3)$$

$$\therefore F(A,B,C) = \pi M(0,1,3)$$

problem: 1

⇒ For the given truth table minimize the pos expression.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

⇒ POS Expression is:

$$Y = (A+\overline{B})\cdot(\overline{A}+\overline{B})$$
$$= \overline{B} + A\overline{A}$$
$$= \overline{B} + 0$$
$$= \overline{B}$$

problem: 2

⇒ Simplify the expression for $Y(A,B,C) = \pi M(0,1,3)$

⇒ POS Expression is:

$$Y = \pi(M_0, M_1, M_3)$$
$$\therefore Y = (\underbrace{A+B}_{X}+C)\cdot(\underbrace{A+B}_{X}+\overline{C})\cdot(A+\overline{B}+\overline{C})$$
$$= (X+C)\cdot(X+\overline{C})\cdot(A+\overline{B}+\overline{C})$$
$$= (X+C\overline{C})\cdot(A+\overline{B}+\overline{C})$$
$$= (X+0)\cdot(A+\overline{B}+\overline{C})$$
$$= (A+B)\cdot(A+\overline{B}+\overline{C})$$
$$= A+B\cdot(\overline{B}+\overline{C})$$
$$= A+B\overline{B}+B\overline{C}$$
$$= A+B\overline{C}$$
$$= (A+B)\cdot(A+\overline{C})$$

problem: 8:—

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

(a) Find the minimize SOP & POS

(b) Simplify the expression for $Y(A,B,C)=\Sigma m(0,2,3,6,7)$

(c) Simplify the expression for $Y(A,B,C)=\pi M(1,4,5)$

(a)

For SOP:—

$$Y=\bar{A}\bar{B}\bar{C}+\bar{A}\cdot B\cdot\bar{C}+\bar{A}\cdot BC+AB\cdot\bar{C}+ABC$$
$$=\bar{A}\bar{B}\bar{C}+\bar{A}B(\bar{C}+C)+AB\cdot(\bar{C}+C)$$
$$=\bar{A}\bar{B}\bar{C}+\bar{A}B+AB$$
$$=\bar{A}\bar{B}\bar{C}+B(\bar{A}+A)$$
$$=\bar{A}\bar{C}+B$$

For POS:—

$$Y=(A+B+\bar{C})\cdot(\underbrace{\bar{A}+B}_{X}+C)\cdot(\underbrace{\bar{A}+B}_{X}+\bar{C})$$
$$=(A+B+\bar{C})\cdot(X+C\bar{C})$$
$$=(A+B+\bar{C})\cdot(\bar{A}+B)$$
$$=B+\bar{A}(A+\bar{C})$$
$$=B+A\bar{A}+\bar{A}\bar{C}$$
$$=(\bar{A}+B)\cdot(B+\bar{C})$$

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Minimal to Canonical Form Conversion:-

For SOP:-

Let, $Y = A + \bar{B}C$

$= A\cdot 1\cdot 1 + \bar{B}\cdot C\cdot 1$

$= A\cdot(B+\bar{B})\cdot(C+\bar{C}) + \bar{B}\cdot C\cdot(A+\bar{A})$

$= (AB + A\bar{B})\cdot(C+\bar{C}) + A\bar{B}C + \bar{A}\bar{B}C$

$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \cancel{A\bar{B}C} + \bar{A}\bar{B}C$

$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C$

Steps:-

Step 1:- Find all the Variable.
Ex:- A, B, C

Step 2:- Find out the Variable absent in each minterm.

Step 3:- Remove the duplicate.

For POS:-

Let, $Y = (A+B+\bar{C})\cdot(\bar{A}+C)$

$\therefore Y = (A+B+\bar{C})\cdot(\bar{A}+C+0)$

$= (A+B+\bar{C})\cdot(\underbrace{\bar{A}+C}_{X} + BB')$

$= (A+B+\bar{C})\cdot(X + BB')$

$= (A+B+\bar{C})\cdot(X+B)\cdot(X+B')$

$= (A+B+\bar{C})\cdot(\bar{A}+C+B)\cdot(\bar{A}+C+\bar{B})$

Steps:-

Step 1:- Find all the variable
Ex: A, B, C

Step 2:- Find out the Variable absent in Maxterm.

Step 3:- Retrieve the absent variable.

KARNAUGH MAP (K-MAP)

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Karnaugh-MAP (K-MAP)

⇒ One way to come up with a minimized SOP or POS formula for a boolean function.

Let, $F = A + B\bar{C}$

	A	B	C	F
m_0	0	0	0	0
m_1	0	0	1	0
m_2	0	1	0	1
m_3	0	1	1	0
m_4	1	0	0	1
m_5	1	0	1	1
m_6	1	1	0	1
m_7	1	1	1	1

Map formation:—

	BC 00	BC 01	BC 11	BC 10
A=0	m_0	m_1	m_3	m_2
A=1	m_4	m_5	m_7	m_6

Note:- This formation is according to the cell adjacency.

⇒ We know:—

SOP expression:—

$$F = \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC$$

$$= \bar{A}B\bar{C} + A\bar{B}(\bar{C}+C) + AB(\bar{C}+C)$$

$$= \bar{A}B\bar{C} + A\bar{B} + AB$$

$$= \bar{A}B\bar{C} + A + B\bar{B}$$

$$= A + B\bar{C}$$

K-map for 3 Variable:

A \ BC	00	01	11	10
0	0 (m_0)	0 (m_1)	0 (m_3)	1 (m_2)
1	1 (m_4)	1 (m_5)	1 (m_7)	1 (m_6)

Grouping: (SOP)

1. No Zero allowed.
2. No diagonals group
3. Groups must contain 1, 2, 4, 8 in general 2^n cells.
4. Groups should be as large as possible.
5. Every one must be in at least one group
7. Overlapping allowed.

Ex:-

A \ BC	00	01	11	10
0	0	0	0	1
1	1	1	1	1

II

I

$\therefore$ $F = I + II$

$F = A + B\overline{C}$

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problem: 1:—

$$F(A,B,C) = \sum m(1,3,5,7)$$

ⓐ Use k-map to minimize the following standard SOP expression.

ⓐ

⇒ Draw the k-map:—

A \ BC	00	01	11	10
0	0 m_0	1 m_1	1 m_3	0 m_2
1	0 m_4	1 m_5	1 m_7	0 m_6

$\therefore F = C$

problem: 2:—

$$F(A,B,C) = \sum m(0,1,2,4,7)$$

ⓐ Use k-map to minimize the SOP expression.

ⓐ

⇒ Draw the k-map:—

A \ BC	00	01	11	10
0	1	1	0	1
1	1	0	1	0

I II III IV

$\therefore F = I + II + III + IV$

$= \bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C} + ABC$

Problem: 3:-

$$F(A,B,C) = \sum m(1,3,6,7)$$

⇒ Draw the K-map:-

$A \backslash BC$	00	01	11	10
0	0	1	1	0
1	0	0	1	1

$$\therefore F = \bar{A}C + AB$$

Problem: 4:-

$$F(A,B,C) = \sum m(0,1,5,6,7)$$

⇒ Draw the K-map:-

$A \backslash BC$	00	01	11	10
0	1	1		
1		1	1	1

I III II′ III′

Case 1:- $F = I + II + III$

$$\therefore F = \bar{A}\cdot\bar{B} + \bar{B}C + AB$$

Case 2:-

$$F = I + II + III'$$
$$= \bar{A}\cdot\bar{B} + AB + AC$$

Note:- The result is minimum but may not be same or unique.

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K-map for 4 Variable:-

	A	B	C	D	F
m0	0	0	0	0	1
m1	0	0	0	1	0
m2	0	0	1	0	1
m3	0	0	1	1	0
m4	0	1	0	0	0
m5	0	1	0	1	0
m6	0	1	1	0	0
m7	0	1	1	1	1
m8	1	0	0	0	0
m9	1	0	0	1	0
m10	1	0	1	0	0
m11	1	0	1	1	1
m12	1	1	0	0	0
m13	1	1	0	1	1
m14	1	1	1	0	1
m15	1	1	1	1	1

Map formation:-

AB \ CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

Problems 1:-

$$F(A,B,C,D) = \sum m(0,2,3,7,11,13,14,15)$$

⇒ Draw the K-maps:-

AB \ CD	00	01	11	10
00	1 m_0	0 m_1	1 m_3	1 m_2
01	0 m_4	0 m_5	1 m_7	0 m_6
11	0 m_{12}	1 m_{13}	1 m_{15}	1 m_{14}
10	0 m_8	0 m_9	1 m_{11}	0 m_{10}

$$\therefore F = \bar{A}\bar{B}\bar{D} + CD + ABD + ABC$$

problem: 2:-

$$F(A,B,C,D) = \sum m(0,2,3,5,7,8,10,11,14,15)$$

⇒ Draw the K-map:-

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	0
11	0	0	1	1
10	1	0	1	1

$$F = CD + AC + \bar{B}\bar{D} + \bar{A}BD$$

problems 3:-

$F(x, y, z, w) = \sum m(1, 5, 7, 9, 11, 13, 15)$

⇒ reduce this by using k-map.

⇒ Draw the k-map:-

AB \ CD	00	01	11	10
00		1		
01		1	1	
11		1	1	
10		1	1	

Let
$A = x$
$B = y$
$z = C$
$w = D$

$\therefore F = \bar{C}D + AD + BD$

$\therefore F = \bar{z}w + xw + yw$

Ans:-

Don't Care in k-Map:-

⇒ To make a group of cells, we can use the "don't care" cells as either 0 or 1. Also If required, we can also ignore that cell. We mainly use the "don't care" cell to make a large group of cells.

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problem:-1:-

$\Rightarrow F(A,B,C) = \sum m(2,3,4,5)$ & $\sum d(6,7)$

reduce the function by using k-map.

$\Rightarrow$ 1st draw the k-map:-

A \ BC	00	01	11	10
0			1	1
1	1	1	X	X

$\therefore F = A + B$

K-MAP Using Max Term:-

$\Rightarrow$ For POS put "0" in blocks of k-map respective to the maxterms. (1's elsewhere)

I II

A \ BC	00	01	11	10
0	0	0	0	1
1	1	1	1	1

Now, For POS:-

$\therefore F = \mathrm{I} \cdot \mathrm{II}$

$= (A+B)\cdot(A+\bar{C})$

$= (\bar{A}\cdot\bar{B})' \cdot (\bar{A}\cdot C)'$

$= (\bar{A}\bar{B} + \bar{A}C)' = \bar{F} \rightarrow POS$

problem: 1:—

$$F(A,B,C,D) = \Sigma m(1,3,4,5,9,11,14,15)$$

ⓐ Find the MAX Term function?

ⓑ Reduce the MAX Term function by using k-map.

ⓐ

$$\therefore F(A,B,C,D) = \Pi M(0,2,6,7,8,10,12,13)$$

ⓑ

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	0	0
11	0	0	1	1
10	0	1	1	0

I

II

III

$$\therefore F = I \cdot II \cdot III$$

$$= (B+D)\cdot(\bar{A}+\bar{B}+C)\cdot(A+\bar{B}+\bar{C})$$

Ans!—

problem : 1 :-

⇒ Map the following expression on K-map :

$$Y = \bar{A} + A\bar{B} + AB\bar{C}$$

⇒ 1st Convert minimal sop to Canonical sop form :-

$$Y = \bar{A} + A\bar{B} + AB\bar{C}$$

$$= \bar{A}(B+\bar{B})\cdot(C+\bar{C}) + A\bar{B}(C+\bar{C}) + AB\bar{C}$$

$$= (\bar{A}B + \bar{A}\bar{B})\cdot(C + \bar{C}) + A\bar{B}C + A\bar{B}\bar{C} + AB\bar{C}$$

$$= \underset{011}{\bar{A}BC} + \underset{010}{\bar{A}B\bar{C}} + \underset{001}{\bar{A}\bar{B}C} + \underset{000}{\bar{A}\bar{B}\bar{C}} + \underset{101}{A\bar{B}C} + \underset{100}{A\bar{B}\bar{C}} + \underset{110}{AB\bar{C}}$$

∴ K-map :-

A \ BC	00	01	11	10
0	1	1	1	1
1	1	1		1

problem : 2 :-

⇒ Map the following expression on K-map

$$Y = \bar{B}\bar{C} + A\bar{B} + AB\bar{C} + \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD$$

⇒ 1st Convert minimal sop to Standard sop form :-

$$Y = \bar{B}\bar{C} + A\bar{B} + AB\bar{C} + \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD$$

$$= \bar{B}\bar{C}(A+\bar{A})\cdot(B+\bar{B}) + A\bar{B}(C+\bar{C})\cdot(D+\bar{D}) + AB\bar{C}(D+\bar{D}) + \bar{A}\bar{B}\bar{C}D$$
$$+ A\bar{B}C\bar{D} + A\bar{B}CD$$

$$\Rightarrow (A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C})\cdot(D+\bar{D}) + (A\bar{B}C + A\bar{B}\bar{C})\cdot(D+\bar{D}) + AB\bar{C}D + AB\bar{C}\bar{D}$$
$$+ \bar{A}\bar{B}\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}CD$$

$$\Rightarrow \underset{1001}{A\bar{B}\bar{C}D} + \underset{1000}{A\bar{B}\bar{C}\bar{D}} + \underset{0001}{\bar{A}\bar{B}\bar{C}D} + \underset{0000}{\bar{A}\bar{B}\bar{C}\bar{D}} + \underset{1011}{A\bar{B}CD} + \underset{1000}{A\bar{B}\bar{C}\bar{D}} + \underset{1101}{AB\bar{C}D} + \underset{1100}{AB\bar{C}\bar{D}}$$
$$+ \underset{0001}{\bar{A}\bar{B}\bar{C}D} + \underset{1010}{A\bar{B}C\bar{D}} + \underset{1011}{A\bar{B}CD}$$

∴ K-map:—

AB \ CD	00	01	11	10
00	1	1		
01				
11	1	1		
10	1	1	1	1

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Problem: 2:-

⇒ Use a K-map to minimize the following standard SOP expression.

(a) $A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C}$

(b) $\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D}$

⇒

(a)

is given:-

$$Y = \underset{101}{A\bar{B}C} + \underset{011}{\bar{A}BC} + \underset{001}{\bar{A}\bar{B}C} + \underset{000}{\bar{A}\bar{B}\bar{C}} + \underset{100}{A\bar{B}\bar{C}}$$

A \ BC	00	01	11	10
0	1	1	1	
1	1	1		

$\therefore Y = \bar{B} + \bar{A}C$

(b)

⇒ 1st Convert it into standard SOP from.

$$Y = \bar{B}\bar{C}\bar{D}(A+\bar{A}) + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D}$$

$$\therefore Y = \underset{1000}{A\bar{B}\bar{C}\bar{D}} + \underset{0000}{\bar{A}\bar{B}\bar{C}\bar{D}} + \underset{0100}{\bar{A}B\bar{C}\bar{D}} + \underset{1100}{AB\bar{C}\bar{D}} + \underset{0011}{\bar{A}\bar{B}CD} + \underset{1011}{A\bar{B}CD} + \underset{0010}{\bar{A}\bar{B}C\bar{D}} + \underset{0110}{\bar{A}BC\bar{D}} + \underset{1110}{ABC\bar{D}} + \underset{1010}{A\bar{B}C\bar{D}}$$

AB \ CD	00	01	11	10
00	1		1	1
01	1			1
11	1			1
10	1		1	1

$\therefore Y = \overline{D} + \overline{B}C$

problem-4:

⇒ Use K-map to minimize the following standard POS expression.

ⓐ $Y = \underset{0\ 0\ 0}{(A+B+C)} \cdot \underset{0\ 0\ 1}{(A+B+\overline{C})} \cdot \underset{0\ 1\ 0}{(A+\overline{B}+C)} \cdot \underset{0\ 1\ 1}{(A+\overline{B}+\overline{C})} \cdot \underset{1\ 1\ 0}{(\overline{A}+\overline{B}+C)}$

⇒

A \ BC	00	01	11	10
0	0	0	0	0
1				0

$\therefore Y = (A+0) \cdot (\overline{B}+C)$

$\therefore Y = A \cdot (\overline{B}+C)$

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problem: 5:

$\Rightarrow$ Use K-map to minimize the following POS expression

$$Y = (B+C+D)\cdot(A+B+\bar{C}+D)\cdot(\bar{A}+B+C+\bar{D})\cdot(A+\bar{B}+C+D)\cdot(\bar{A}+\bar{B}+C+D)$$

$\Rightarrow$ 1st convert the given POS to standard POS from.

$$\therefore Y = (B+C+D+AA')\cdot(A+B+\bar{C}+D)\cdot(\bar{A}+B+C+\bar{D})\cdot(A+\bar{B}+C+D)\cdot(\bar{A}+\bar{B}+C+D)$$

$$\Rightarrow (B+C+D+A)\cdot(A'+B+C+D)\cdot(A+B+\bar{C}+D)\cdot(\bar{A}+B+C+\bar{D})\cdot(A+\bar{B}+C+D)\cdot(\bar{A}+\bar{B}+C+D)$$

$$\Rightarrow \underset{0000}{(A+B+C+D)}\cdot\underset{1000}{(A'+B+C+D)}\cdot\underset{0010}{(A+B+\bar{C}+D)}\cdot\underset{1001}{(\bar{A}+B+C+\bar{D})}\cdot\underset{0100}{(A+\bar{B}+C+D)}\cdot\underset{1100}{(\bar{A}+\bar{B}+C+D)}$$

AB \ CD	00	01	11	10
00	0			0
01	0			
11	0			
10	0	0		

$$\therefore Y = (C+D)\cdot(\bar{A}+B+C)\cdot(A+B+D)$$

Ans:-

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ADDER CIRCUIT

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Adder:

⇒ A digital circuit that performs addition of Numbers.

There are two types:-

1. Half Adder
2. Full Adder.

Half Adder:-

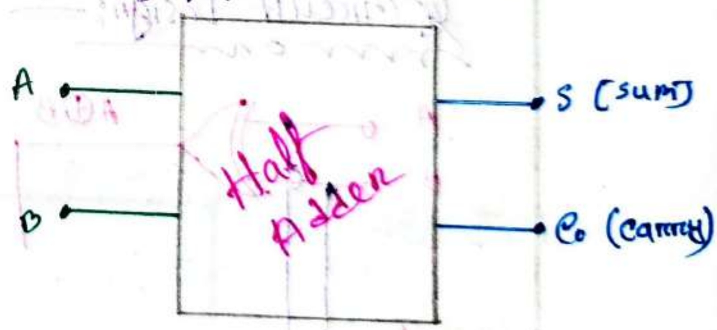
⇒ It is a digital logic circuit that performs binary addition of two single-bit (1 bit) binary numbers.

A	B	S	C_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

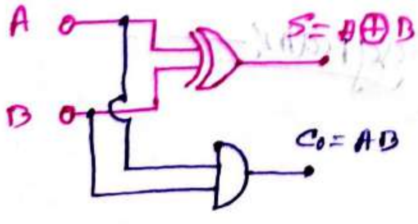
$\therefore S = A \oplus B$

$C_o = AB$

Block Diagram:-



Logic Circuit Design:-



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Full Adder:

A	B	C_{in}	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

For Sum (S):-

A \ BC_{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$\therefore S = A \oplus B \oplus C_{in}$

For Carry (C_o)

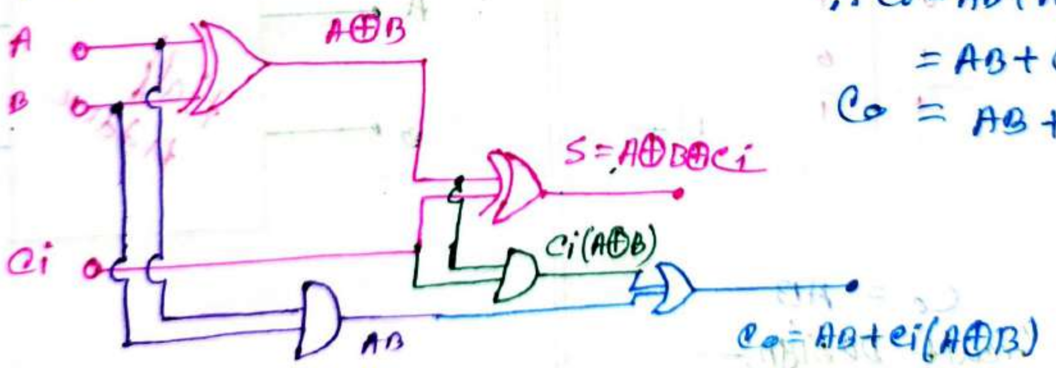
A \ BC_{in}	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$\therefore C_o = AB + A\bar{B}C_{in} + \bar{A}BC_{in}$

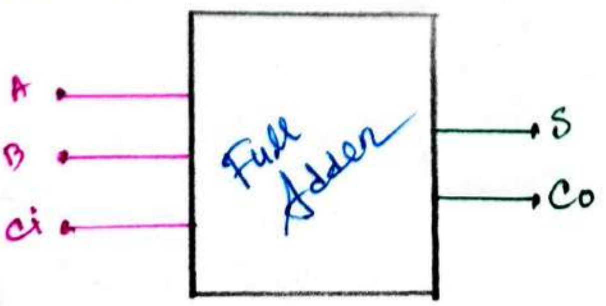
$= AB + C_{in}(A\bar{B} + \bar{A}B)$

$C_o = AB + C_{in}(A \oplus B)$

Circuit Design:-



Standard design:/ ~~Block~~ Diagram:-



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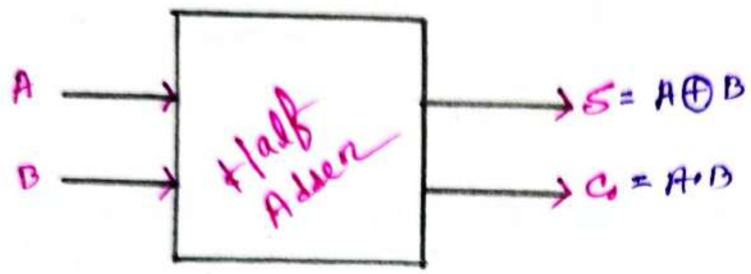
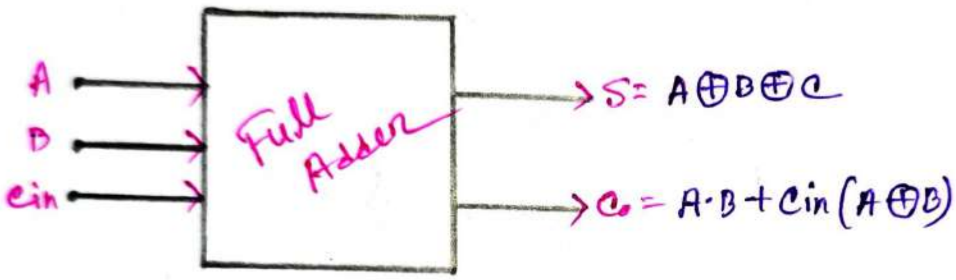
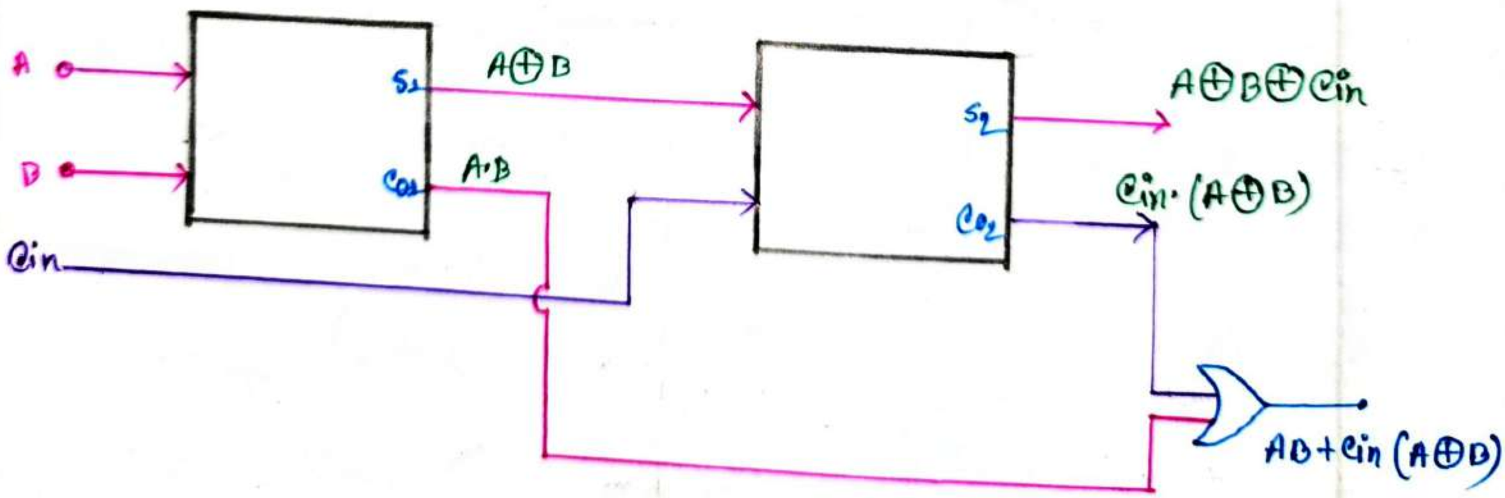
Full Adder Using Half Adder:-

we know:-





Now we need two circuit of Half Adder to impliment a full Adder circuit.



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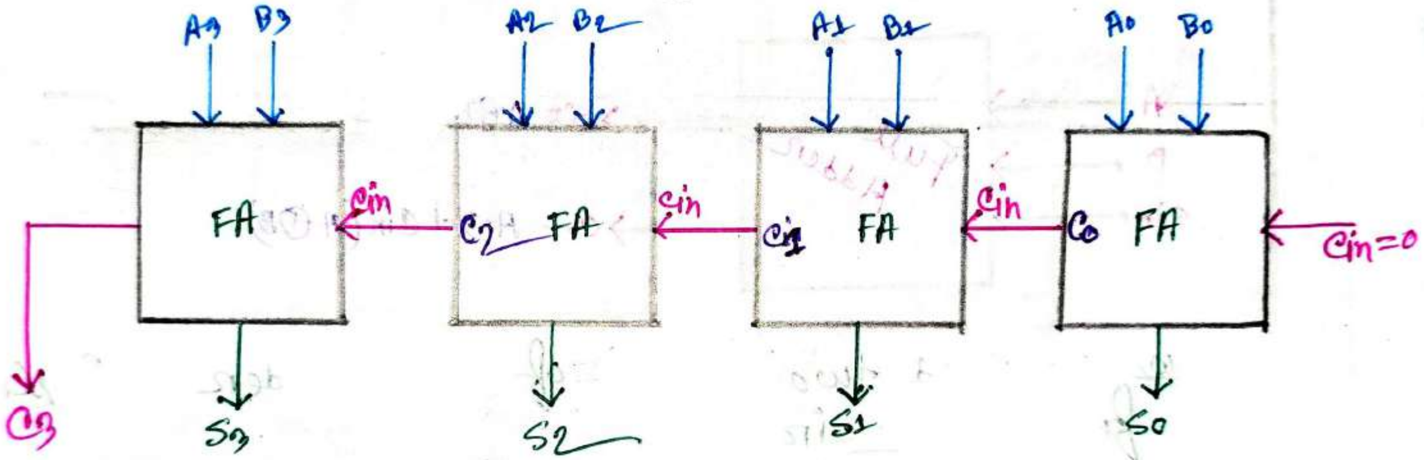
4 Bit parallel Adder using Full Adders:-

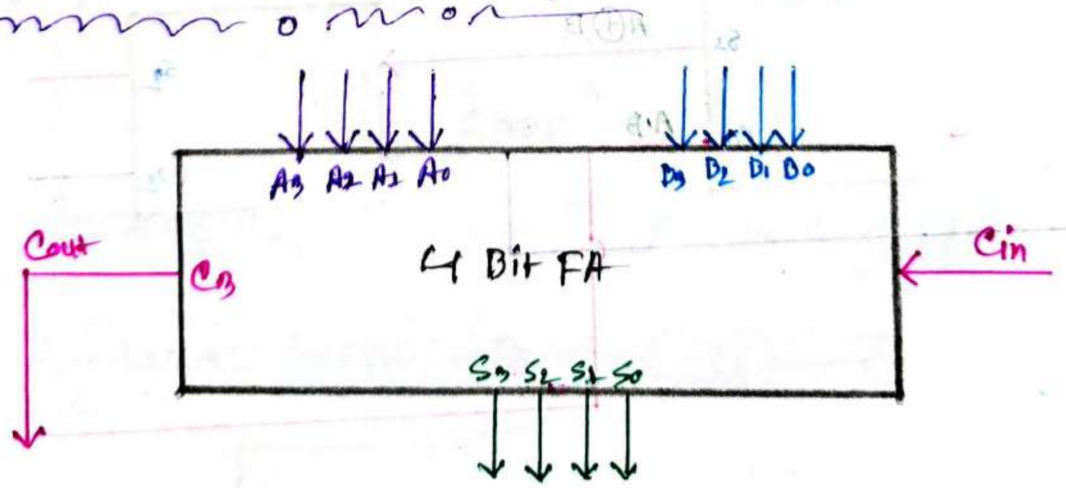
Let, 4 Bit numbers are:-

$$
\begin{array}{rcccc}
A = & A_3 & A_2 & A_1 & A_0 \\
+B = & B_3 & B_2 & B_1 & B_0 \\
\hline
\therefore S = C_3 & S_3 & S_2 & S_1 & S_0 \\
 & C_3 & C_2 & C_1 & C_0
\end{array}
$$

Cin Cin Cin Cin Cin=0



It Also represent like:-



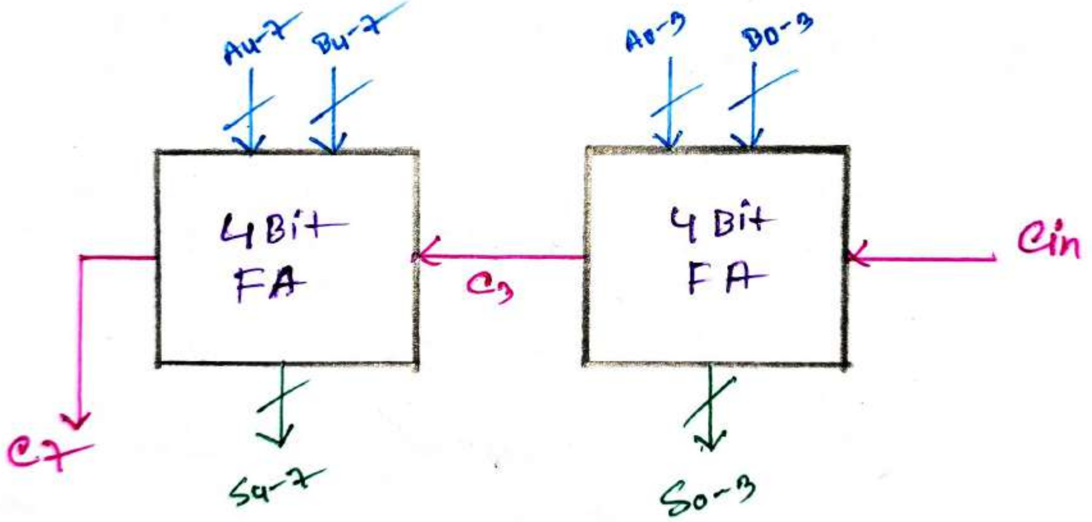
8 Bit Full Adder Using 4 Bit Full Adder:-

Let, 8 Bit numbers are:-

$C_{in} = 0$

$A =$		A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
$B =$		B_7	B_6	B_5	B_4	B_3	B_2	B_1	B_0
$\therefore S =$	C_7	S_7	S_6	S_5	S_4	S_3	S_2	S_1	S_0
		C_7	C_6	C_5	C_4	C_3	C_2	C_1	C_0

Now, we need two 4 bit full adder to impliment 8 bit full adder.



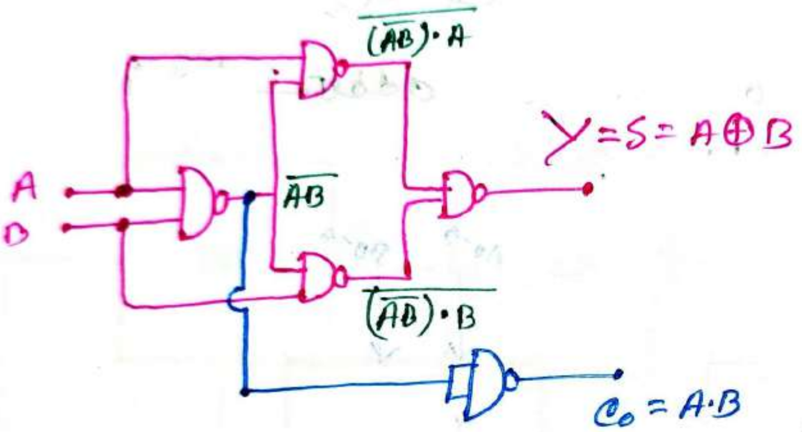
Half Adder Using NAND Gate only:-

We know:-

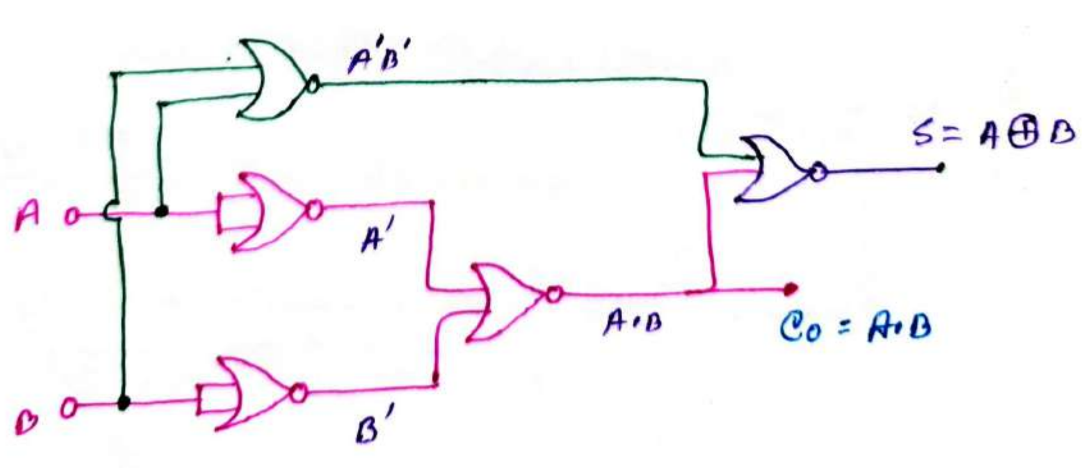
For Half Adder:-

$S = A \oplus B$

$C_o = AB$



Half Adder Using NOR gate only:-



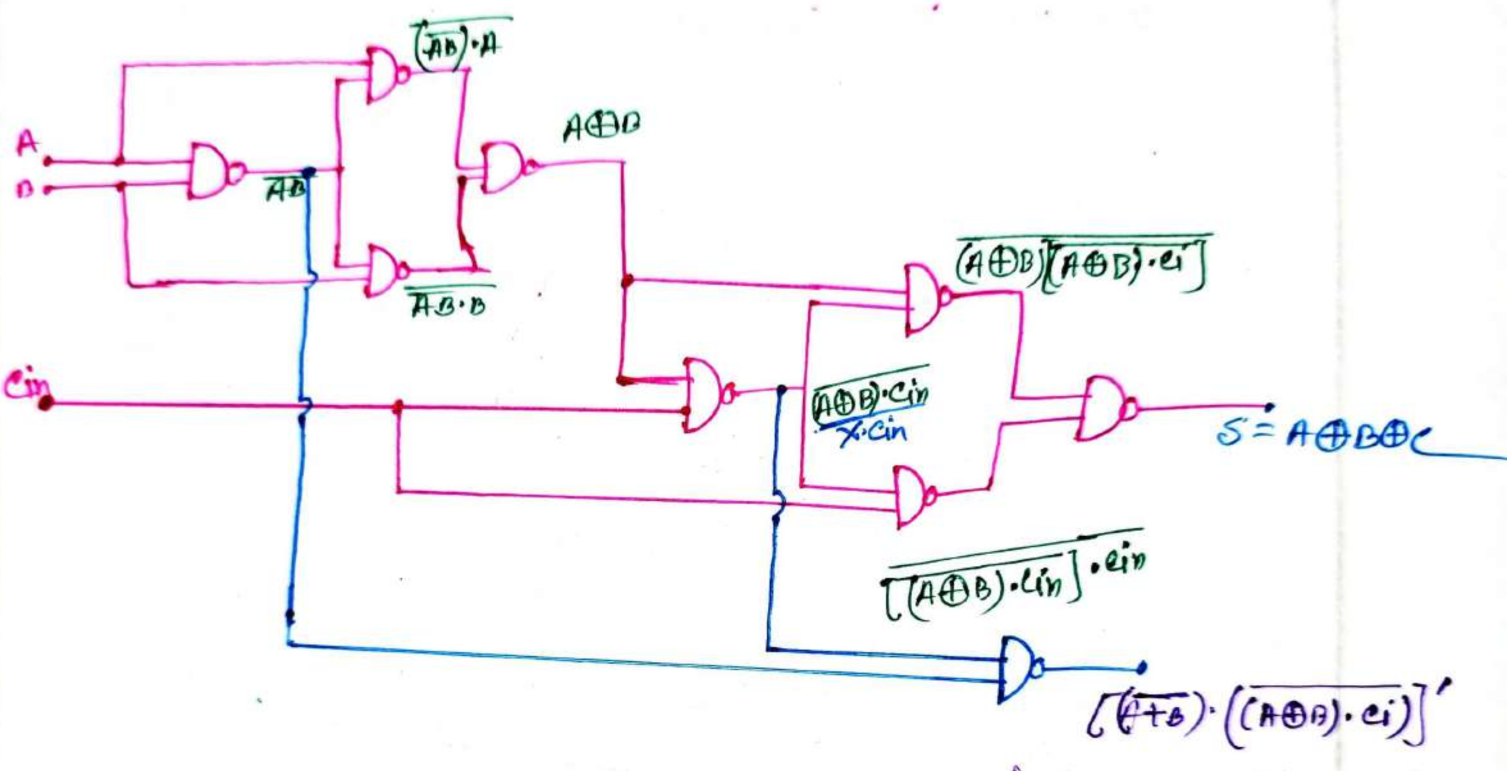
Date : / /

Full Adder Using NAND gate:-

We know:-

$S = A \oplus B \oplus C$

$C_o = A \cdot B + C_{in}(A \oplus B)$



$\therefore C_o = A \cdot B + C_{in}(A \oplus B)$

Full Adder Using NOR only:

Do it yourself.

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Subtractor Circuit

Subtractor:-

⇒ A digital circuit that performs subtraction of Numbers.

There are two types:-

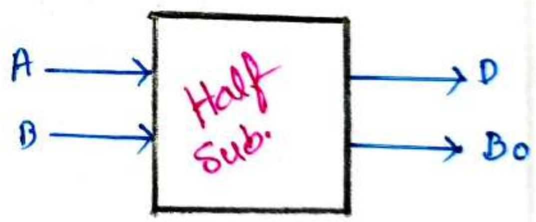
1. Half Subtractor
2. Full Subtractor.

Half Subtractor:-

⇒ It is a digital logic circuit that performs binary subtraction of single bit (1-bit) Binary Numbers.

A	B	D	B_o
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

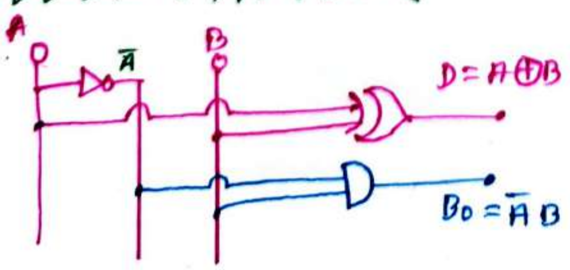
Block-Diagram:-



$\therefore D = A \oplus B$

$B_o = \bar{A} B$

Logic circuit Design:-



Full Subtractors (F.S)

A	B	C	D	Bo
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

For Diff :- (D)

A \ BC	00	01	11	10
0	0	1	0	1
1	1	0	1	

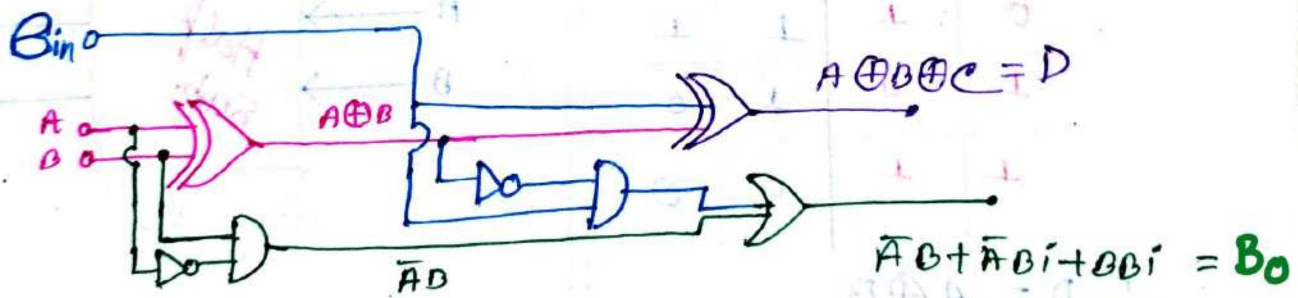
$\therefore D = A \oplus B \oplus C$

For Borrow (Bin)

A \ BBin	00	01	11	10
0	0	1	1	1
1	0		1	0

$\therefore B_0 = \bar{A}B + \bar{A}B_i + BB_i$

Logic Circuit for F.S



Block Diagram:-

Half Subtractor Using

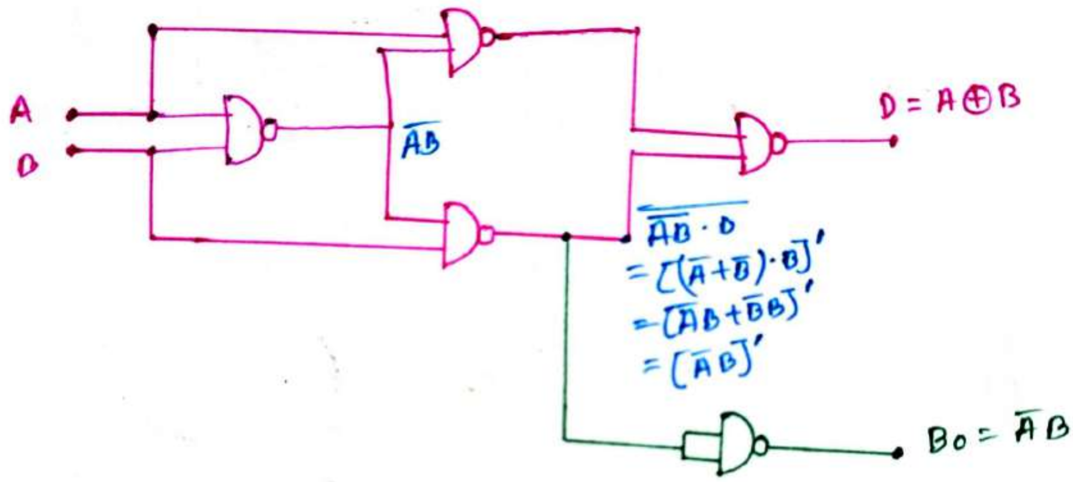
(i) NAND Gate Only:—

We know:—

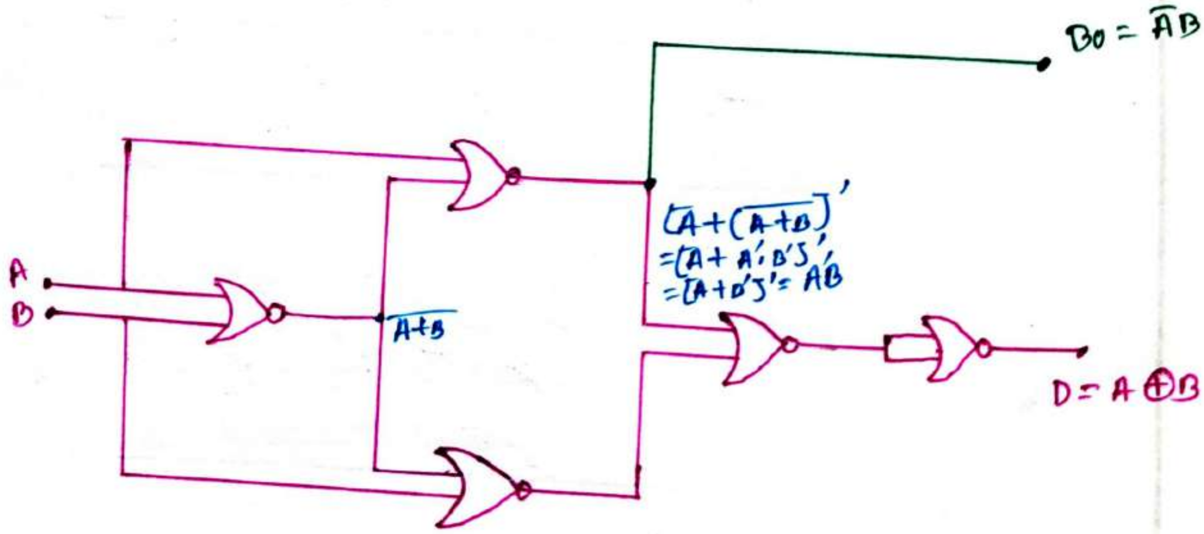
For Half Subtractor:—

$D = A \oplus B$

$B_o = A'B$



(ii) NOR Gate Only:—



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Full Subtractor Using NAND Gate Only:-

We know:-

For Full Subtractor:-

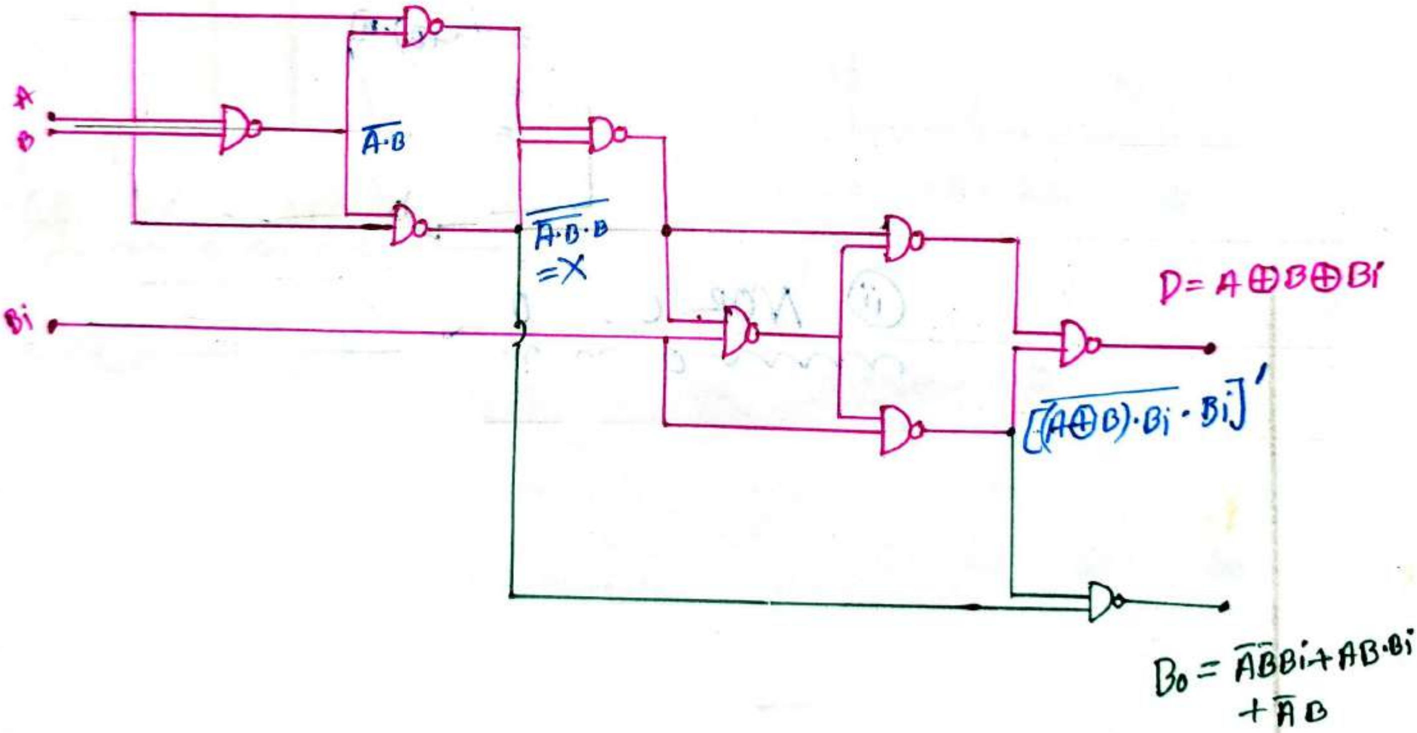
$$D = A \oplus B \oplus B_i$$

$$B_o = \bar{A}B + \overline{A \oplus B}\, B_i + ABB_i$$

OR $B_o = \bar{A}B + \bar{A}B_i + BB_i$

A \ BB_i	00	01	11	10
0	0	1	1	1
1	0	0	1	0

$\therefore B_o = \bar{A}\bar{B}B_i + ABB_i + \bar{A}B$



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#COMPARATOR

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Comparator:

⇒ A comparator circuit compares two voltages and outputs either a 1 or a 0 to indicate which is larger.

1-bit Comparator:

⇒ A comparator used to compare two bits is called a single-bit comparator.

The truth table:

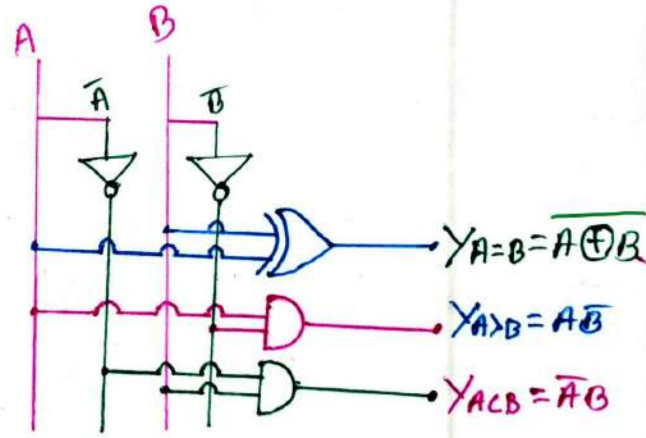
A	B	A = B	A > B	A < B
0	0	1	0	0
0	1	0	0	1
1	0	0	1	0
1	1	1	0	0

$\therefore Y_{A=B} = \overline{A \oplus B}$

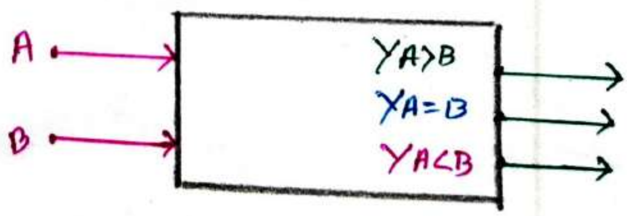
$Y_{A>B} = A\bar{B}$

$Y_{A<B} = \bar{A}B$

Logic Circuit Design:



Block Diagram:



#2-bit-Comparators

⇒ A comparator used to compare two binary numbers each of two bits called a 2-bit magnitude Comparator. It Consists a 2-bits of four inputs & three outputs.

⇒ A digital Comparator is a combinational circuit designed to compare two n-bit binary words.

⇒ Comparators has three outputs.

Truth Table :-

A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

For A<B:-

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$\therefore Y_{A<B} = \bar{A_1}\bar{A_0}B_0 + \bar{A_1}B_1 + \bar{A_0}B_1B_0$$

For A=B:-

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$\therefore Y_{A=B} = (A_1 \odot B_1)\cdot(A_0 \odot B_0)$$

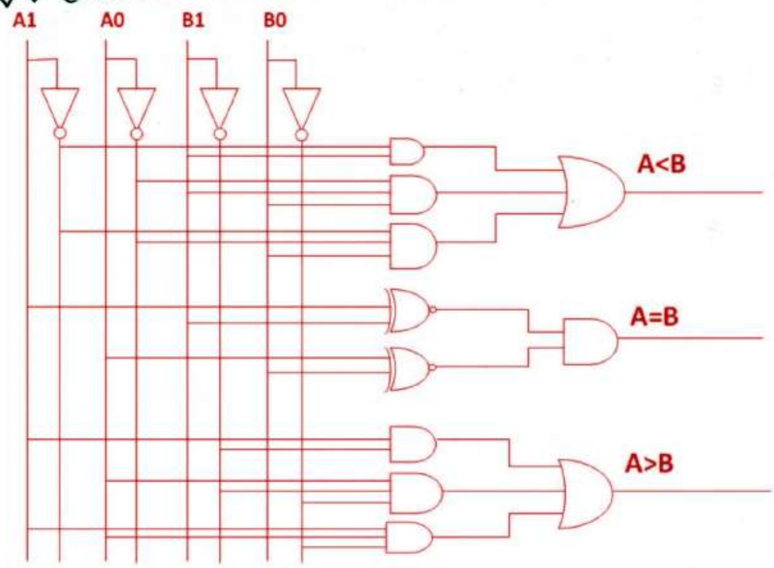
$$\therefore Y_{A=B} = (A_1 \oplus B_1)'\cdot(A_0 \oplus B_0)'$$

For A>B:-

$A_1A_0 \backslash B_1B_0$	00	01	11	10
00				
01	1			
11	1	1		1
10	1	1		

$$\therefore Y_{A>B} = A_0\bar{B_1}\bar{B_0} + A_1A_0\bar{B_0} + A_1\cdot\bar{B_1}$$

Logic Circuit Design:-



2-bit - Comparator Using 1 bit - blocks:-

⇒ Let,

Consider 2 words:-

Word $A = A_1 A_0$

Word $B = B_1 B_0$

∴ Now, For $A=B$:-

if $(A_1 = B_1)$ && $(A_0 = B_0)$

Then $A = B$

For $A>B$:-

if $(A_1 > B_1)$ then $(A>B)$ ||

if $(A_1 = B_1)$ && $(A_0 > B_0)$ Then $(A > B)$

For $A<B$:-

if $(A_1 < B_1)$ then $(A<B)$ ||

if $(A_1 = B_1)$ && $(A_0 < B_0)$

then $(A<B)$

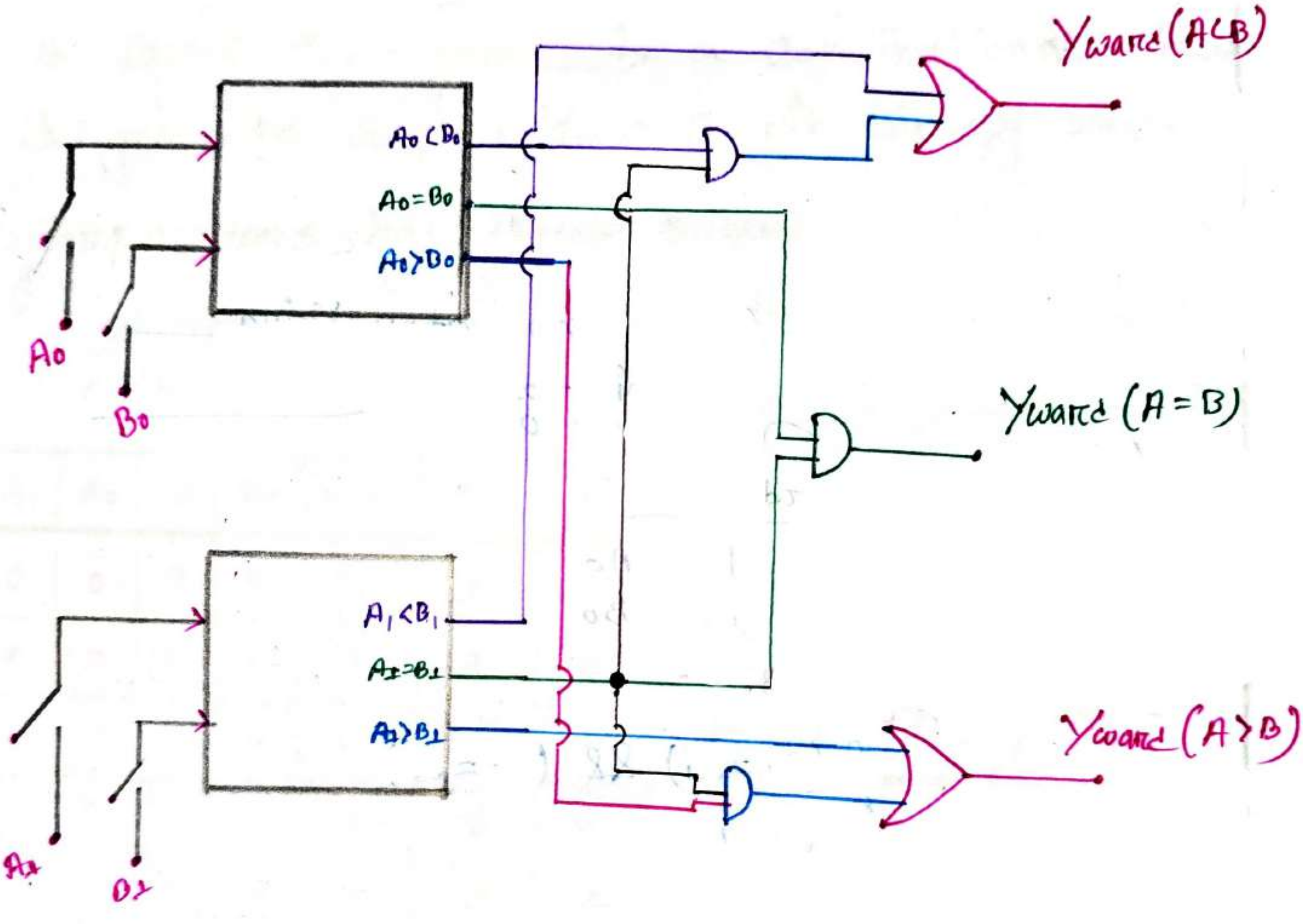


Fig:- 2-Bit Comparator Using
1-Bit Comparator Block

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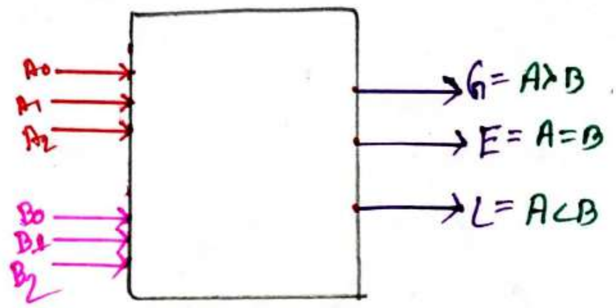
3 bit - Comparator:-

Block Diagram:-



Case: 1:-

$$E = A_2 \odot B_2 + A_1 \odot B_1 + A_0 \odot B_0$$

For this:-

$A_2 = B_2$; $A_1 = B_1$; $A_0 = B_0$

Case: 2:-

$$G = A_2 \cdot \bar{B}_2 + (A_2 \odot B_2)(A_1 \cdot \bar{B}_1) + (A_2 \odot B_2)(A_1 \odot B_1)(A_0 \cdot \bar{B}_0)$$

For this Subcase:-

Subcase 1:-

if $A_2 > B_2$ then $A > B$ OR

Subcase 2:-

if $A_2 = B_2$ && $A_1 > B_1$ then $A > B$ OR

Subcase 3:-

if $A_2 = B_2$ && $A_1 = B_1$ && $A_0 > B_0$ then $A > B$

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Case: 3:-

$$L = \bar{A}_2 B_2 + (A_2 \odot B_2)(\bar{A}_1 \cdot B_1) + (A_2 \odot B_2)(A_1 \odot B_1)(\bar{A}_0 \cdot B_0)$$

For this subcase:-

Subcase 1:-

⇒ if $A_2 < B_2$ then $A < B$ OR

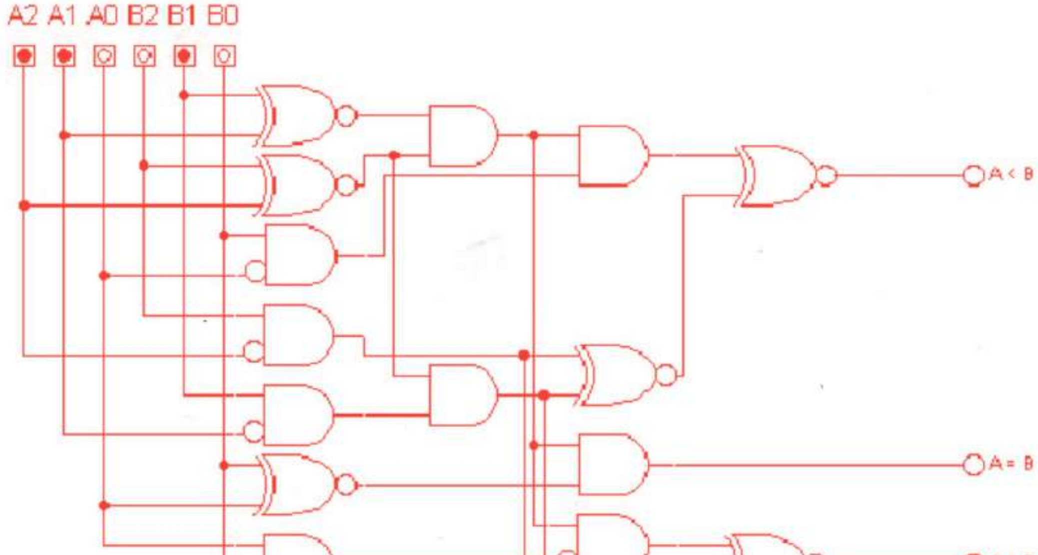
Subcase: 2:-

⇒ if $A_2 = B_2$ && $A_1 < B_1$ then $A < B$ OR

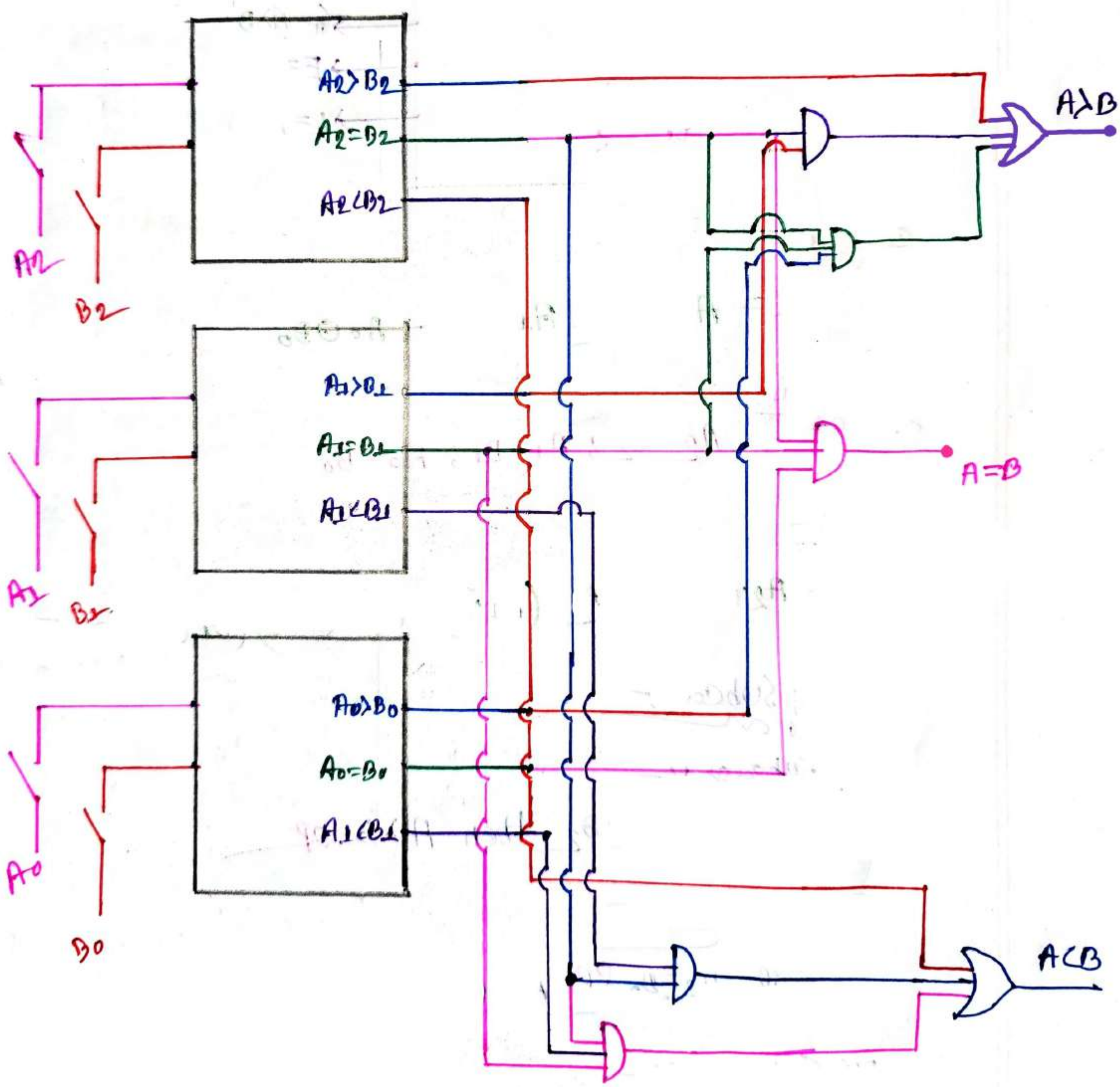
Subcase: 3:-

⇒ if $A_2 = B_2$ && $A_1 = B_1$ && $A_0 < B_0$ then $A < B$

Logic circuit:-



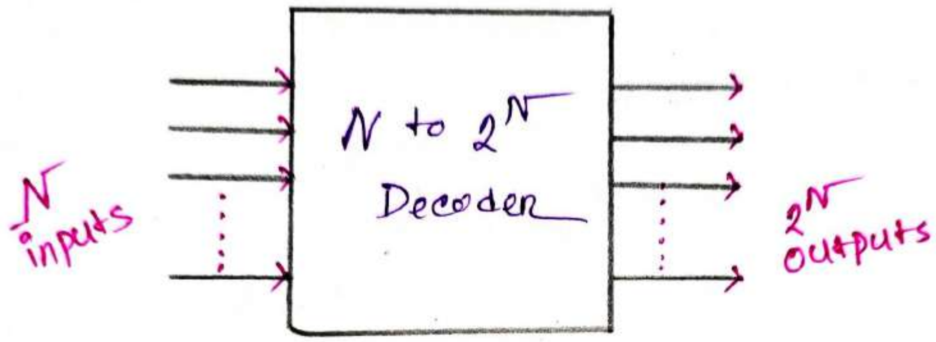
3-bit - Comparator Using 1 bit - blocks:-



#DECODERS:-

Decoder:-

⇒ The combinational circuit that change the binary information into 2^N output lines in known as Decoders.



2 to 4 line decoder:-

⇒ In this decoder. There is total two inputs. A_0, A_1 and four Outputs $\{(2^2) = 4\}$, Y_0, Y_1, Y_2, Y_3

A_1	A_0	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

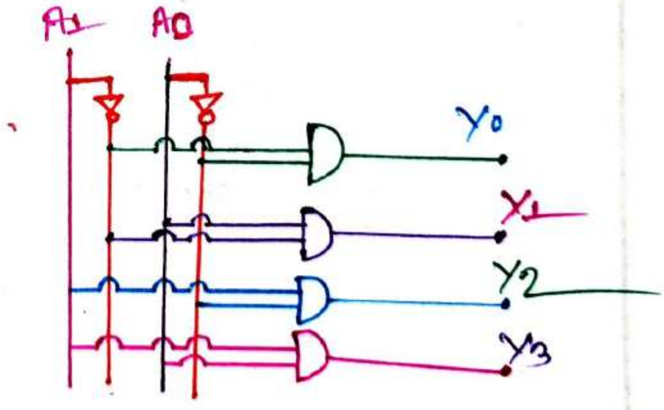
$\therefore Y_0 = \overline{A_0} \cdot \overline{A_1}$

$Y_1 = A_0 \cdot \overline{A_1}$

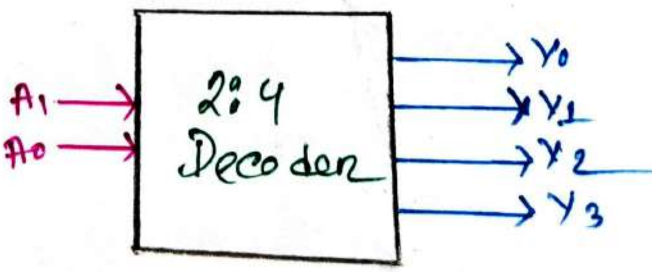
$Y_2 = \overline{A_0} \cdot A_1$

$Y_3 = A_0 \cdot A_1$

Logic circuit Design:-



Block Diagram:-



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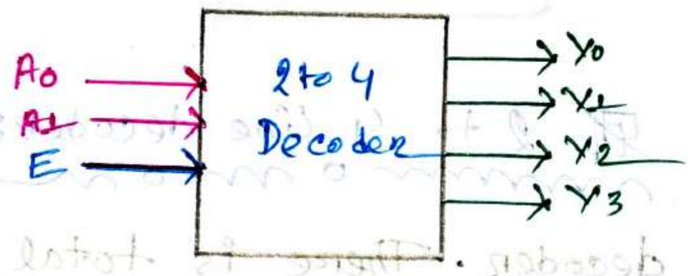
2 to 4 Line Decoder using Enable Inputs:-

⇒ In this decoder there is a total of three input A_0, A_1, E & Four Outputs Y_0, Y_1, Y_2, Y_3. For each Combination of inputs when the Enable (E) is set to 1. One of these four outputs will be 1.

Block Diagram:-



Truth table:-

Enable	Inputs		Outputs			
E	A_1	A_0	Y_0	Y_1	Y_2	Y_3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

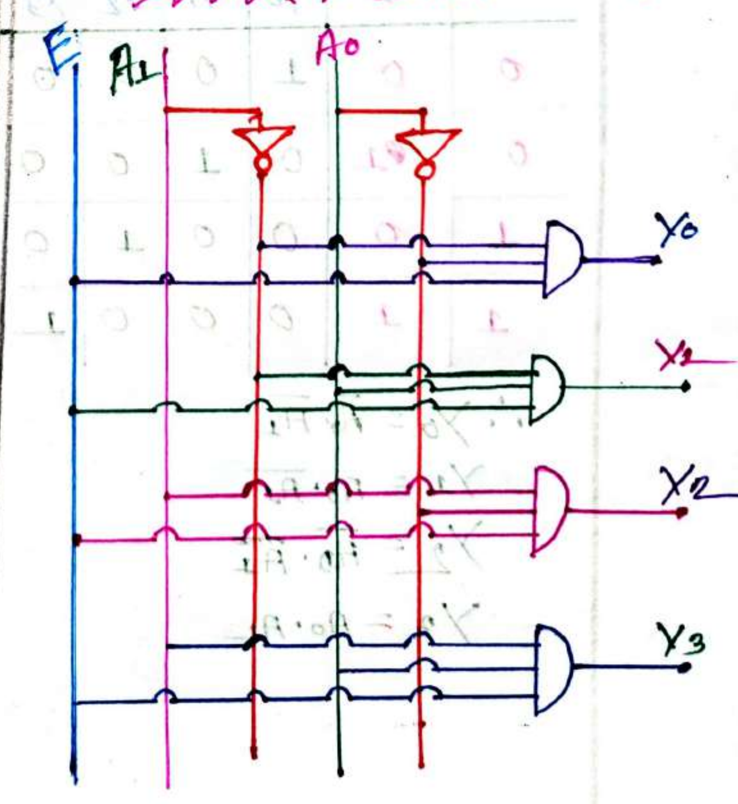
$\therefore Y_0 = \overline{A_1}\,\overline{A_0} \cdot E$

$Y_1 = \overline{A_1} A_0 \cdot E$

$Y_2 = A_1 \overline{A_0} \cdot E$

$Y_3 = A_1 A_0 E$

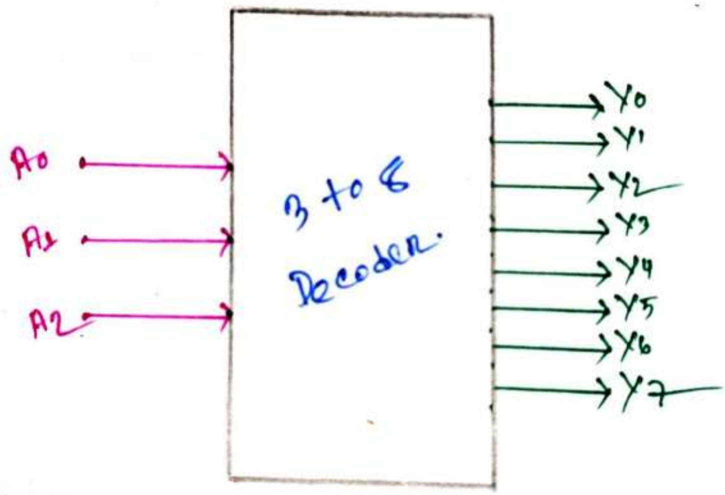
Logic Circuit Design:-



3 to 8 Line Decoder:-

⇒ This decoder Circuit Also known as Binary to Octal Decoder. In this decoder there is a total of three inputs A_0, A_1, A_2 & 8 outputs $Y_0, Y_1 \cdots, Y_7$

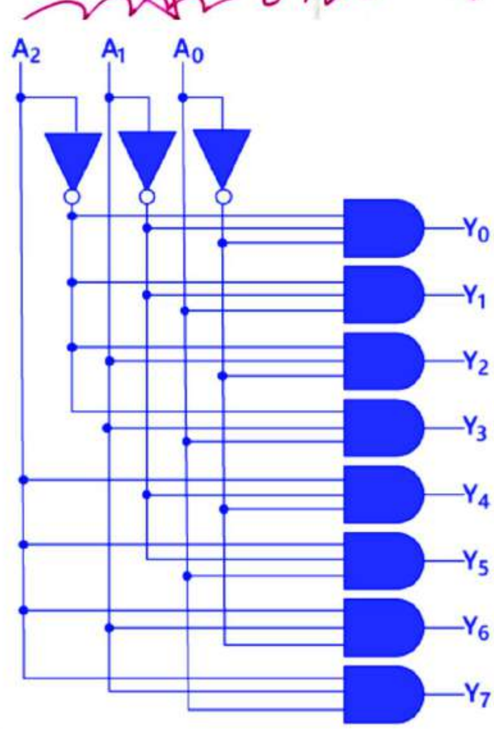
Block Diagram:-



Truth table:-

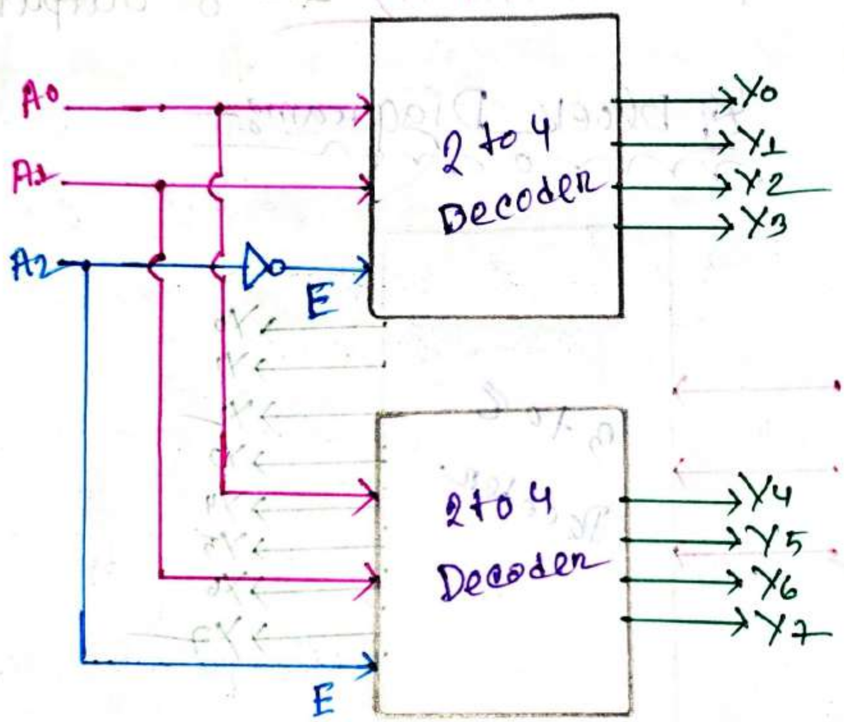
Inputs			Outputs							
A_2	A_1	A_0	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Logic Circuit Design:-

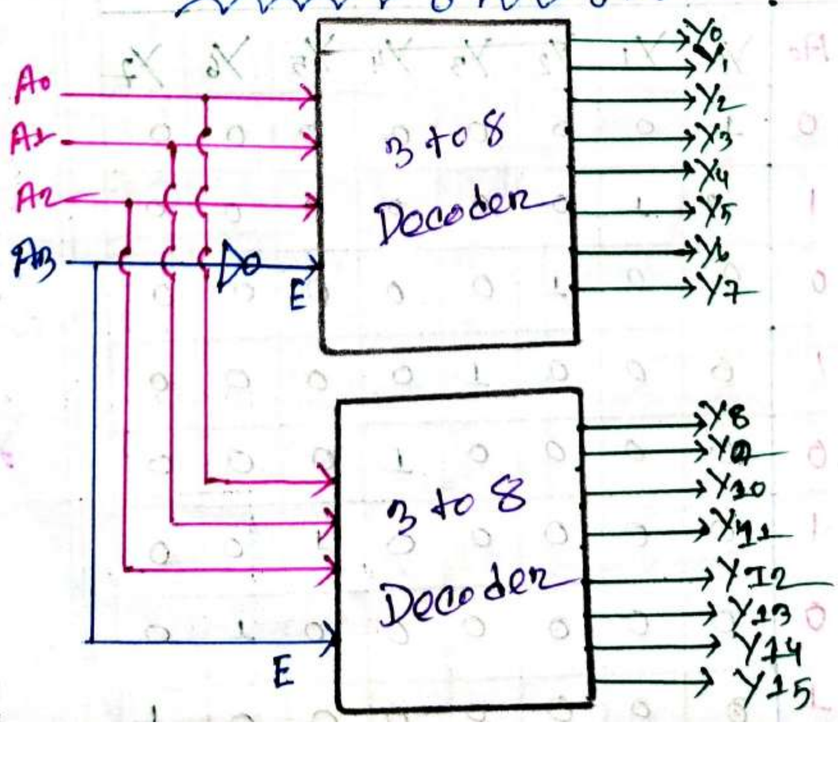


3 to 8 Decoder Using 2 to 4 Decoder:-

⇒ For this we need two(2), 2 to 4 decoder.



4 to 16 Decoder Using 3 to 8 Decoder



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Full Adder Implementation Using Decoder:—

Truth Table:—

A	B	Cin	S	Co	
0	0	0	0	0	m_0
0	0	1	1	0	m_1
0	1	0	1	0	m_2
0	1	1	0	1	m_3
1	0	0	1	0	m_4
1	0	1	0	1	m_5
1	1	0	0	1	m_6
1	1	1	1	1	m_7

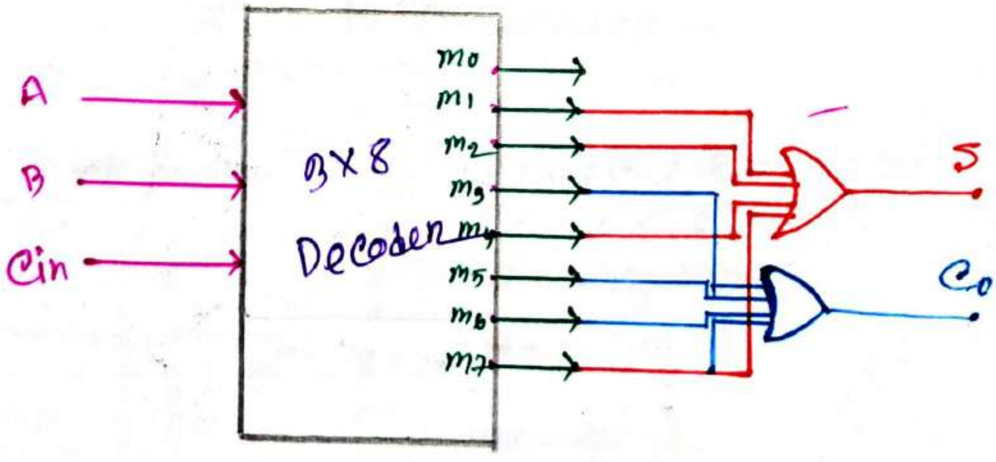
∴ minterm for Sum (S)

$F_S = \sum(m_1, m_2, m_4, m_7)$

minterm for Carry (C_0)

$F_{C_0} = \sum(m_3, m_5, m_6, m_7)$

Now Design Full Adder



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Encoders

Encoders:-

⇒ An Encoder is a digital circuit that Converts a set of binary inputs into a Unique binary Code.

Notes:- It's a combinational circuit that performs the reverse operation of a Decoder.

If input $= 2^n$

Then output $= n$

Types of Encoders —

1. 4 to 2 Encoder
2. 8 to 3 Encoder (Octal to Binary Encoder)
3. Decimal to BCD Encoder.
4. Priority Encoder.

4 to 2 Encoder:-

Truth table:-

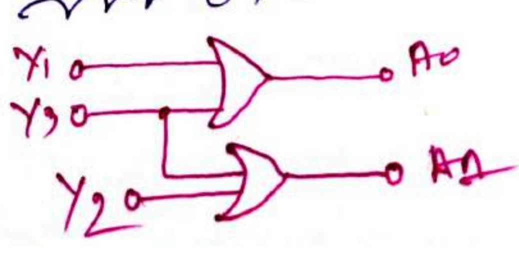
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Logical Expression:-

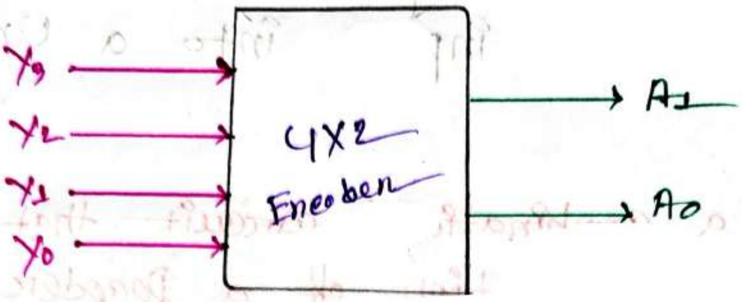
$A_1 = Y_3 + Y_2$

$A_0 = Y_1 + Y_3$

Logic Circuit:-



Block Diagram:-



8 to 3 Encoder:-

Truth table:-

Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

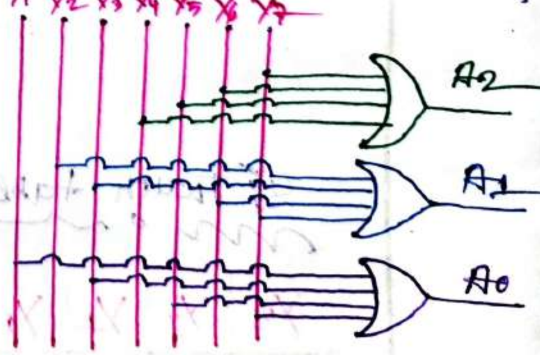
Logical Expression:-

$A_2 = Y_4 + Y_5 + Y_6 + Y_7$

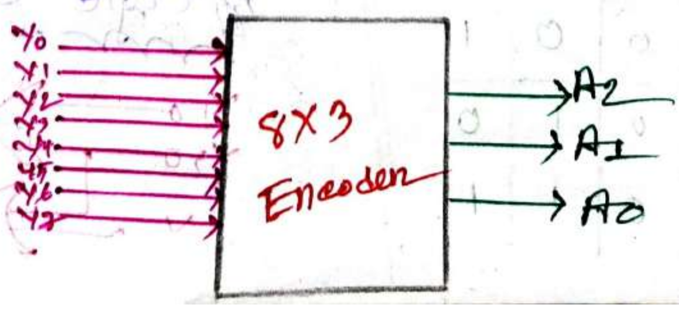
$A_1 = Y_3 + Y_2 + Y_6 + Y_7$

$A_0 = Y_1 + Y_3 + Y_5 + Y_7$

Logic Circuit

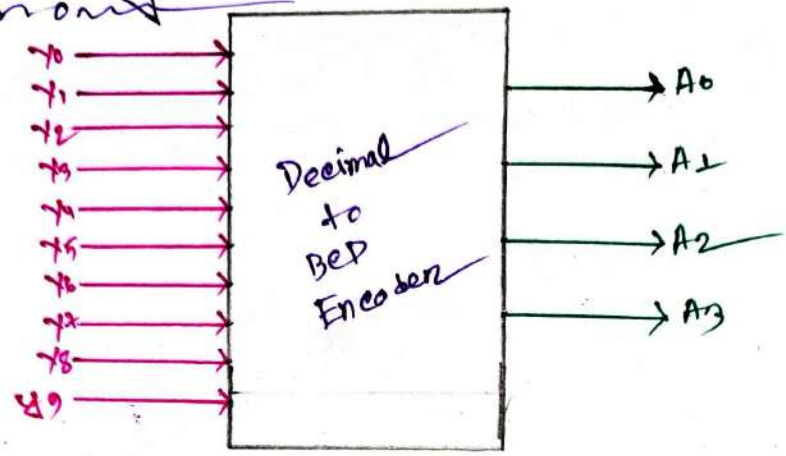


Block Diagram:-



Decimal to BCD Encoder:—

Block Diagram:—



Logical Expression:—

$A_3 = Y_9 + Y_8$

$A_2 = Y_7 + Y_6 + Y_5 + Y_4$

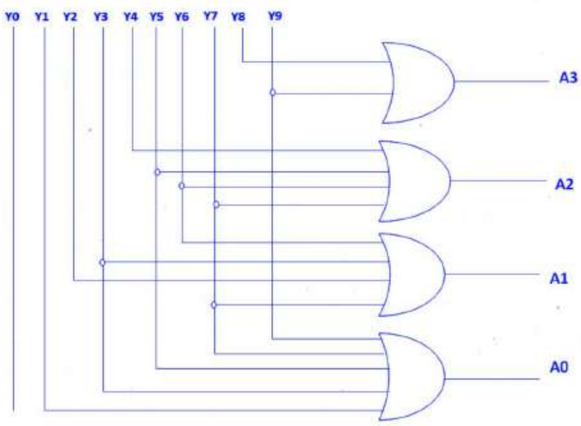
$A_1 = Y_7 + Y_6 + Y_3 + Y_2$

$A_0 = Y_9 + Y_7 + Y_5 + Y_3 + Y_1$

Truth Table:—

INPUTS								OUTPUTS		
Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	A2	A1	A0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Logical Circuit:—



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Priority Encoder:-

⇒ A circuit that compress multiple binary inputs into a smaller number of outputs.

4 to 2 priority Encoder:-

1. Given that:

priority : $Y_2 > Y_1 > Y_0 > Y_3$

Truth table:-

Y_3	Y_2	Y_1	Y_0	A_1	A_0
X	1	X	X	1	0
X	0	1	X	0	1
X	0	0	1	0	0
1	0	0	0	1	1

Logical Expression:-

$$A_1 = Y_2 + Y_3\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}$$
$$= Y_2 + Y_3\overline{Y_1}\,\overline{Y_0}$$

$$A_0 = \overline{Y_2}Y_1 + Y_3\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}$$
$$= \overline{Y_2}(Y_1 + Y_3\overline{Y_1}\,\overline{Y_0})$$
$$= \overline{Y_2}(Y_1 + Y_3\overline{Y_0})$$

2. Given that:-

priority : $Y_3 > Y_2 > Y_1 > Y_0$

Truth table:-

Y_3	Y_2	Y_1	Y_0	A_1	A_0
1	X	X	X	1	1
0	1	X	X	1	0
0	0	1	X	0	1
0	0	0	1	0	0

Logical Expression:-

$$A_1 = Y_3 + \overline{Y_3}Y_2 = Y_3 + Y_2$$

$$A_0 = Y_3 + \overline{Y_3}\,\overline{Y_2}Y_1 = Y_3 + \overline{Y_2}Y_1$$

8 to 3 Priority Encoder:-

∴ Given that the priority:-

$Y_7 < Y_6 < Y_5 < Y_4 < Y_3 < Y_2 < Y_1 < Y_0$

Truth table:-

Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
X	X	X	X	X	X	X	1	0	0	0
X	X	X	X	X	X	1	0	0	0	1
X	X	X	X	X	1	0	0	0	1	0
X	X	X	X	1	0	0	0	0	1	1
X	X	X	1	0	0	0	0	1	0	0
X	X	1	0	0	0	0	0	1	0	1
X	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Logical Expression:-

$$A_2 = Y_4 \underbrace{\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}}_{A} + Y_5 \overline{Y_4} \underbrace{\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}}_{A} + Y_6 \overline{Y_5}\,\overline{Y_4} \underbrace{\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}}_{A}$$
$$+ Y_7 \overline{Y_5}\,\overline{Y_4} \underbrace{\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}}_{A} \overline{Y_6}$$
$$= Y_4 A + Y_5 \overline{Y_4} A + Y_6 \overline{Y_5} A + Y_7 \overline{Y_6}\,\overline{Y_5}\,\overline{Y_4} A$$
$$= Y_4 A + Y_5 A + Y_6 A + Y_7 \overline{Y_6}\,\overline{Y_4} A$$
$$= Y_4 A + Y_5 A + Y_6 A + Y_7 A$$
$$= (Y_4 + Y_5 + Y_6 + Y_7) A$$
$$\therefore A_2 = (Y_4 + Y_5 + Y_6 + Y_7) \cdot (\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0})$$

$$A_1 = Y_2 \underbrace{\overline{Y_1}\,\overline{Y_0}}_{A} + Y_3 \overline{Y_2} \underbrace{\overline{Y_1}\,\overline{Y_0}}_{A} + Y_6 \overline{Y_5}\,\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2} \underbrace{\overline{Y_1}\,\overline{Y_0}}_{A} +$$
$$Y_7 \overline{Y_6}\,\overline{Y_5}\,\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2} \underbrace{\overline{Y_1}\,\overline{Y_0}}_{A}$$
$$= Y_2 A + Y_3 A + Y_6 \underbrace{\overline{Y_5}\,\overline{Y_4}}_{B} \overline{Y_3}\,\overline{Y_2} A + Y_7 \underbrace{\overline{Y_5}\,\overline{Y_4}}_{B} \overline{Y_3} A \overline{Y_2}$$
$$= Y_2 A + Y_3 A + Y_6 B A + Y_7 B A$$

$$A_1 = (Y_2 + Y_3 + Y_6 B + Y_7 B) A$$

$$= \{Y_2 + Y_3 + (Y_6 + Y_7) B\} A$$

$$\therefore A_1 = \overline{Y_1}\,\overline{Y_0}\{\overline{Y_5}\,\overline{Y_4}(Y_6 + Y_7) + Y_3 + Y_2\}$$

$$A_0 = Y_1\overline{Y_0} + Y_3\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0} + Y_5\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0} + Y_7\overline{Y_6}\,\overline{Y_5}\,\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2}\,\overline{Y_1}\,\overline{Y_0}$$

$$= Y_1\overline{Y_0} + Y_3\overline{Y_2}\,\overline{Y_0} + Y_5\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2}\,\overline{Y_0} + Y_7\overline{Y_6}\,\overline{Y_5}\,\overline{Y_4}\,\overline{Y_3}\,\overline{Y_2}\,\overline{Y_0}$$

$$= Y_1\overline{Y_0} + Y_3\overline{Y_2}\,\overline{Y_0} + Y_5\overline{Y_4}\,\overline{Y_2}\,\overline{Y_0} + Y_7\overline{Y_6}\,\overline{Y_4}\,\overline{Y_2}\,\overline{Y_0}$$

$$= \overline{Y_0}(Y_1 + \overline{Y_2}\,\overline{Y_0}(Y_3 + \overline{Y_4}(Y_5 + Y_7\overline{Y_6})))$$

$$\therefore A_0 = \overline{Y_0}(Y_1 + \overline{Y_2}\,\overline{Y_0}(Y_3 + \overline{Y_4}(Y_5 + Y_7\overline{Y_6})))$$

3: Given that the priority

$$Y_7 > Y_6 > Y_5 > Y_4 > Y_3 > Y_2 > Y_1 > Y_0$$

Truth table:

Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
1	X	X	X	X	X	X	X	1	1	1
0	1	X	X	X	X	X	X	1	1	0
0	0	1	X	X	X	X	X	1	0	1
0	0	0	1	X	X	X	X	1	0	0
0	0	0	0	1	X	X	X	0	1	1
0	0	0	0	0	1	X	X	0	1	0
0	0	0	0	0	0	1	X	0	0	1
0	0	0	0	0	0	0	1	0	0	0

Logical Expression:-

$$A_0 = Y_7 + Y_5\overline{Y_6}\,\overline{Y_7} + Y_3\overline{Y_4}\,\overline{Y_5}\,\overline{Y_6}\,\overline{Y_7} + Y_1\overline{Y_2}\,\overline{Y_3}\,\overline{Y_4}\,\overline{Y_5}\,\overline{Y_6}\,\overline{Y_7}$$

$$= Y_7 + Y_5\overline{Y_6} + Y_3\overline{Y_4}\,\overline{Y_6} + Y_1\overline{Y_2}\,\overline{Y_4}\,\overline{Y_6}$$

$$= Y_7 + \overline{Y_6}(Y_5 + Y_3\overline{Y_4} + Y_1\overline{Y_2}\,\overline{Y_4})$$

$$= Y_7 + \overline{Y_6}(Y_5 + \overline{Y_4}(Y_3 + Y_1\overline{Y_2}))$$

$$A_1 = Y_7 + Y_6\overline{Y_7} + Y_3\overline{Y_4}\,\overline{Y_5}\,\overline{Y_6}\,\overline{Y_7} + Y_2\overline{Y_3}\,\overline{Y_4}\,\overline{Y_5}\,\overline{Y_6}\,\overline{Y_7}$$

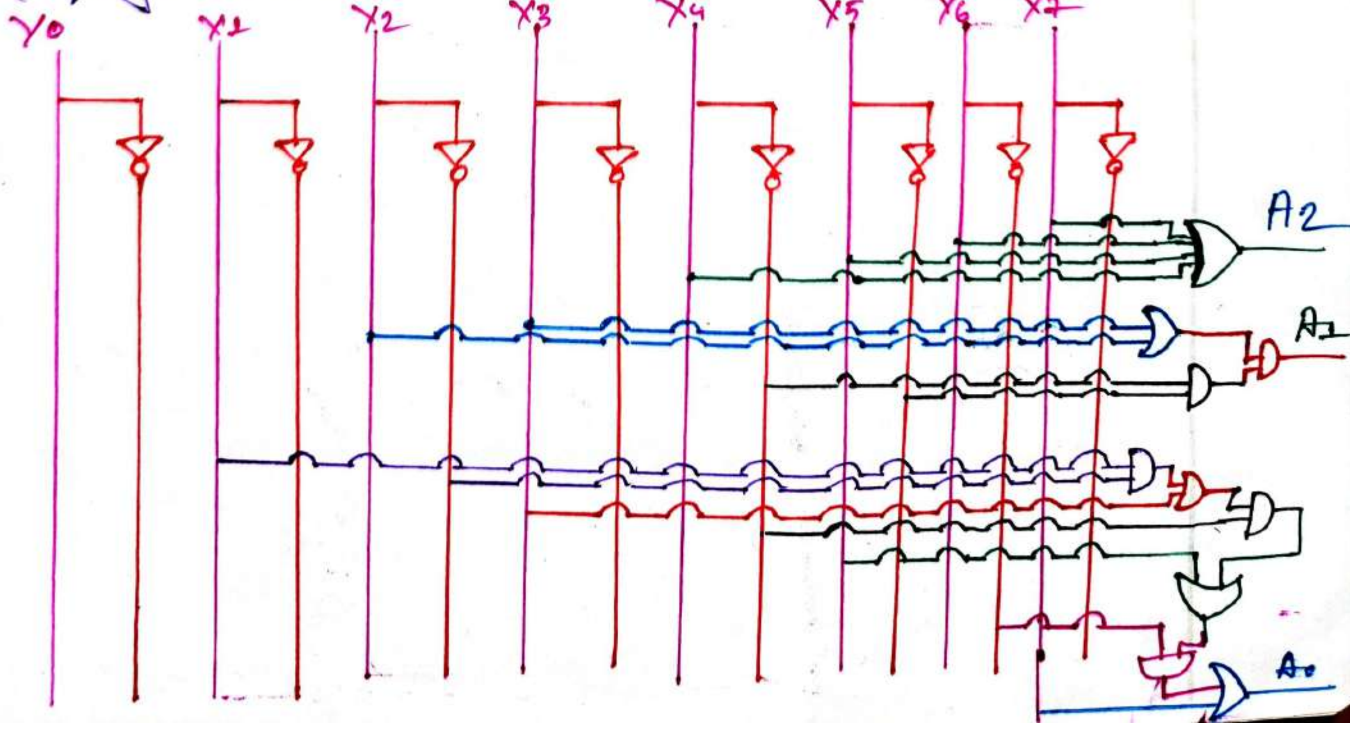
$$= Y_7 + Y_6 + Y_3\overline{Y_4}\,\overline{Y_5} + Y_2\overline{Y_4}\,\overline{Y_5}$$

$$= Y_7 + Y_6 + \overline{Y_4}\,\overline{Y_5}(Y_3 + Y_2)$$

$$A_2 = Y_7 + Y_6\overline{Y_7} + Y_5\overline{Y_6}\,\overline{Y_7} + Y_4\overline{Y_5}\,\overline{Y_6}\,\overline{Y_7}$$

$$= Y_7 + Y_6 + Y_5 + Y_4$$

Logical Circuit:-



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MULTIPLEXERS (MUX)

Multiplexers: (MUX)

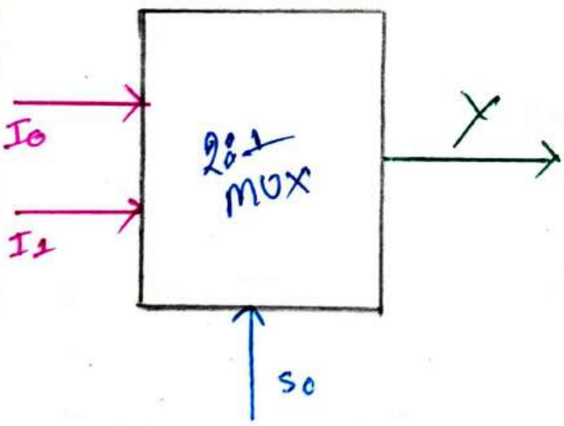
⇒ A multiplexer is a combinational circuit that has many data inputs and one output.

Types of MUX:—

1. 2X1 MUX
2. 8X1 MUX
3. 4X1 MUX
4. 32X1 MUX

2:1 MUX:—

Block Diagram:—



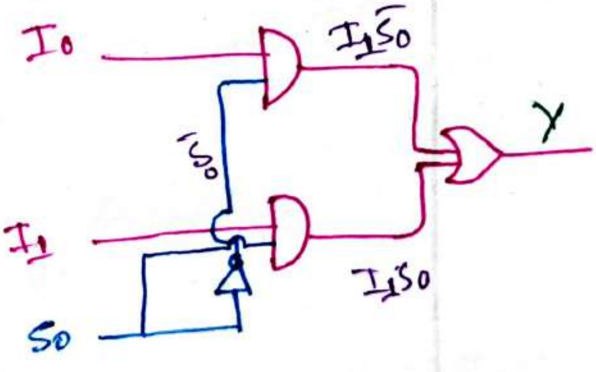
Truth Table.

S_0	S_0	I_0	I_1	Y
	0	0	X	0
	0	1	X	1
	1	X	0	0
	1	X	1	1

Logical Expression:—

$$Y = \bar{S}_0 I_0 + S_1 I_1$$

Logical circuit



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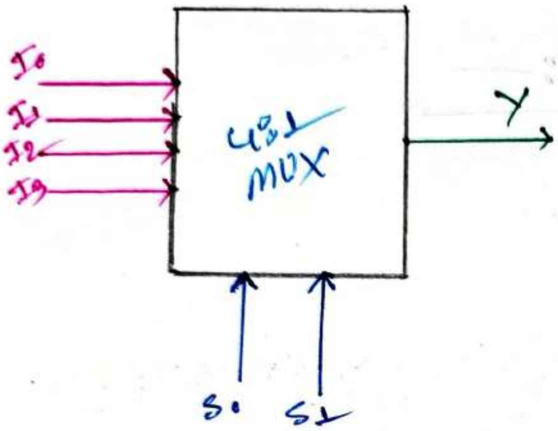
4:1 Multiplexer:-

Block Diagram:-



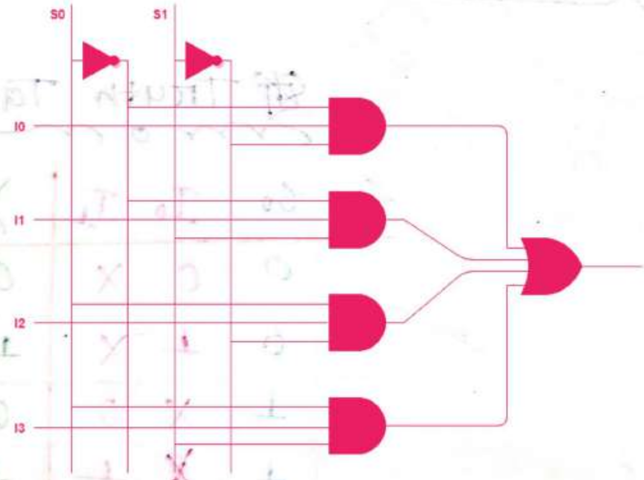
Truth table:-

S_0	S_1	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Logical Expression:-

$$Y = \bar{S}_0\bar{S}_1 I_0 + \bar{S}_0 S_1 I_1 + S_0\bar{S}_1 I_2 + S_0 S_1 I_3$$

Logical Circuit:-



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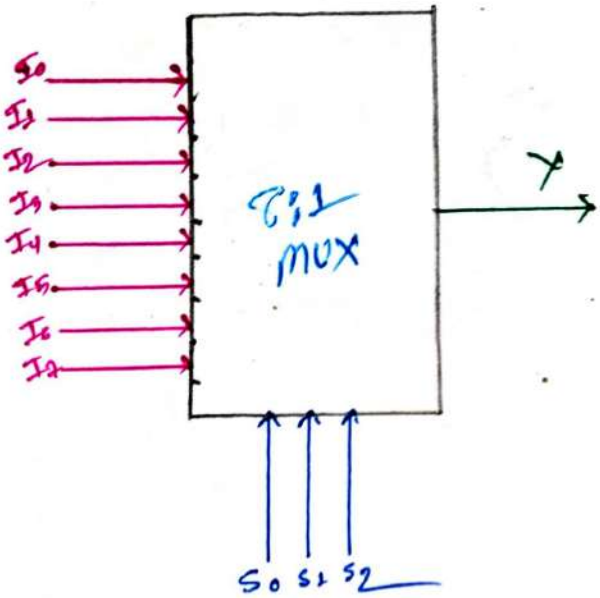
8:1 MUX:-

Block Diagram:-



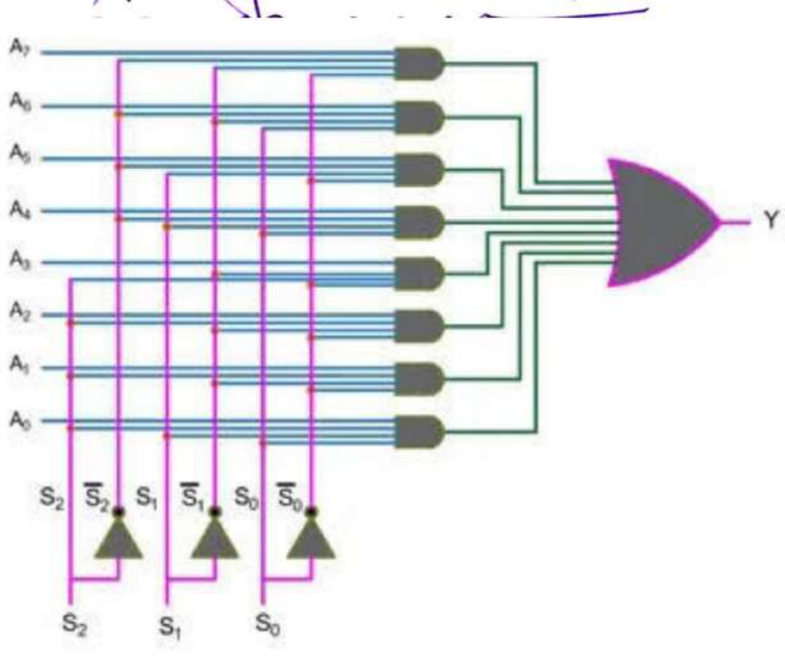
Truth table:-

S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

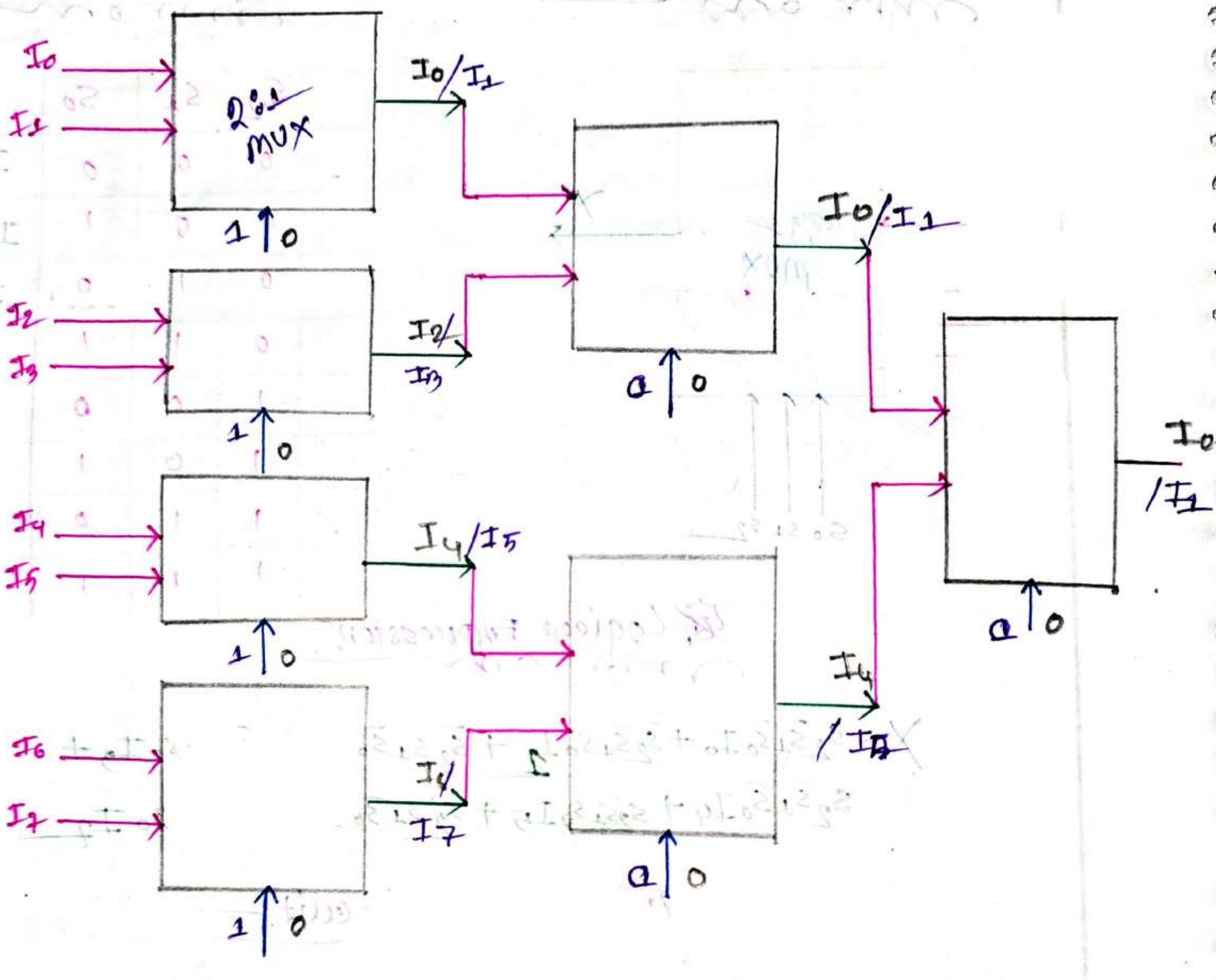
Logical Expression:-

$$Y = \bar{S}_2\bar{S}_1\bar{S}_0 I_0 + \bar{S}_2\bar{S}_1 S_0 I_1 + \bar{S}_2 S_1 \bar{S}_0 I_2 + \bar{S}_2 S_1 S_0 I_3 + S_2\bar{S}_1\bar{S}_0 I_4 + S_2\bar{S}_1 S_0 I_5 + S_2 S_1 \bar{S}_0 I_6 + S_2 S_1 S_0 I_7$$

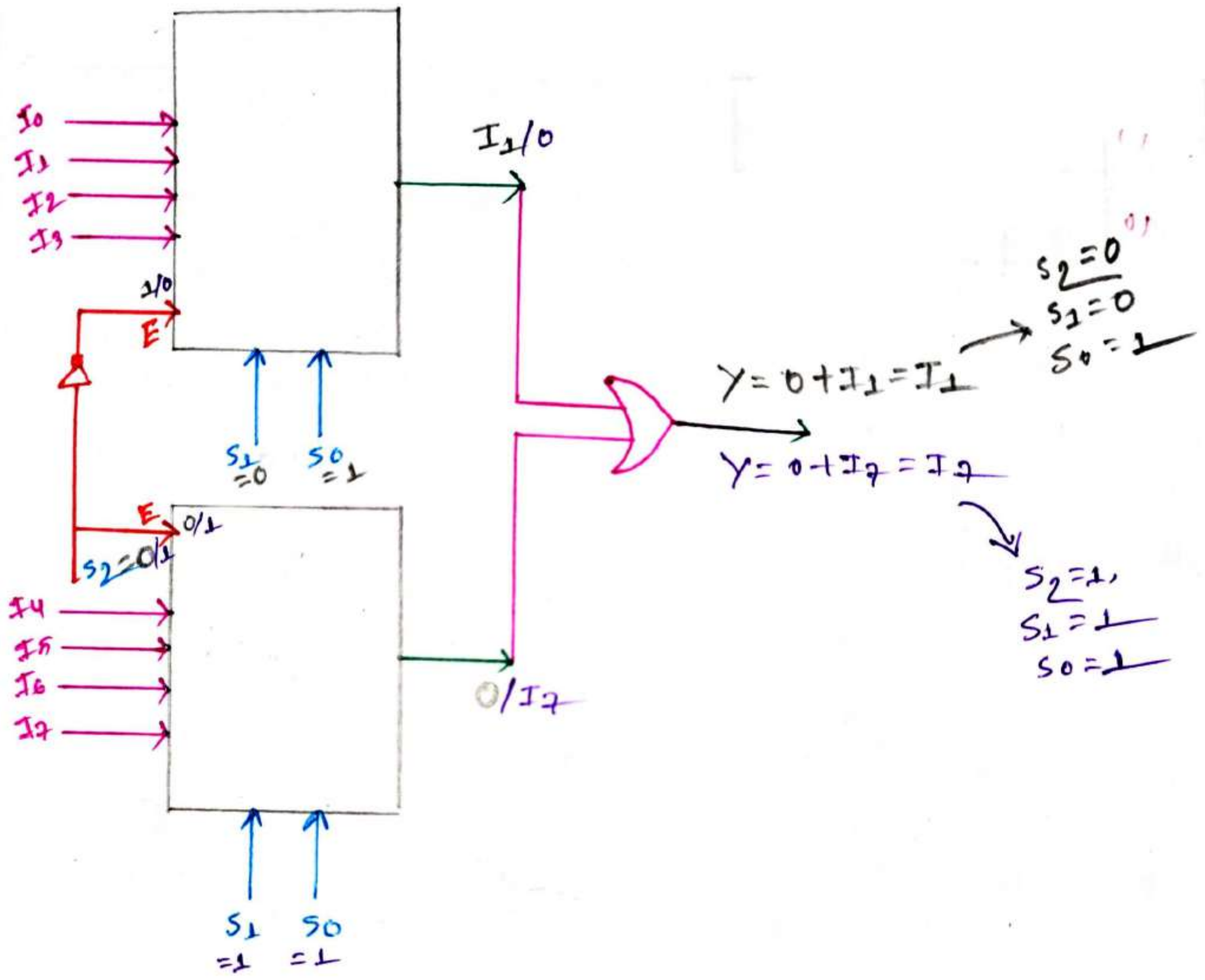
Logical circuit:-



8:1 MUX Using 2:1 MUX



8X1 MUX Using 4X1 MUX



Implementation of Boolean function Using multiplexer (MUX)

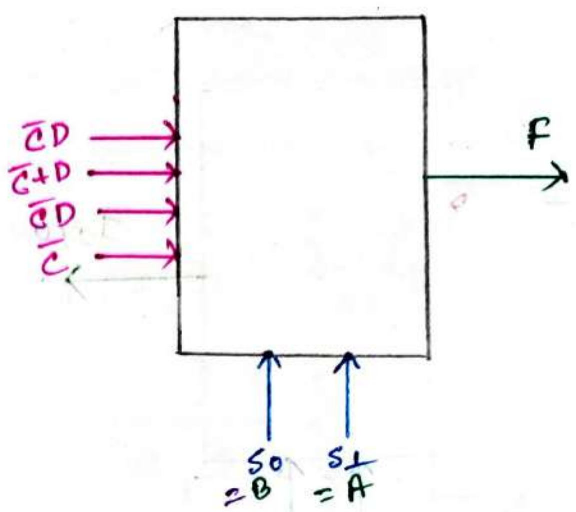
problem 1:—

⇒ Implement $F(A,B,C,D) = \sum m(1,4,5,7,9,12,13)$ using 4X1 MUX.

Step 1:—

AB \ CD (S_1S_0)	00	01	11	10	
00		1			$\bar{C}D$
01	1	1	1		$\bar{C}+D$
11	1	1			$\bar{C}$
10		1			

Step 2



Step 3:—

Assume:—

$S_0 = B$

$S_1 = A$

Step 4:—

S_1	S_0	Y
0	0	$I_0 = \bar{C}D$
0	1	$I_1 = \bar{C}+D$
1	0	$I_2 = \bar{C}D$
1	1	$I_3 = \bar{C}$

Full adder implementation using 4X1 MUX:—

Truth table:—

A	B	Cin	S	C_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

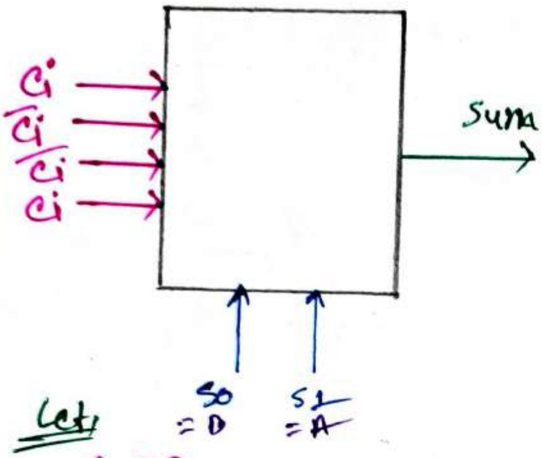
K-map: (S)

A/S_1 \ BC (S_0)	00	01	11	10
0	For 0,0 S_1S_0	0,0 1 S_1S_0	For 0,1	For 0,1 1
1	For 1,0 1	For 1,0	1 For 1,1	For 1,1

For Carry (C_0)

A \ BC	00	01	11	10
0			1	
1		1	1	1

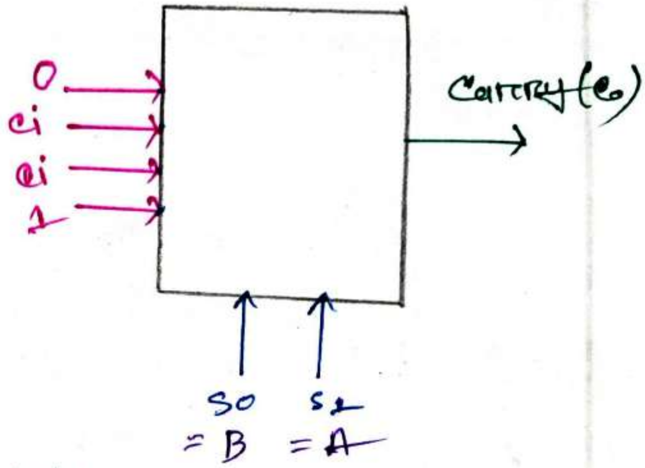
4X1 MUX for sum:-



Let,
$S_0 = B$
$S_1 = A$

S_1	S_0	Sum(S)
0	0	$I_0 = C_i$
0	1	$I_1 = \overline{C_i}$
1	0	$I_2 = \overline{C_i}$
1	1	$I_3 = C_i$

4X1 MUX for Carry (C_0)

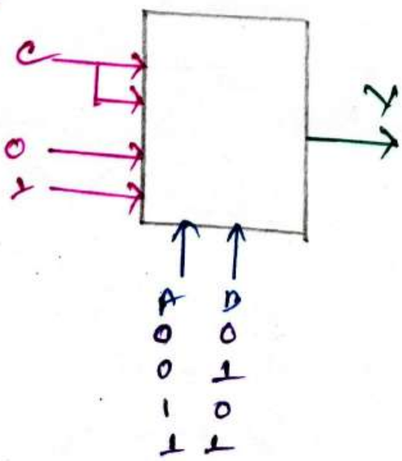


Let,
$S_0 = B$
$S_1 = A$

S_1	S_0	Carry (C_0)
0	0	$I_0 = 0$
0	1	$I_1 = C_{in}$
1	0	$I_2 = C_{in}$
1	1	$I_3 = 1$

Logical Expression from MUX:-

(i)



$$Y = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\cdot 0 + AB\cdot 1$$
$$= \bar{A}\bar{B}C + \bar{A}BC + 0 + AB$$
$$= \bar{A}\bar{B}C + BC + AB$$
$$= C + BC + AB$$

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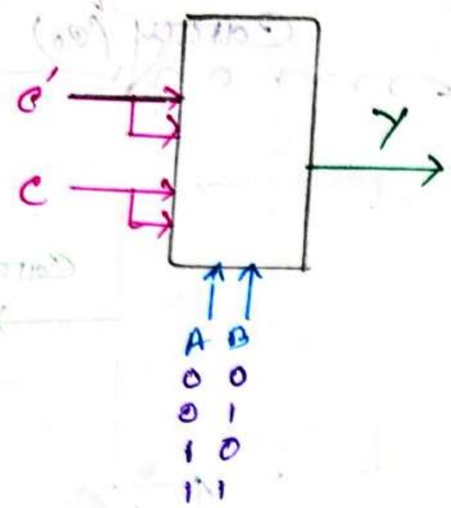
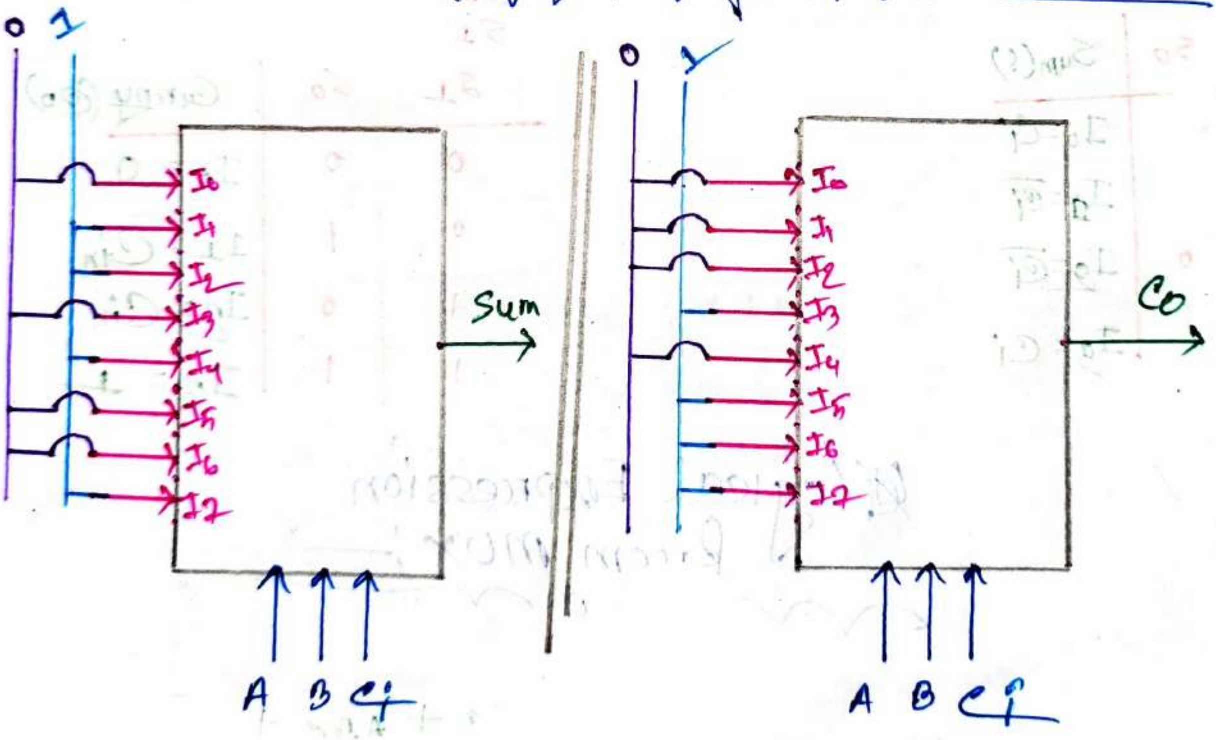
(ii)



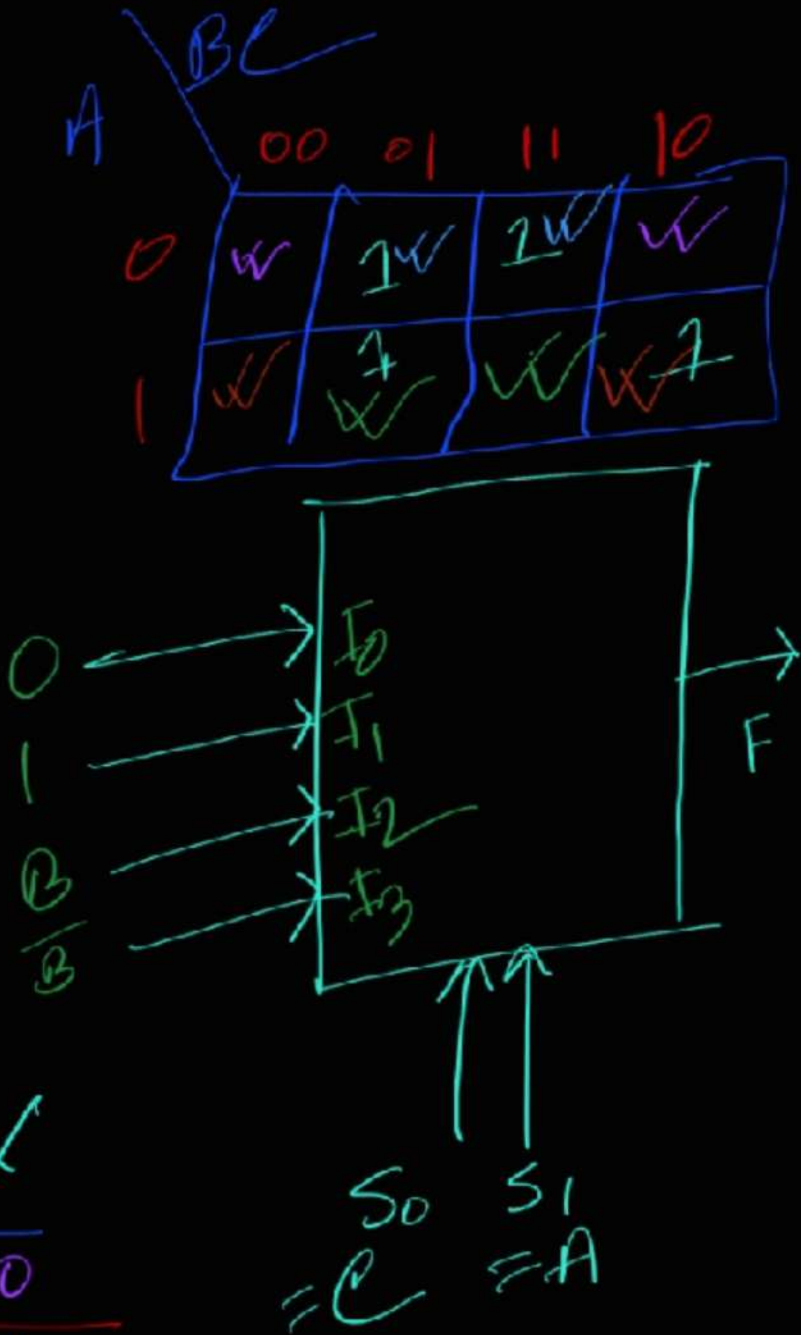
$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + ABC$$
$$= \bar{A}(\bar{B}\bar{C} + B\bar{C}) + AC(\bar{B} + B)$$
$$= \bar{A}(B\bar{C} + \bar{C}) + AC$$
$$= \bar{A}B\bar{C} + \bar{A}\bar{C} + AC$$
$$= \bar{A}\bar{C}(B+1) + AC$$
$$= \bar{A}\bar{C} + AC$$
$$= A \odot C$$

Full adder Implement using 8:1 MUX



Implementing Boolean function using multiplexer using 4:1 MUX

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



A	C	Y
0	0	$I_0 = 0$
0	1	$I_1 = 1$
1	0	$I_2 = B$
1	1	$I_3 = \bar{B}$

Let,
$S_0 = C$
$S_1 = A$

Date : / /

DeMultiplexer (DeMUX)

Date : / /

DeMultiplexer (DEMUX)

⇒ One input and many output

⇒ Reverse operation of multiplexer

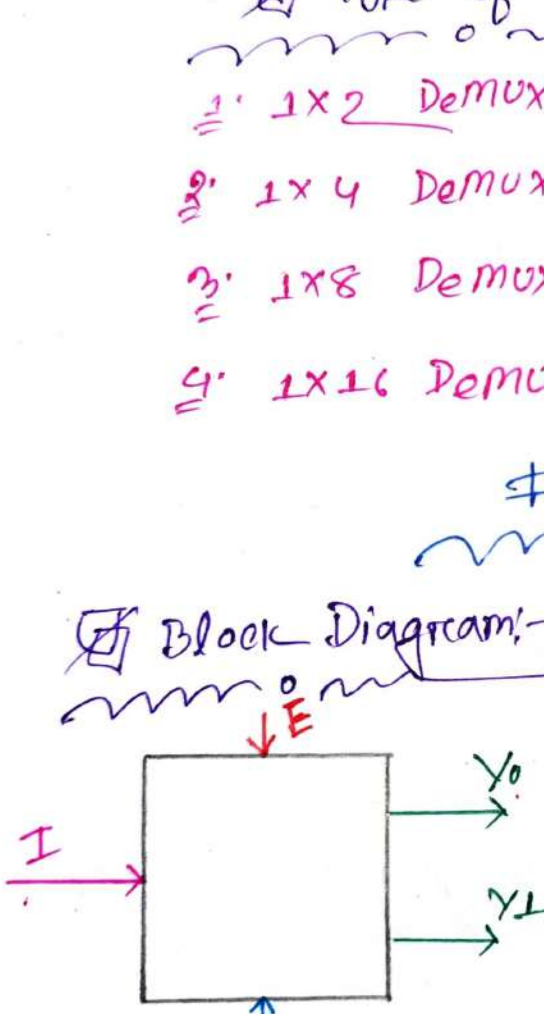
⇒ One to many circuit or Data distributor.

Type of DeMUX:-

1. 1×2 DeMUX
2. 1×4 DeMUX
3. 1×8 DeMUX
4. 1×16 DeMUX

1×2 DeMUX

Block Diagram:-



Truth table:-

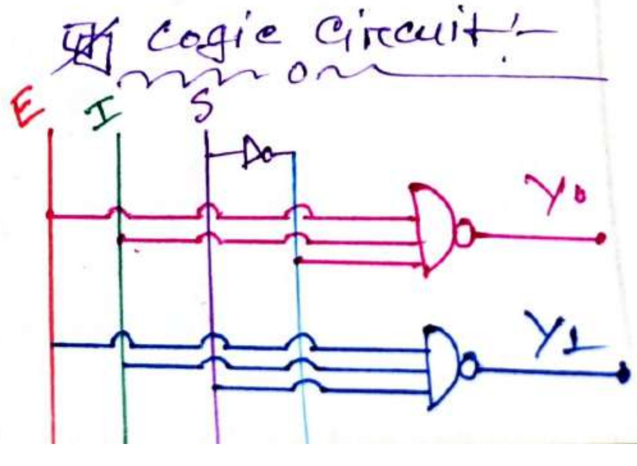
E	S_0	Y_0	Y_1
0	0 X	0	0
0	1 X	0	0
1	0	I	0
1	1	0	I

Logical Expression:-

$Y_0 = E\bar{S}_0 I$

$Y_1 = E S_0 I$

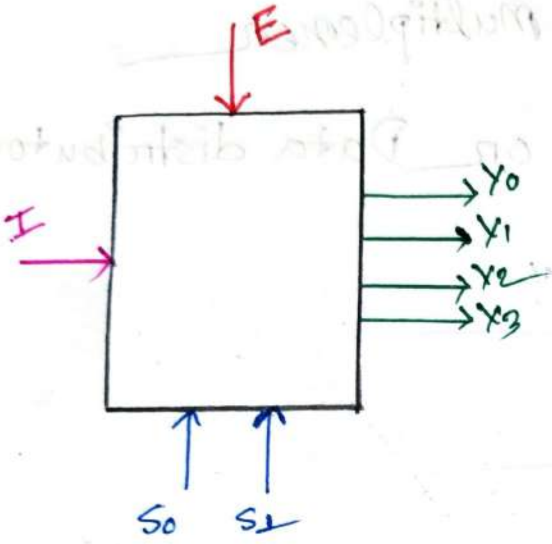
Logic Circuit:-



Subject ________ Date : / /

1×4 DeMUX:-

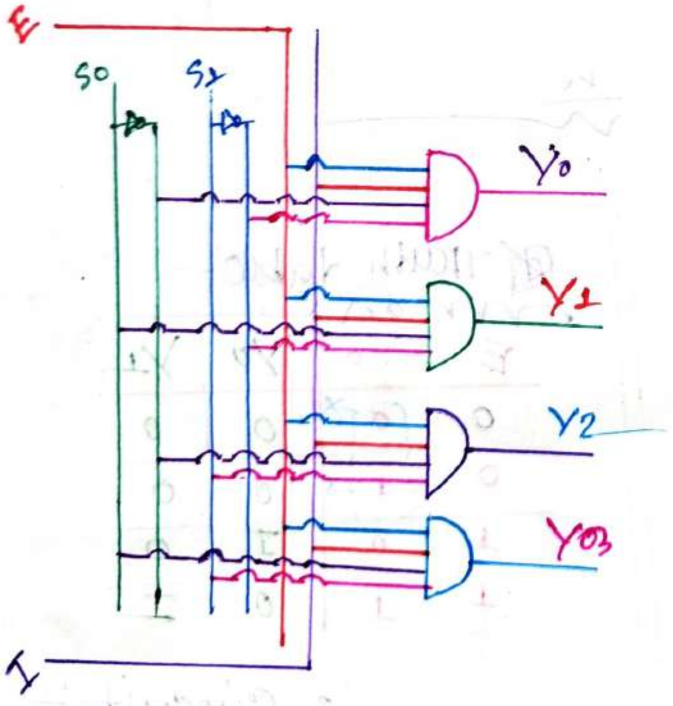
Block Diagram:-



Truth Table:-

E	S_1	S_0	Y_0	Y_1	Y_2	Y_3
0	X	X	0	0	0	0
1	0	0	I	0	0	0
1	0	1	0	I	0	0
1	1	0	0	0	I	0
1	1	1	0	0	0	I

Logical Circuit:-



Logical Expression

$Y_0 = E\bar{S_1}\bar{S_0}I$

$Y_1 = E\bar{S_1}S_0I$

$Y_2 = ES_1\bar{S_0}I$

$Y_3 = ES_1S_0I$

Full Subtractor Using 1:8 DeMUX

⇒ From Subtractor truth table:-

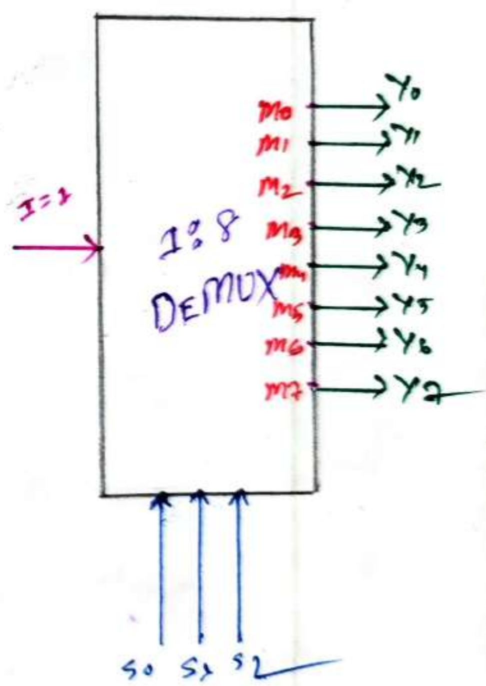
$$D(A, B, Bi) = \sum m(1, 2, 4, 7)$$

$$Bo(A, B, Bi) = \sum m(1, 2, 3, 7)$$

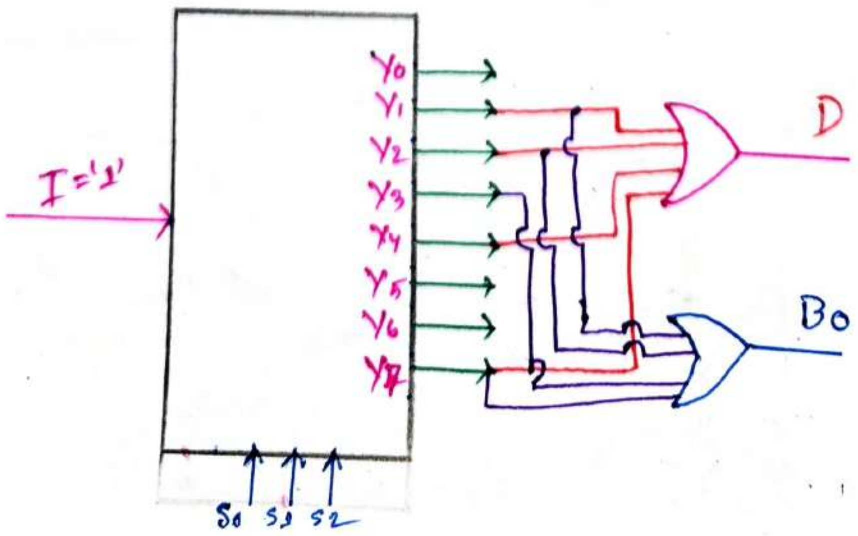
Truth table for 1:8 DeMUX:-

S_2	S_1	S_0	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Block Diagram



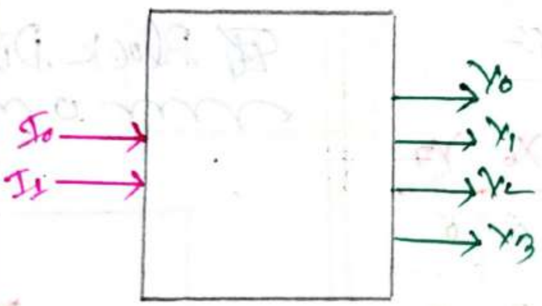
Now Implement Subtractor



Demux as Decoders—

1:4 Demux As 2:4 Decoder:—

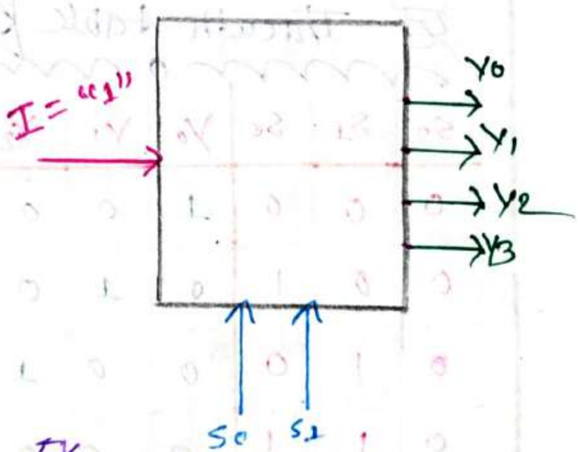
2:4 Decoder:—



Truth table:—

I_1	I_0	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

1:4 Demux:—



Truth table:—

S_1	S_0	Y_0	Y_1	Y_2	Y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Note:- For 1:8 Demux As 3:8 Decoder we have to follow the same procedure

Date :   /   /

Digital Logic Family

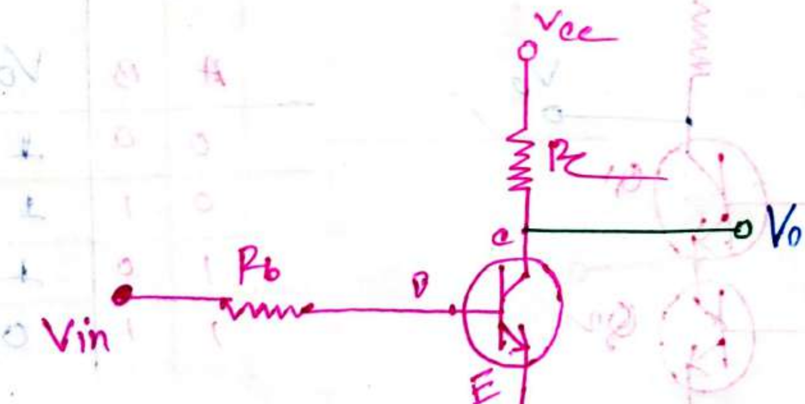
Subject: ________ Date : / /

Digital Logic Family:-

1. "RTL" → Resistor - Transistor Logic
2. "DTL" → Diode - Transistor Logic
3. "ECL" → Emitter Collector Logic
4. "TTL" → Transistor - Transistor Logic
5. "MOS" → Metal Oxide Semiconductor
6. "CMOS" → Complementary MOS

Resistor - Transistor Logic:- (RTL)

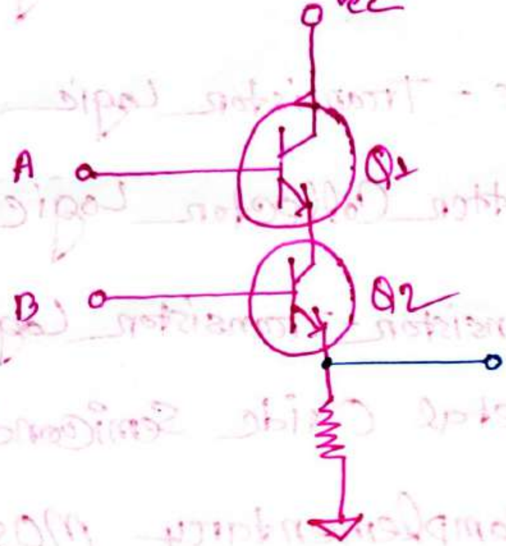
NOT Gate Using RTL:-



Truth Table

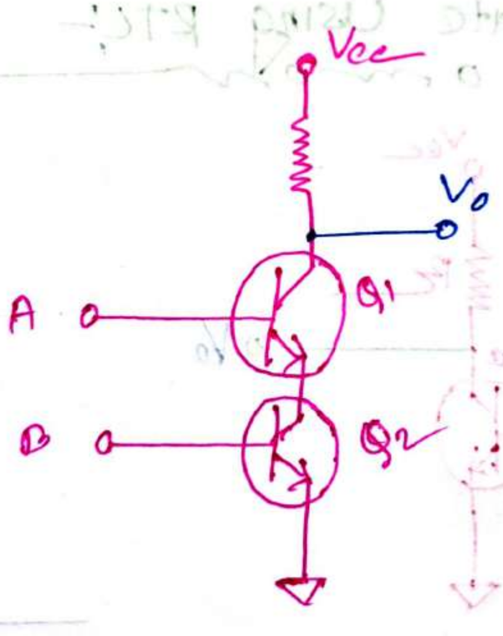
Vin	Vo
0	1
1	0

AND Gate Using RTL:-



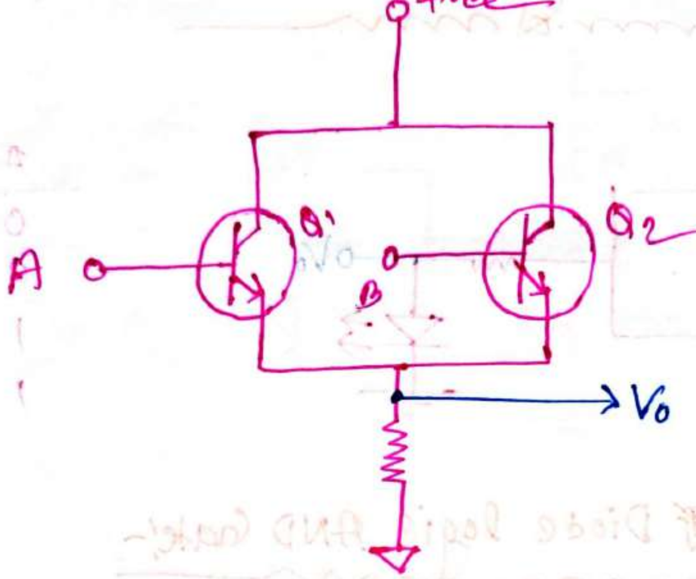
A	B	Vo
0	0	0
0	1	0
1	0	0
1	1	1

NAND Gate Using RTL:-



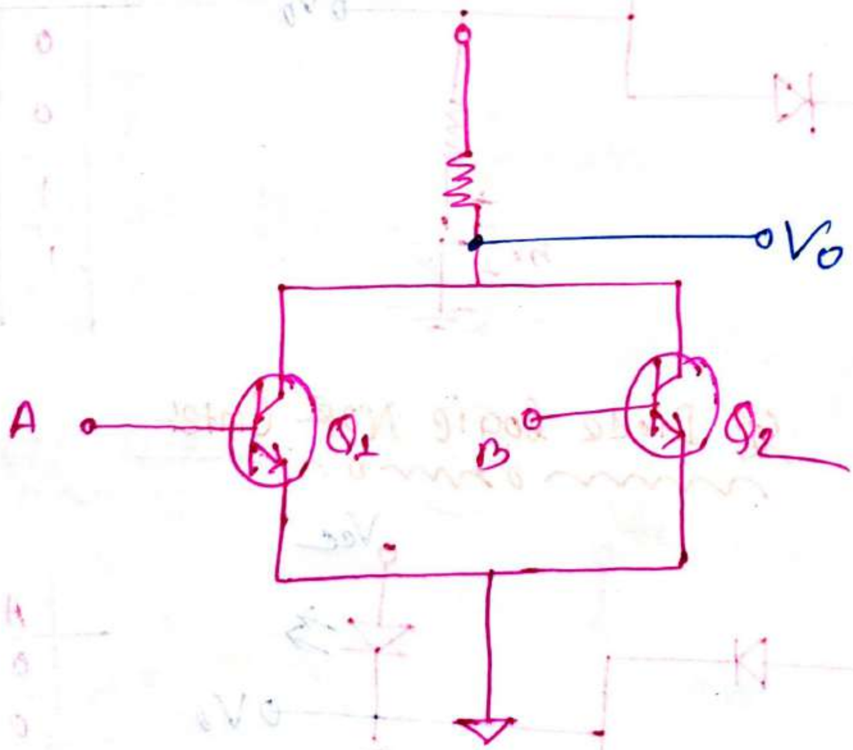
A	B	Vo
0	0	1
0	1	1
1	0	1
1	1	0

OR Gate Using RTL:-



A	B	V_o
0	0	0
0	1	1
1	0	1
1	1	1

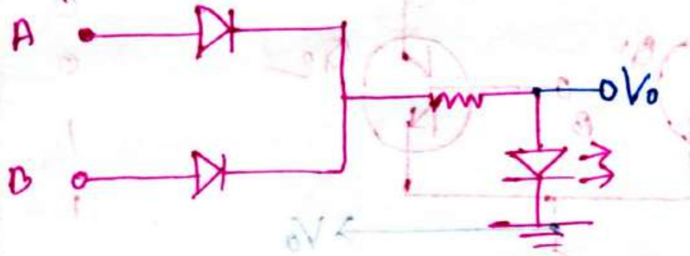
NOR Gate Using RTL:-



A	B	V_o
0	0	1
0	1	0
1	0	0
1	1	0

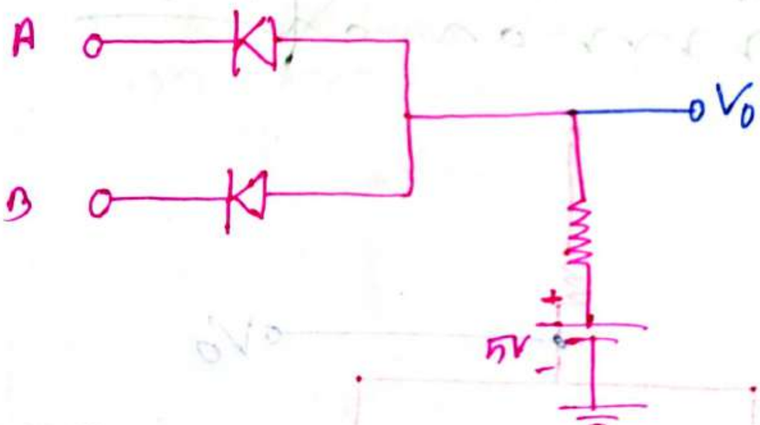
Diode Transistor Logic (DTL)

Diode Logic OR Gate:-



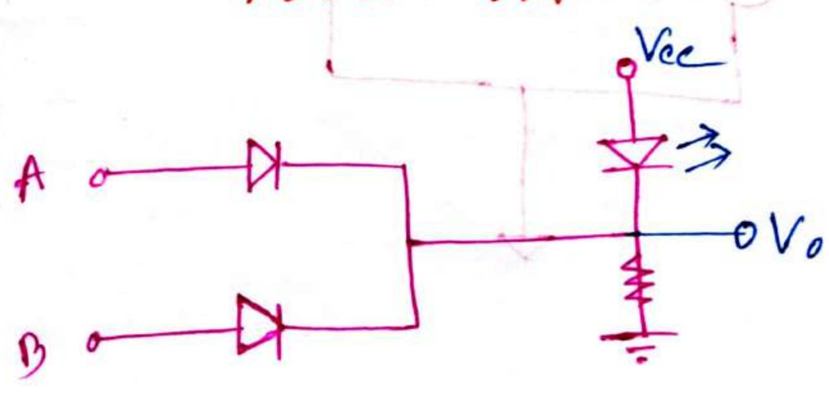
A	B	V_o
0	0	0
0	1	1
1	0	1
1	1	1

Diode logic AND Gate:-



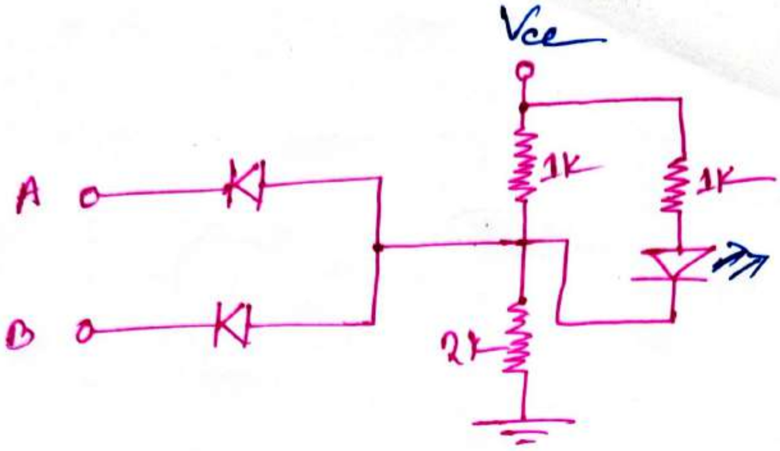
A	B	V_o
0	0	0
0	1	0
1	0	0
1	1	1

Diode logic NOR Gate:-



A	B	V_o
0	0	1
0	1	0
1	0	0
1	1	0

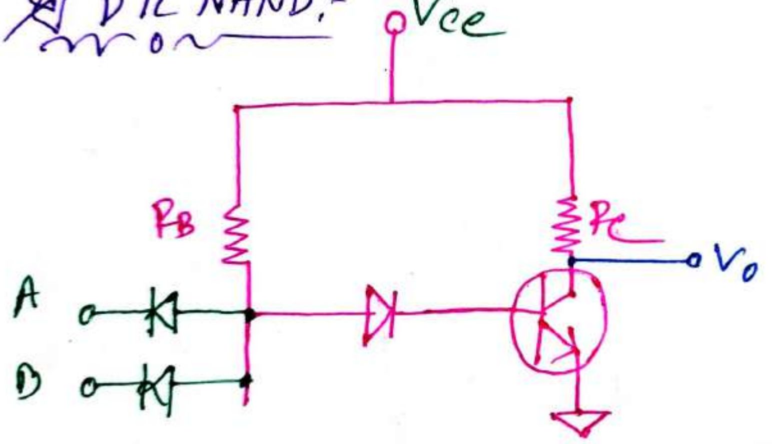
Diode NAND Gate:-



A	B	LED
0	0	1
0	1	1
1	0	1
1	1	0

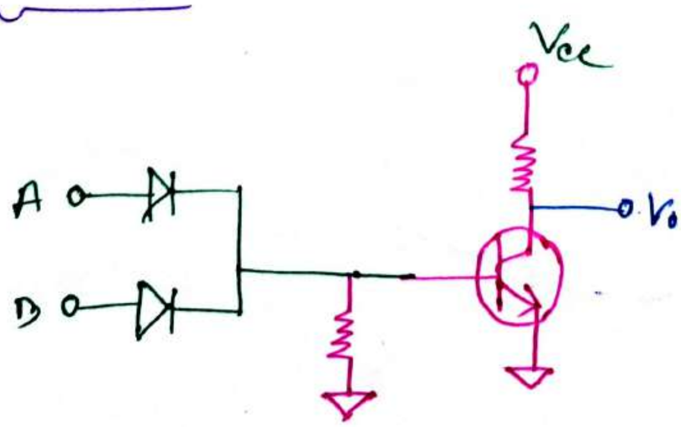
Diode transistor Logic:-(DTL):-

DTL NAND:-



A	B	Vo
0	0	1
0	1	1
1	0	1
1	1	0

DTL NOR:-

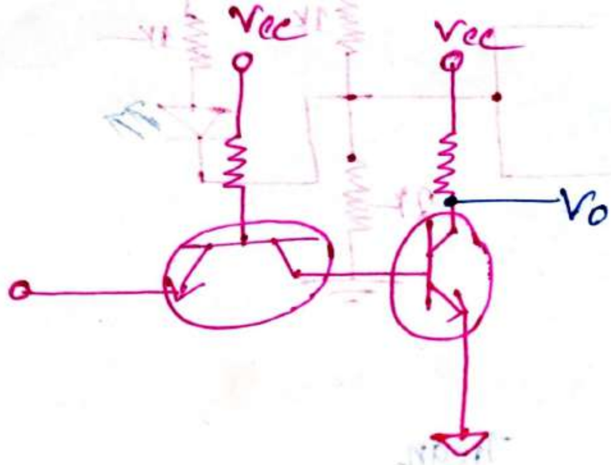


A	B	Vo
0	0	1
0	1	0
1	0	0
1	1	0

Subject ______ Date : / /

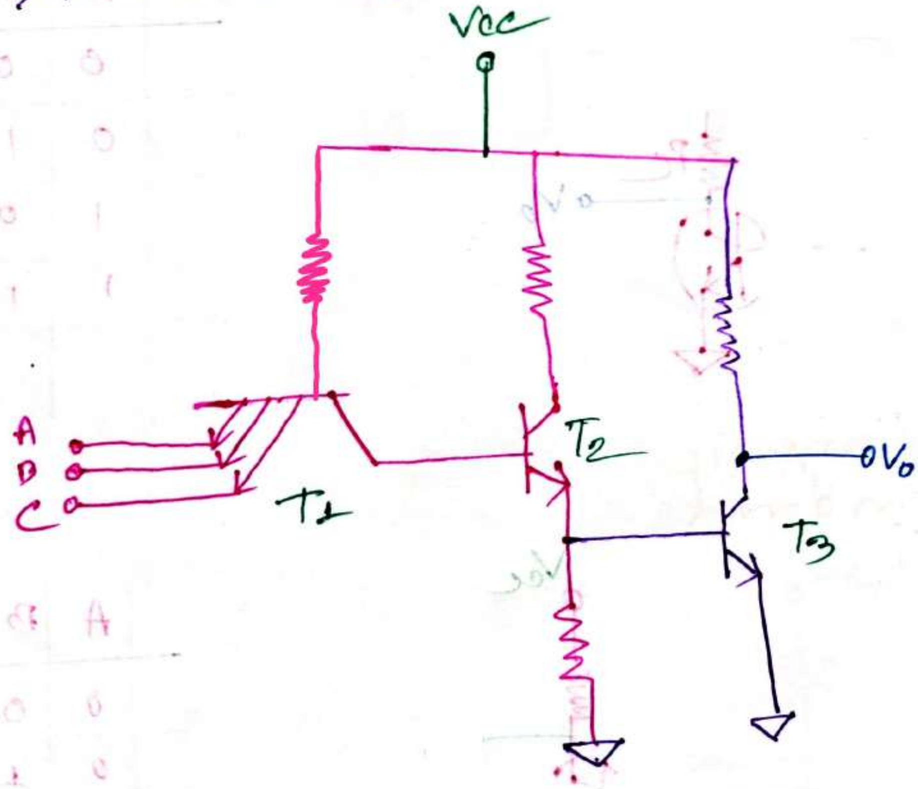
Transistor - Transistor Logic (TTL):-

TTL Inverter:-



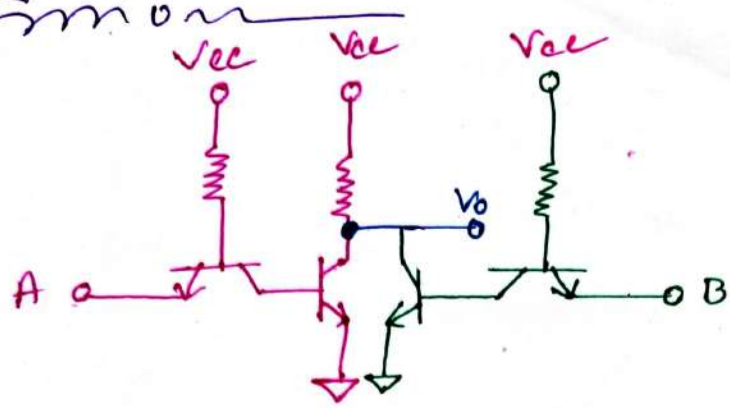
A	V_o
0	1
1	0

TTL NAND:-



A	B	C	V_o
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

TTL NOR:-



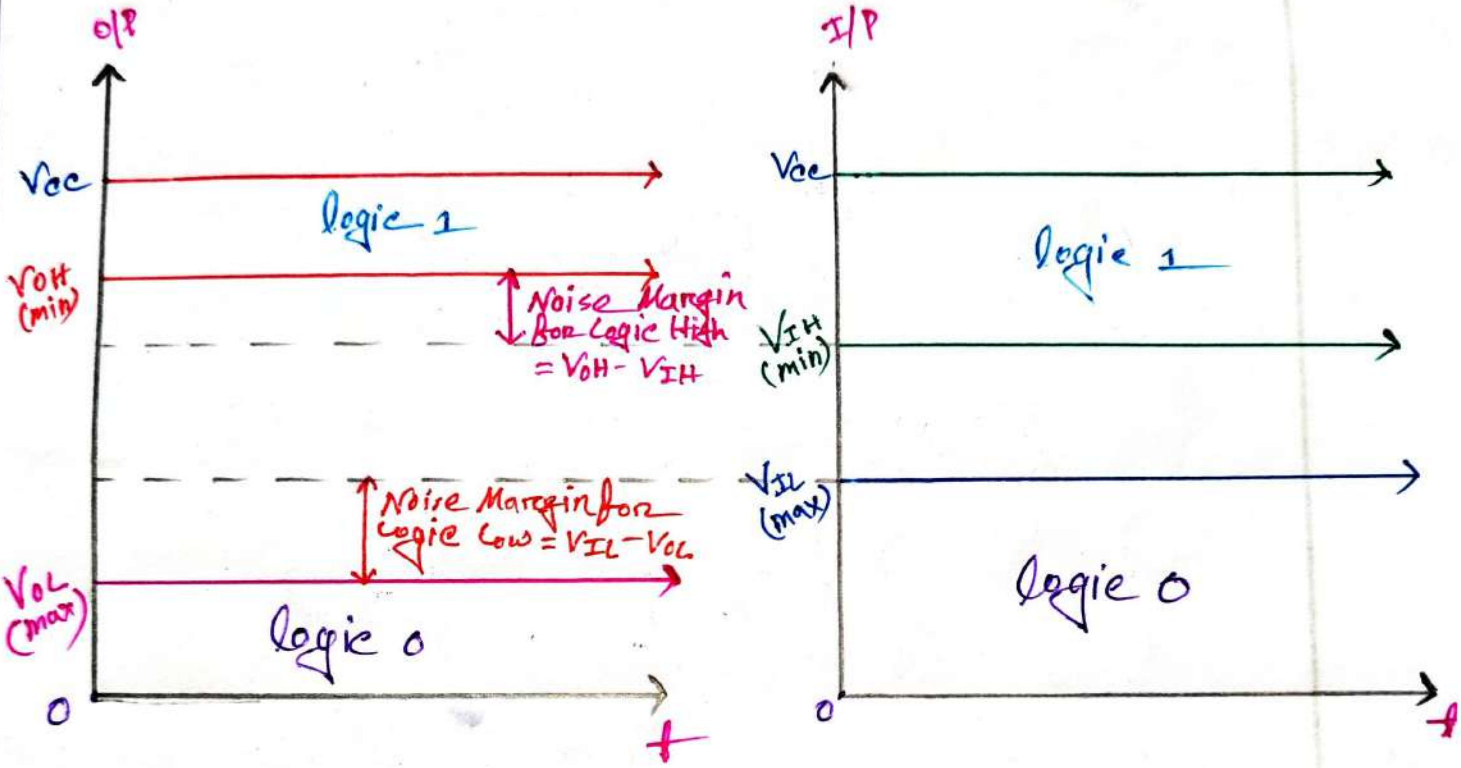
A	B	V_0
0	0	1
0	1	0
1	0	0
1	1	0

INTRODUCTION TO LOGIC LEVEL PARAMETERS

Subject ______ Date : / /

Digital IC Parameters:—

Voltage, Current, & Noise Margin:—



⇒ V_{OH} (min) → It is min o/p volt for Logic 1

⇒ V_{OL} (max) → It is max o/p volt for Logic 0

⇒ V_{IH} (min) → It is min I/P volt for Logic 1

⇒ V_{IL} (max) → It is max I/P volt for Logic 0

⇒ I_{OH} → High Level O/P Current.

⇒ I_{OL} → Low Level O/P Current

⇒ I_{IH} → High Level I/P Current

⇒ I_{IL} → Low Level I/P Current.

Date : / /

Performance Parameters:-

⇒ the most important parameters that are evaluted and Compared are.

⇒ 1. Fan-Out
2. Fan-In
3. power-Dissipation.
4. propagation Delay.
5. speed-power product
6. Noise Margin.

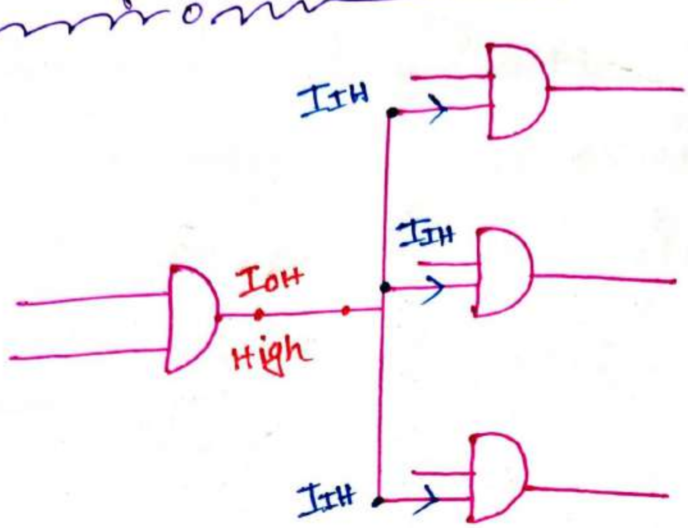
Fan-Out

⇒ It refers to how many outputs or devices a ~~Sig~~ signle Source can drive without performance Loss.

⇒ It should be as high as possible, so that we Can drive Large numbers of gates by Single gate.

⇒ It defines Current supply by given gate.

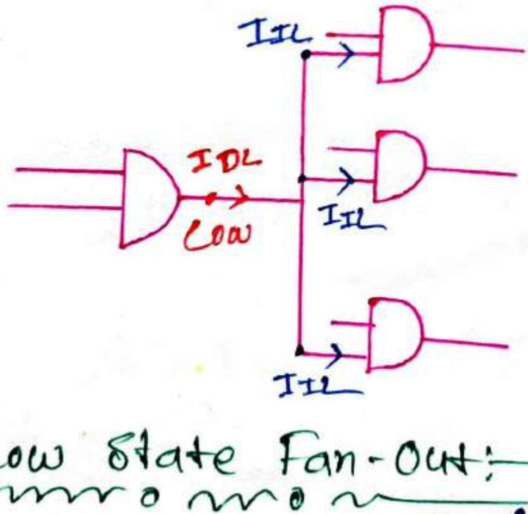
For High state Fan out:-



∴ high state Fan-Out:-

$$\text{High FanOut} = \frac{I_{OH\,(max)}}{I_{IH}}$$

For Low state FanOut:-



∴ Low State Fan-Out:-

$$\text{Low FanOut} = \frac{I_{OL(max)}}{I_{IL}}$$

∴ $$\boxed{\text{DC Fanout} = \min\left(\frac{I_{OH}}{I_{IH}}, \frac{I_{OL}}{I_{IL}}\right)}$$

problem: 1:-

⇒ Calculating the fan-out of a driving gate with the following parameters.

$$I_{OH} = 400\mu A \; ; \; I_{OL} = 40\mu A$$
$$I_{IH} = 16mA \; ; \; I_{IL} = 2mA$$

⇒ We know:-

$$FO = min\left(\frac{I_{OH}}{I_{IH}}, \frac{I_{OL}}{I_{IL}}\right)$$
$$= min\left(\frac{400\mu}{16m}, \frac{40\mu}{2m}\right)$$
$$= min(25m, 20m)$$
$$= 20m \text{ Units}.$$

problem:2:-

⇒ Calculating the fan out of a driving gate with the following parameters.

$$I_{OH} = 400\mu A \; ; \; I_{OL} = 16mA$$
$$I_{IH} = 40\mu A \; ; \; I_{IL} = 2mA$$

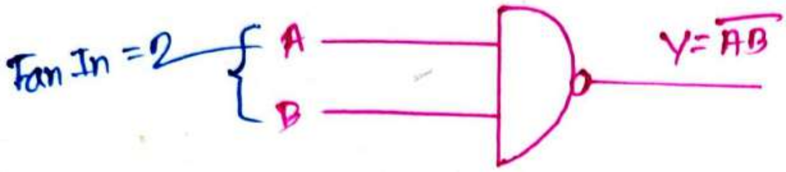
⇒ We know:-

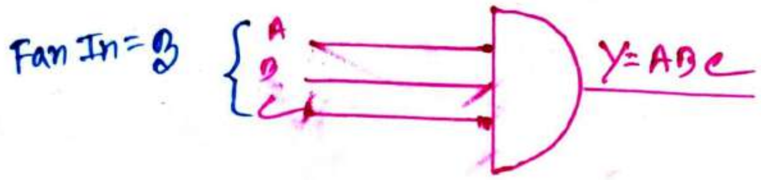
$$FO = min\left(\frac{I_{OH}}{I_{IH}}, \frac{I_{OL}}{I_{IL}}\right)$$
$$= min\left(\frac{400\mu A}{40\mu A}, \frac{16mA}{2mA}\right)$$
$$= min(10, 8)$$
$$= 8 \text{ Units}.$$

Ans:-

Fan In :-

⇒ Fan in of Logic gate is defined as number of inputs that the gate is designed to handle.



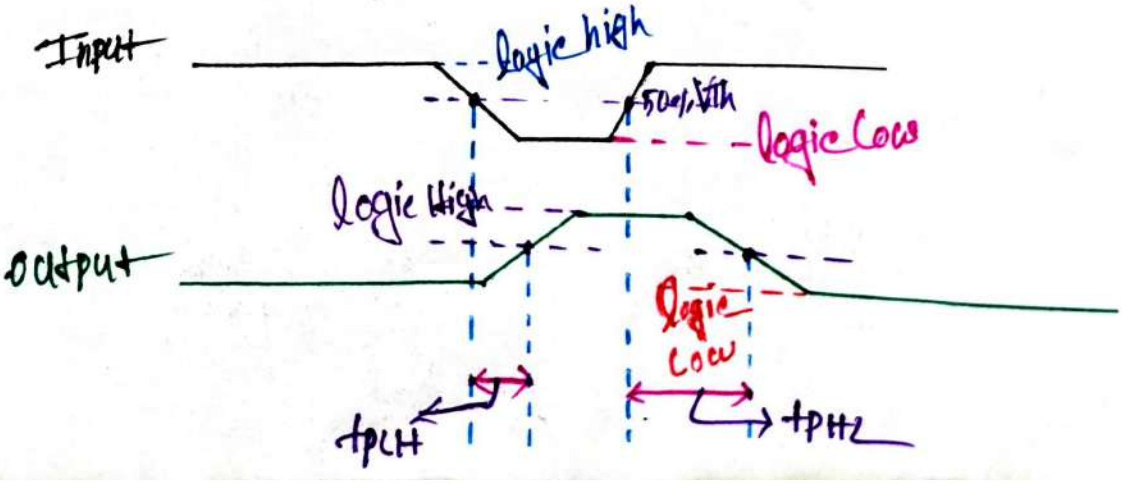


Propagation Delay :- (t_p)

⇒ It is average transition time of signal from I/p to O/p.

⇒ It define operating speed of IC

⇒ It is measure in terms of "n" sec.



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$$\therefore \boxed{t_p = \frac{t_{pLH} + t_{pHL}}{2}}$$

t_{pLH} = Delay for Logic Low to Logic High.

t_{pHL} = Delay for Logic High to Logic Low.

⇒ problems 1:—

⇒ Find the propagation Delay where $t_{pLH} = 5 nsec$, $t_{pHL} = 7 nsec$

⇒ we know:—

$$t_p = \frac{t_{pLH} + t_{pHL}}{2}$$

$$= \frac{7n + 5n}{2}$$

$$= 6n \text{ sec.}$$

Power Dissipation:—

⇒ It is the amount of power dissipated in an IC in form of heat.

∴ Total power Dissipated (P_d)

$$P_d = V_{cc} \times I_{cc (Avg)}$$

$$I_{cc} = \frac{I_{ccH} + I_{ccL}}{2}$$

I_{ccH} → I_{cc} Current with logic High at output

Time: Date: / /

I_{ccL} = I_{cc} Current with logic low at output.

∴ power Dissipation per logic gates:-

$$\therefore P_{dn} = \frac{V_{cc} \times I_{cc\,(Avg)}}{n}$$

n = no. of logic gate.

$V_{DD} = V_{cc}$ = Supply voltage.

problems 1:- Find the power dissipation (P_d) if $V_{cc} = 5V$, $I_{ccH} = 0.5mA$, $I_{ccL} = 1mA$.

⇒ We know:-

$$I_{cc} = \frac{I_{ccH} + I_{ccL}}{2}$$

$$= \frac{0.5mA + 1mA}{2}$$

$$= 0.75mA$$

$$\therefore P_d = V_{cc} \times I_{cc}$$

$$= 5V \times 0.75mA$$

$$= 3.75mW$$

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Speed Power Product

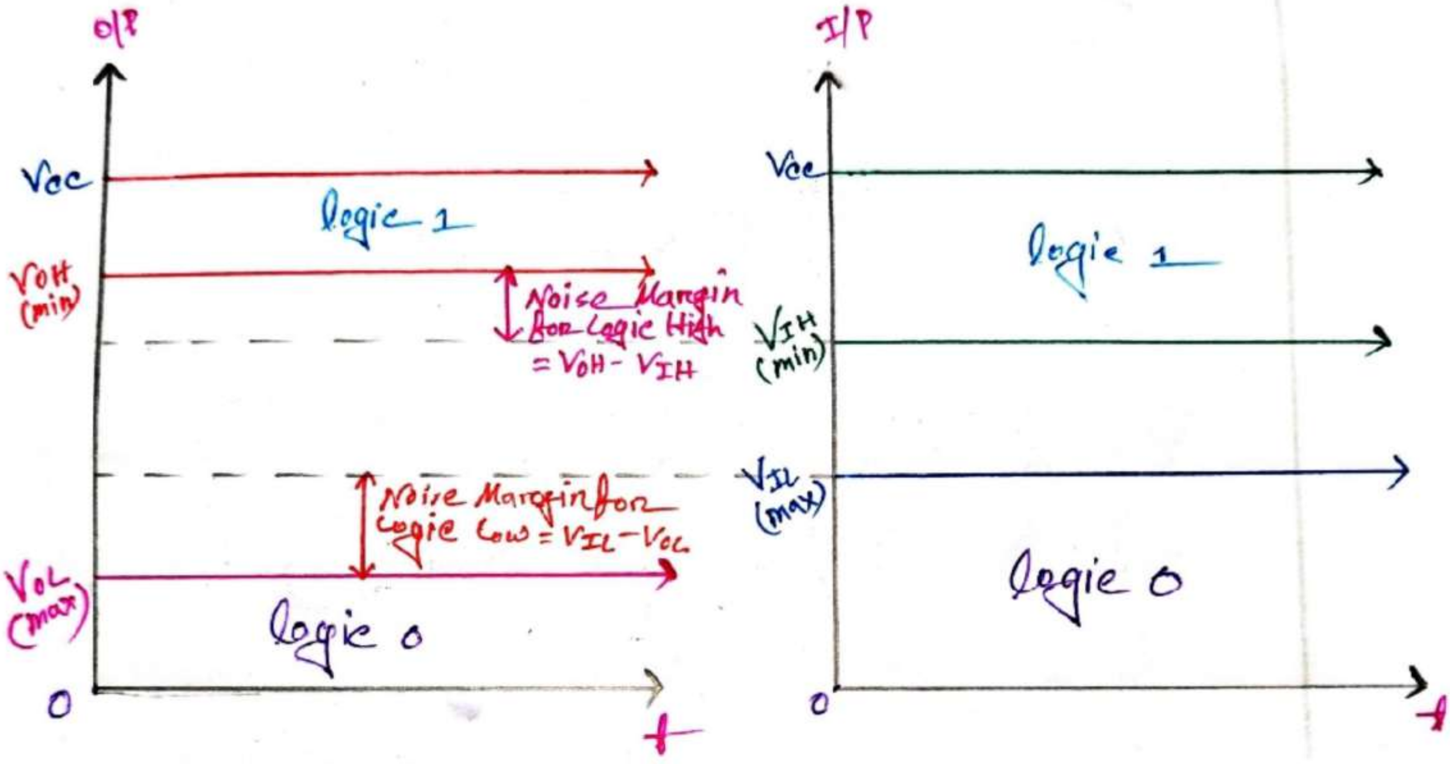
⇒ It is the product of "power Disspation" & "propagation Delay".

⇒ It also known as "Figure of Merits".

$$\boxed{SPP = \text{power Dissipation} \times \text{propagation delay}}$$

Note:- Good IC have Lower SPP

Noise Margin:-



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Noise Margin for Logic High (V_{NH})

∴ The High Level noise margin

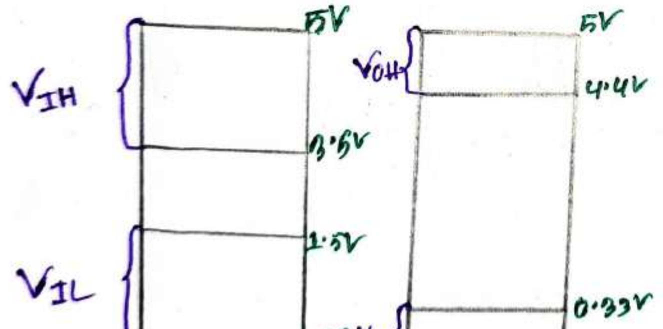
$$\boxed{V_{NH} = V_{OH} - V_{IH}}$$

Noise Margin for Logic Low (V_{NL}):—

∴ The Low Level noise margin:—

$$\boxed{V_{NL} = V_{IL} - V_{OL}}$$

problem 1:—



Find

(a) High level Noise Margine
(b) Low level Noise Margin.

(a)

Is given:—

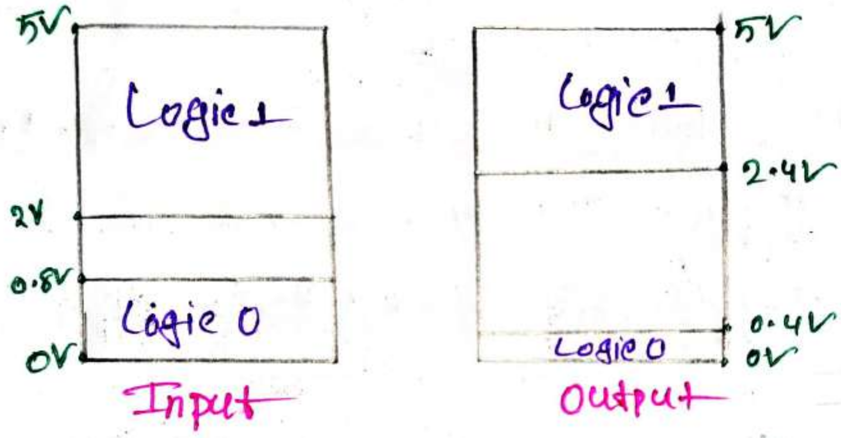
$V_{OH} = 4.4V$ | $V_{OL} = 0.33V$
$V_{IH} = 3.5V$ | $V_{IL} = 1.5V$

Sub: Time: Date: / /

We know—

$$V_{NH} = V_{OH} - V_{IH}$$
$$= 4.4 - 3.5$$
$$= 0.9V$$
$$V_{NL} = V_{IL} - V_{OL}$$
$$= 1.5 - 0.33$$
$$= 1.17V$$

problem 2:—



⇒ Find

(i) High level Noise margine.

(ii) Low level Noise margine.

Is given:—

$V_{OH} = 2.4V$; $V_{IH} = 2V$

$V_{OL} = 0.4V$; $V_{IL} = 0.8V$

(a)

$$V_{NH} = V_{OH} - V_{IH} = (2.4 - 2)\,V = 0.4V$$

(b)

$$V_{NL} = V_{IL} - V_{OL} = (0.8 - 0.4V) = 0.4V$$

Ans:—

Sub : Day Time : Date : / /

Problem 3:—

	$V_{OH(min)}$	$V_{OL(max)}$	$V_{IH(min)}$	$V_{IL(max)}$
Gate A	2.4V	0.4V	2V	0.8V
Gate B	3.5V	0.2V	2.5V	0.6V
Gate C	4.2V	0.2V	3.2V	0.8V

⇒ Find the

(i) High level Noise margine for each gate.

(ii) Low level Noise margine for each gate.

(a)

For Gate A:—

$$V_{NH} = V_{OH} - V_{IH} = 2.4 - 2 = 0.4V$$

$$V_{NL} = V_{IL} - V_{OL} = 0.8 - 0.4 = 0.4V$$

For Gate B:—

$$V_{NH} = V_{OH} - V_{IH} = 3.5 - 2.5 = 1V$$

$$V_{NL} = V_{IL} - V_{OL} = 0.6 - 0.2 = 0.4V$$

For Gate C:—

$$V_{NH} = V_{OH} - V_{IH} = 4.2 - 3.2 = 1V$$

$$V_{NL} = V_{IL} - V_{OL} = 0.8 - 0.2 = 0.6V$$

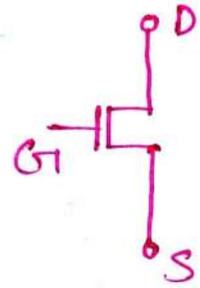
CMOS Logic Circuit

#CMOS:-

⇒ A CMOS circuit Consists 1-NMOS transistor & 1-PMOS transistor Connected in a Complementary Configuration.

Basic Information:-

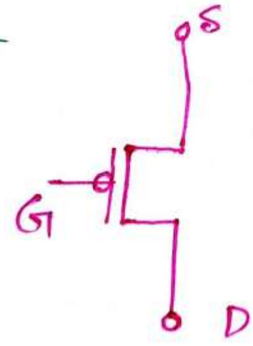
About N-MOS:-

n-mos → 

→ if G=0, off, (D to S as O.C)

→ if G=1, ON, (D to S as S.C)

About P-MOS:-

p-mos → 

→ if G=0, ON, (S to D as S.C)

→ if G=1, off, (S to D as O.C)

For Logic operation:-

For "AND" Operation

→ pmos → parallel.

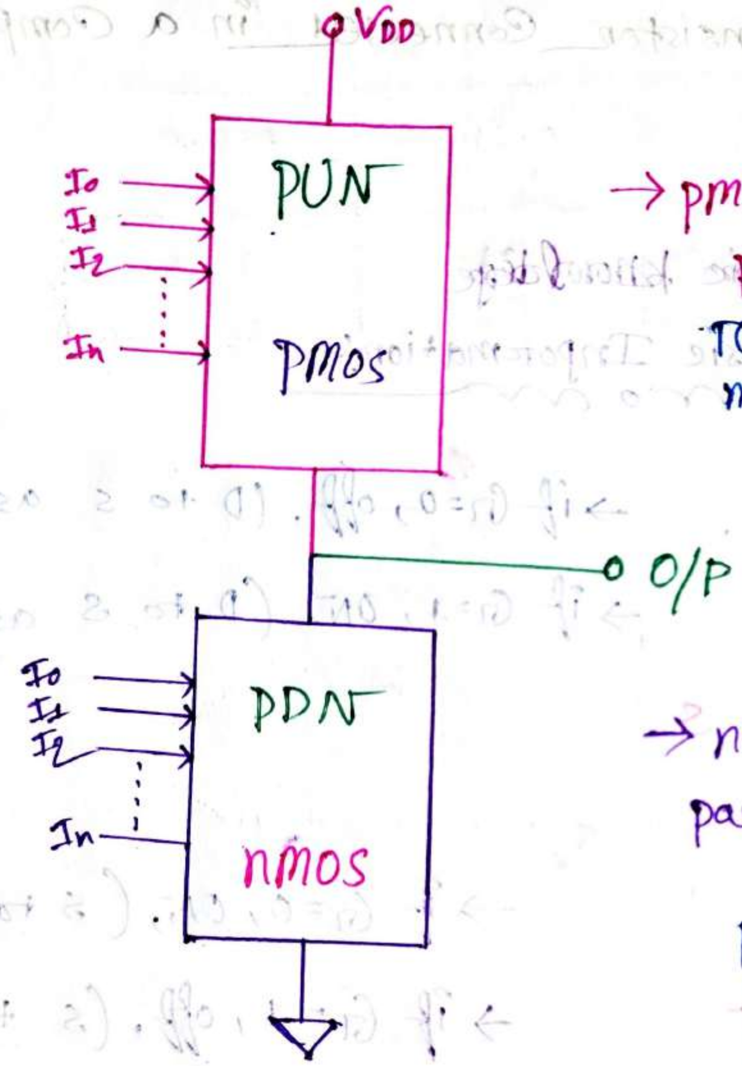
→ nmos → Series

For "OR" operation:-

→ pmos → Series

→ nmos → parallel.

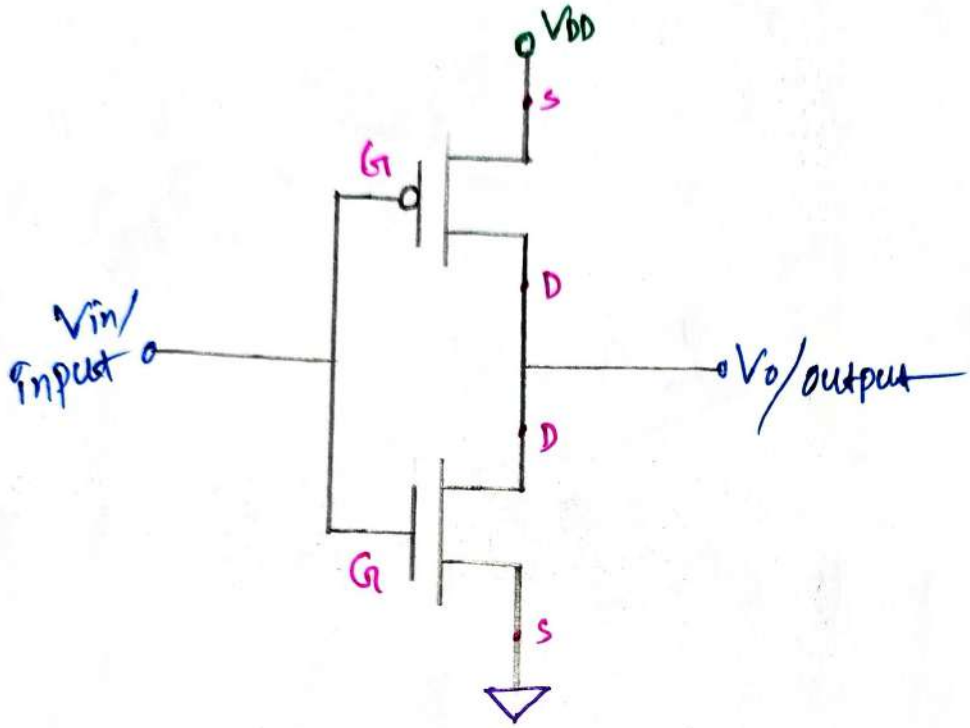
CMOS - Logic Generic Architecture:-



→ pmos Transistor can easily pass logic High. That the reason pull-up network (PUN) made by pmos only

→ n-mos Transistor can easily pass logic Low. That's why pull-down network (PDN) made by n-mos only

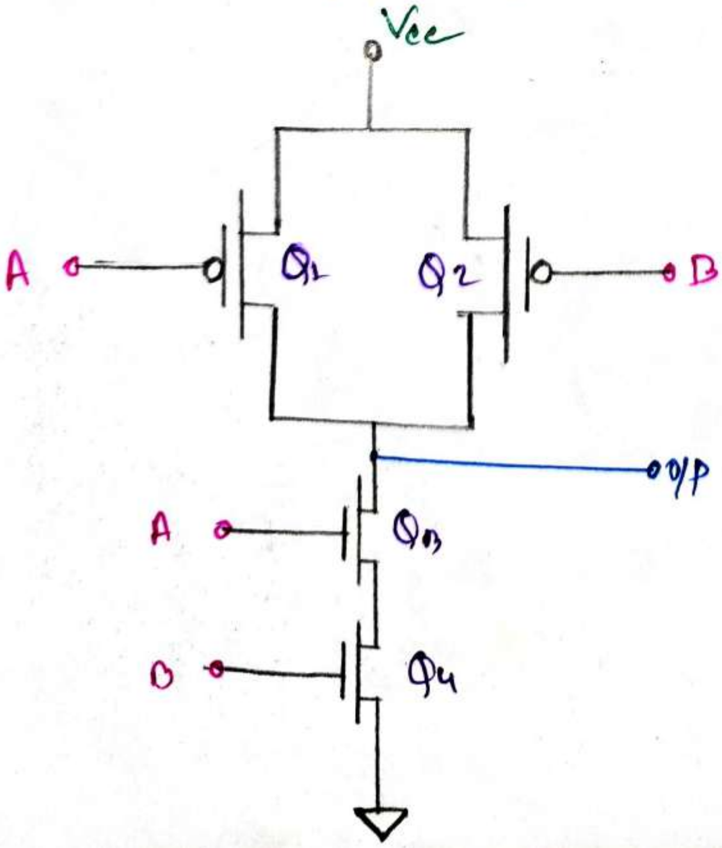
CMOS Logic - NOT Gate:-



Truth table:-

Vin	Vout
0	1
1	0

CMOS · Logic - NAND Gate:-



Truth table:-

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Logical Expression

$Y = \overline{A \cdot B}$

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Now the Logic Gate is:-

For CMOS NAND

A	B	Q_1 (PMOS)	Q_2 (PMOS)	Q_3 (NMOS)	Q_4 (NMOS)	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

Mathematical Approach

⇒ For "PDN", $\overline{Y} = A \cdot B$

$\therefore Y = \overline{A \cdot B}$

So, A & B should be Connected in Series for PDN

Since, AND operation

⇒ For, "PUN", $Y = \overline{A} \cdot \overline{B} + \overline{A} \cdot B + A \cdot \overline{B}$

$= \overline{A}(B + \overline{B}) + A \cdot \overline{B}$

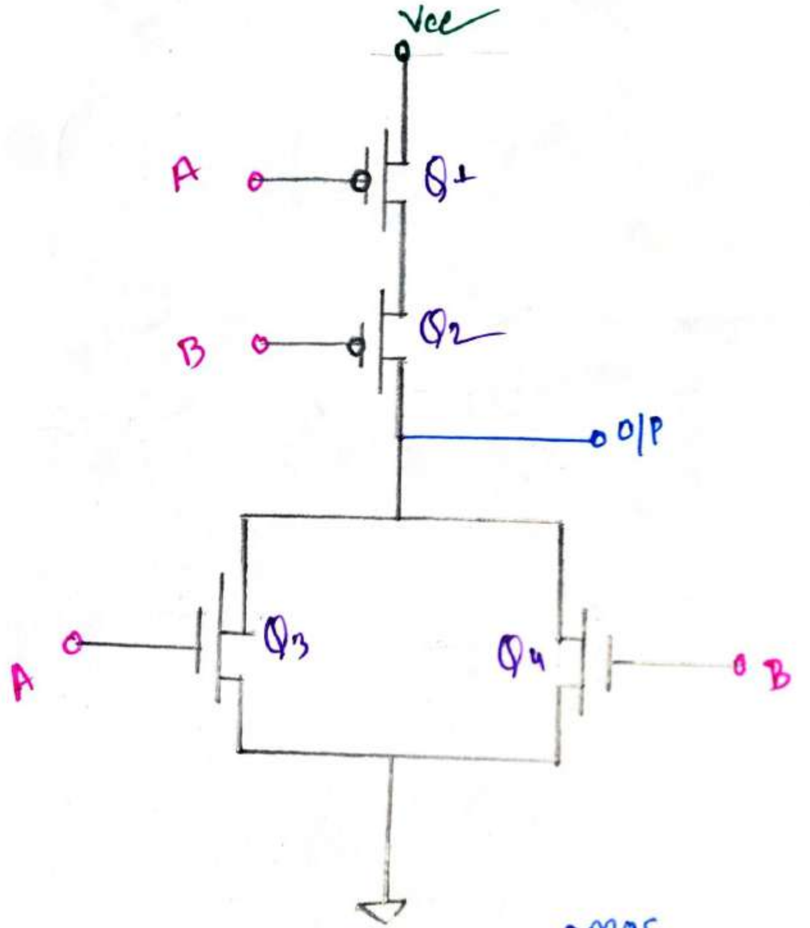
$= \overline{A} + A \cdot \overline{B}$

$= \overline{A} + \overline{B} = \overline{A \cdot B}$

So, A & B should be Connected in parallel for PUN.

Since AND operation.

C-mos Logic N-OR Gate:-



Truth table:-

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Logical Expression:-

$Y = \overline{A + B}$

A	B	Q1	Q2	Q3	Q4	Y
		pmos		n-mos		
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	0
1	0	OFF	ON	ON	OFF	0
1	1	OFF	OFF	ON	ON	0

Mathematical Approach:-

⇒ For "PDN", $\bar{Y} = \bar{A}\cdot B + A\cdot\bar{B} + A\cdot B$

$\bar{Y} = A + B$

So, $Y = \overline{A+B}$

So, A & B should be connected in parallel for "PDN"

⇒ For "PUN", $Y = \bar{A}\cdot\bar{B}$

$Y = \overline{A+B}$

So, A & B should be connected in series for "PUN"

Boolean Function Implementation Using CMOS:-

The key factor:-

→ For AND (·) operation:-

→ nmos → series.

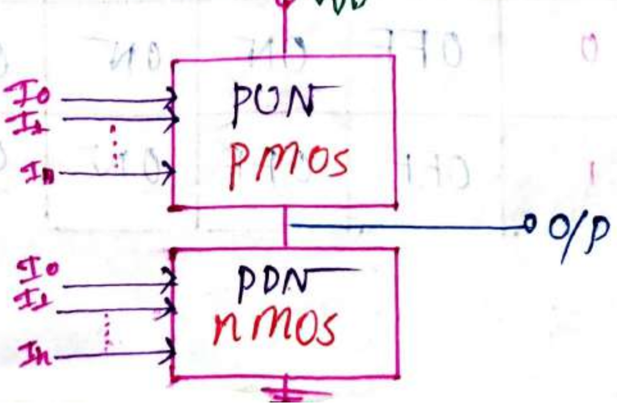
→ pmos → parallel.

→ For OR (+) operation:-

→ nmos → parallel.

→ pmos → series

Basic CMOS Circuit



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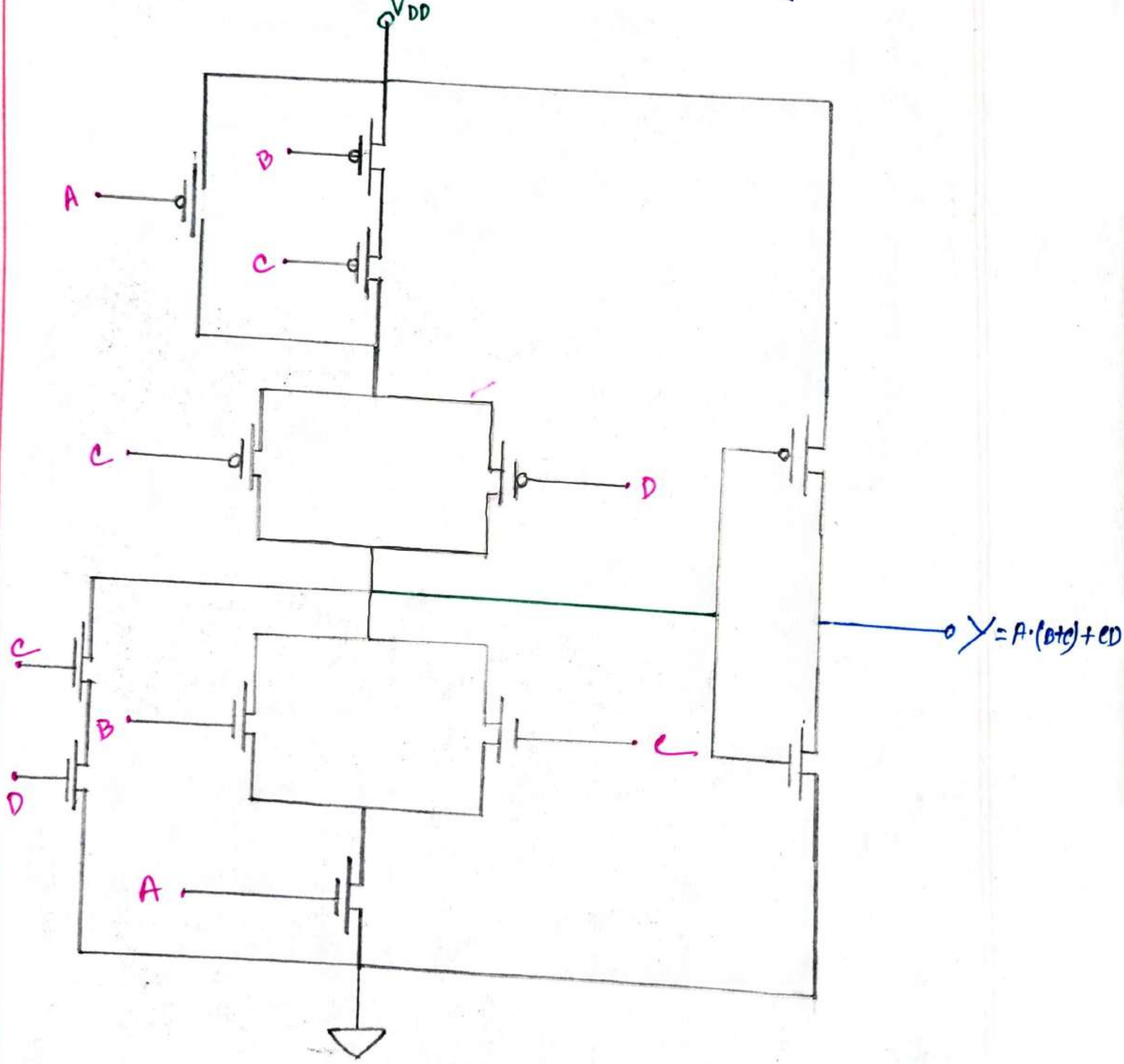
problem: 1!-

=> Impliment $Y = A \cdot (B+C) + CD$ using cmos transistor



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problems 2:—

⇒ Implement $F = \overline{A + B \cdot C}$ in CMOS Logic